



Folang Programming Language

Folang is a general-purpose programming language designed to be **expressive, consistent, and extensible**, merging functional fluency with object-centric abstractions.

Normative Status

This document is the authoritative and normative definition of the FoLang programming language.

All FoLang syntax, semantics, type-system rules, name-resolution rules, execution behavior, and other language requirements are defined by this document.

Every addition, correction, clarification, modification, or extension to the FoLang language must be recorded in this document before it becomes part of the language.

A compiler or other language implementation may use any internal architecture, parser, intermediate representation, runtime, or backend. However, to be identified as a conforming FoLang implementation, its externally observable behavior must adhere to the requirements defined in this document.

When an implementation conflicts with this specification, the specification governs unless the discrepancy is formally accepted and incorporated as a specification revision.

Implementation-specific extensions must be clearly identified as extensions and must not be represented as standard FoLang features unless they have been incorporated into this specification.

Design Overview

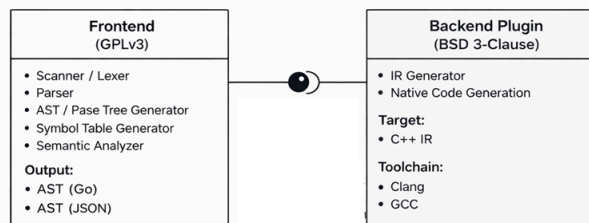


Figure: Frontend and Backend are independent components

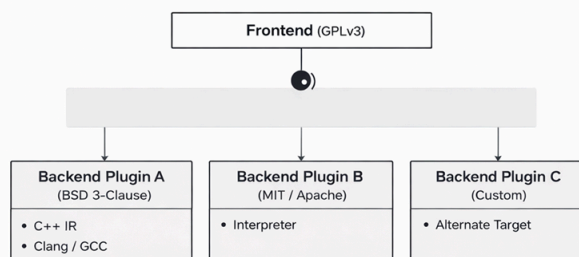


Figure: Multiple backend implementations can coexist or be swapped without modifying the Frontend.

Folang follows a deliberately different approach from conventional programming language designs. The system is structured to ensure **clear separation of concerns, license isolation, and extensibility through well-defined integration boundaries**.

Compiler and Backend

1. Frontend

The Frontend is responsible for source-level analysis and semantic processing of FoLang programs.

Components

- Scanner / Lexer
- Parser
- AST / Parse Tree Generator
- Symbol Table Generator
- Semantic Analyzer

Implementation

- Implemented in **Go**
- Generates AST representations in **Go structures** or **plain JSON**

License

- **GNU General Public License v3 (GPLv3)**
-

2. Backend

The Backend is responsible for transforming validated frontend output into executable artifacts.

Default Backend should be downloaded/build separately. They are not bundled with Frontend Binary

Components

- Intermediate Representation (IR) Generator
- Native Binary Executable Generation

Implementation

Implemented in any language. The frontend emits HIR over an IPC boundary in the declared wire format. Config declares the full protocol, schema, and wire format.

Default Backend

- Backend orchestration implemented in **Go**
- Code generation target is **C++**
- Uses **Clang** or **GCC** to generate native binaries from generated C++ IR

License

3rd Party Backends can have their own licensing terms and implementation choices. Default backend has the following license. Default backend is not part of complete compiler binary and is separate should be downloaded and configured using configuration file.

- **BSD 3-Clause License *****

Configuration File Structure

Informs the frontend how to generate IR to be consumed by backend. This process is not different for default backend.

```
{
  "protocol": "folang-plugin/1.0",
  "hir_schema": "folang-hir/1",
  "wire": "protobuf",
  "output-folder": "<absolute-path>",
}
```

Licensing Summary

Layer	Responsibility	Implementation	License
Frontend	Parsing and semantic analysis	Go	GPLv3
Backend (default)	IR processing and native code generation	Go (orchestration) + C++ (target)	BSD 3-Clause

The copyrightable material in the [FoLang Language Definition and Documentation](#), including its syntax, grammar, and semantic-rule descriptions, is licensed separately under [CC BY 4.0](#). ***

3. Capability Security Model

FoLang’s compiler ships with all language features compiled in but **systems and FFI features are disabled by default**. The compiler has no hardcoded keys — capability configuration happens entirely at install time. This moves authorization from source code (developer-controlled) to the compiler installation (organization-controlled).

Feature Tiers

Tier	Features	Default State
<code>application</code>	All standard language features, <code>co.net</code> , <code>co.core</code> , <code>co.encoding</code> , <code>co.crypto</code> , etc.	✔ Always enabled
<code>system</code>	Raw pointers, pointer arithmetic, <code>co.sys.unsafe</code> , MMIO, heap allocators	🔒 Disabled — requires install-time configuration
<code>ffi</code>	<code>@co.dap.native</code> , <code>co.sys.ffi</code> , extern types, <code>co.lang.void</code> pointers, C ABI	🔒 Disabled — requires install-time configuration

Quick Start

Hello World

```
// hello.fol – entry file, no annotation needed
co.out.println("Hello FoLang!")
```

Or with an alias for shorter form:

```
@co.ddap.alias(co.out, as="out")

out.println("Hello FoLang!")
```

Variables

```
// typed
name co.lang.string = "Rao";
age co.lang.int      = 30;

// inferred from value – := errors if already declared
name := "Rao";
age  := 30;

// define infer and assign if not defined, otherwise assing new value
name ?= "Kumar";
```

`co.lang.string` and `co.lang.int` are Builtin Data types to know more about Builtin Data types, ples refer seccion [Builtin Data Types](#)

`=`, `:=` and `?=` are built in operators to know more about Built in operators please refer section [Builtin Operators](#)

Single Source Application File

FoLang developers can create a complete executable program in one source file. A **single-source application** is an application whose entry file contains the complete program and does not depend on user package source files.

A single-source application file and an application entry file use the same entry-file grammar, context, and restrictions. This section presents the allowed constructs, so a developer can start programming without first reading the complete specification.

Allowed Constructs

The application file may contain:

- built-in directives that are valid for an entry file
- package and library imports
- import aliases declared with `as=`
- file-local aliases for `co.*` paths declared with `@co.ddap.alias`
- type aliases declared with `co.lang.type`
- new types declared with `co.lang.newtype`
- opaque types declared with `co.lang.opaquetype`
- dependent-type aliases and dependent-type usages that do not declare a type-constructor function
- subtype declarations
- supertype declarations
- non-capturing entry-local function-pattern groups
- capturing entry-local `let` function-pattern groups
- variable declarations, initialization, assignment, and mutation
- calls to built-in methods and functions
- calls to imported package and library APIs
- ordinary expressions
- comprehensions
- ternary expressions
- conditions and loops
- containment operations such as `contains`
- iterators such as `each`
- pattern matching and destructuring

```
a co.lang.int = 10;
b co.lang.int = 20;

co.out.println( a + b );
```

`co` is a keyword/reservedword in `foLang` for more details please refer section [Reserved Words](#)

`co` is a built in package in `foLang` for more details please check [Built in Packages](#)

`a + b` is an expression in `foLang`

More about Expression rules please refer section [Expressions](#).

Built-in and Imported Names

All `co.*` paths are always available.

A developer may use the complete built-in path:

```
co.out.println("Hello");
co.core.list.of(1, 2, 3);
```

A developer may optionally create a file-local alias:

```
@co.ddap.alias(co.out, as="out")
@co.ddap.alias(co.core.list, as="list")

out.println("Hello");
values := list.of(1, 2, 3);
```

Creating an alias does not hide the complete `co.*` name; both forms remain valid in that file.

Third party packages User packages and libraries are not automatically available. They must be imported using `@co.ddap.import`. When `as=` is present, the imported API is accessed through that alias. When `as=` is omitted, the complete imported package or library path must be used.

```
@co.ddap.import(package="hr.employee", as="emp")
first := emp.EmployeeService.find(1001);

@co.ddap.import(package="finance.payroll")
second := finance.payroll.calculate(request);
```

A developer needs to import the package to use

`@co.ddap.import` and `@co.ddap.alias` are built in directives to know more about built in directives please refer section [Built-in Directives](#)

for more details on `foLang` import directive please check [Import details](#)

`println` is a built in method in `foLang` for more details please check [Built in Methods](#)

Variable Kinds

`foLang` supports different kind of variables for different purposes, even though language provides, developers cannot use at random places. For more information where they are supported refer section [Variable Kind support](#)

Simple Variable Declaration

```
someVar co.lang.int;
someString co.lang.string;
```

With Initialization

```
someBool co.lang.bool = co.const.true;
someInt co.lang.int = 42;
```

With Type Inference

```
someVal := "Hello, World!";
someNum := 3.14; // if not defined, define and initialize; else throws error
someR ?= "Kamesh"; // if not defined, define and initialize; else reassign
```

Pointer Declaration

```
somePtr co.lang.int->>(*);
someDblPtr co.lang.int->(**);
```

Array Declaration

```
someArray      co.lang.int->([5]); // single dimension
someDb1Array   co.lang.int->([2,3]); // multi dimension
someJaggedArray co.lang.int->([2][3]); // jagged
someV1Array    co.lang.int->([...]); //varialbe length
someZeroLA     co.lang.int->([0]); //zero length array
someZeroDimA   co.lang.int->([.]); // zerodimension array
```

Array Declaration with Initialization

```
someInitializedArray  co.lang.int->([3]) = [1, 2, 3];
someInitializedArray1 co.lang.int->([ ]) = [1, 2, 3];
someInitializedDb1Array co.lang.int->([,]) = [[1, 2], [3, 4]];
```

Reference Declaration

```
someRef      co.lang.int->>(&); // reference
someLValueRef co.lang.int->>(&&); // LValue reference
someHpRef     co.lang.int->(~); // heap allocated reference
someAddr      co.lang.int->(@); // address
someThunk     co.lang.int->(^); // thunk
someSlice     co.lang.int->([:]); // slice
```

Range Declaration

```
// Typed range variable declaration
someRange co.lang.int->(..);

// Inferred range declarations
rangeI := 1..10; // [1, 10] ExcludeStart=false, ExcludeEnd=false
rangeS := 0<..5; // (0, 5] ExcludeStart=true, ExcludeEnd=false
rangeL := 0..<100; // [0, 100) ExcludeStart=false, ExcludeEnd=true
rangeB := 0<..
```

Auto and Dynamic Variable Declaration

```
someAutoVar co.lang.auto = "Hello"; // type inferred from value; initialization required
someDynamicVar co.lang.dynamic; // dynamic typing
```

Lazy

```
@co.dap.lazy
x = add(1, 2); //on doing some thing on x calls add(1,2) till that time add function on right hand side is not
invoked
```

Bind Variables

```
$[0-9]*
```

Discard / Wildcard Variable

```
_
```

Comma and Grouping

```
// Comma
x co.lang.int = 10, y co.lang.string = "Hello", z co.lang.bool = co.const.true;

// Grouping
(x co.lang.int = 10, y co.lang.string = "Hello", z co.lang.bool = co.const.true);
```

Fat Pointers

```
x co.lang.int->>(*, kind="", meta={});

co.lang.int->>(*, meta={});

y co.lang.int->>(*, meta={len:co.lang.usize, vtab:somepkg.VTable->(*)})

z co.lang.int->>(*kind=region, meta={})
```

```
Pointer
├─ base_type: T
├─ kind: <FatKind>
│   ├── thin
│   ├── slice
│   ├── relative
│   ├── trait
│   ├── buffer
│   ├── view
│   ├── opaque
│   ├── custom
│   ├── mem
│   ├── nullptr
│   ├── sptr
│   ├── uptr
│   ├── ptrdiff
│   ├── usize
│   ├── ssize
│   └─ (region) ← optional syntactic sugar
└─ meta:
    ├── region: heap | stack | global | numa(N) | mmio | constant | ...
    └─ len, cap, vtab, bits, endian, ...
```

Pointers for address manipulation

```
y co.lang.word->(repr=intptr);
z co.lang.word->(sign=unsigned, repr=uintptr);
p co.lang.word->(repr=ptrdiff);
n co.lang.word->(sign=unsigned, repr=usize);
m co.lang.word->(repr=isize);
o co.lang.void->(repr=nullptr);
```

Relative Pointers

```
z co.lang.int->>(*kind=relative, meta={})
```

Note Other than normal variable declaration rest have restrictions

Conditions, Loops and Iterators

Conditions

```
(boolean truth).do({
}).otherwise(boolean truth).do({
}).otherwise.do({
});
```

Loops

```
(boolean truth).loop({
}).otherwise(boolean truth).loop({
}).otherwise.loop({
});
```

Condition and Loop Mix

```
(boolean truth).do({
}).otherwise(boolean truth).loop({
}).otherwise(boolean truth).do({
}).otherwise.loop({
});
```

Ternary Operator

```
s = (boolean truth).return(some var/value).otherwise.return(some val/var);
s = (boolean truth).return(some var/val).otherwise(boolean truth).return(some var/val).otherwise.return(some var/val);
```

Looping Arrays / Lists / Maps / Ranges

```
arr co.lang.int->([5]) = [6,7,8,9,10];
arr.each(idx, val).do({
  co.out.print(idx);
  co.out.print(" :: ");
  co.out.println(val);
});

arr.each(_, val).do({
  co.out.println(val);
});
```

Array / List / Map / Range — Contains Element

```
arr co.lang.int->([5]) = [35,57,96,81,31];
k co.lang.int = 31;
arr.contains(k).do({
  co.out.println(k);
}).otherwise.do({
  co.out.println("Not Found");
});
```

Comprehensions *(planned)*

```
k := (1..10).filter(|x| => x % 2 == 0).map(|x| => x * x);

result := for (x <- List(1,2,3)).yield(x * 2)           // List(2, 4, 6)
result := for (x <- Set(1,2,3)).yield(x * 2)           // Set(2, 4, 6)
result := for (x <- Some(5)).yield(x * 2)             // Some(10)
result := for (x <- fetchData()).yield(x.process())  // Future

ages := {"A":30,"B":40,"c":66,"e":88};
upper := for ((name, age) <- ages).yield(name.toUpperCase, age)
```

Pattern Matching

```
x co.lang.int = 10;

x.match.case(n: n > 10 => { n = n+100; "GT" }).case( n: n < 10 => "LT").default("EQ");
x.match.case(n: n > 10 => { n = n+100; "GT" }).case( n: n < 10 => "LT").case(_=>"EQ");
x.match(co.pattern.Type).case(co.lang.int => ...).case(co.lang.float => ...);
x.match(co.pattern.Value).case(0 => ...).case(1 => ...);
x.match(co.pattern.Instance).case(xx.CAT => ...).case(xx.DOG => ...).default("Animal");
x.match(co.pattern.Object).case(xx.Ball => "Ball").case(xx.CAT => "CAT").default("Unknown");
x.match(co.pattern.Shape).case(Point{x, y} => ...).default(...);

x.match(co.pattern.Any).case(co.lang.int => ...).case(co.lang.float => ...).case(0 => ...).default( ...);

x.match(PositiveEvenMatcher).case(0 => "Neither even nor odd").case(2 => "First Even Prime").default(...);
```


Object vs Instance in FoLang: Instance is from types of class/structs. Objects are anything — functions, classes, structs, types, etc.

`_` is a special discard/wildcard variable usable only inside pattern matching, contains, and iterator constructs. Elsewhere `_` must be accompanied by an ASCII letter or number.

`PostiveEvenMatcher` is custom matcher for more details about creating custom matcher please refer to section [Custom Matcher](#)

Type Declarations

```
// Alias
x co.lang.type = co.lang.int;

// New
x co.lang.newtype = co.lang.int;

// Opaque
x co.lang.opaquetype = co.lang.int; //opaque types act like alias with base type from which they declared and
act like distinct type (new types) for others even when two opaque types derived from same base types

// ADT (tagged union)
y co.lang.type = co.lang.int | co.lang.char;

// Subtype / covariant
test co.lang.subtype = co.lang.int;

// Supertype / contravariant
test co.lang.supertype = co.lang.int;
```

Let Bindings

```
y co.lang.int = let({x = 10}).in({x + 1});
y co.lang.int = let({$ = 10}).in({$ + 1}); // $ refers to the value being defined

x co.lang.int = (x + 1).where(x = 10);
x co.lang.int = ($ + 1).where($ = 10);

offset := 100;

let adjust(0) = offset
let adjust(n) = n + offset
```

`$` is a special identifier usable in ordinary `let` binding expressions for recursive or self-referential expressions.

Ordinary `let` value-binding expressions remain available in language contexts that permit them, but they are forbidden directly in the application entry file. In the entry file, `let` is reserved exclusively for a named function-pattern group that captures at least one surrounding runtime binding. It cannot introduce an anonymous function, a general closure value, or a curried function.

Function Pattern

```
f(Some(x)) => { x + 1 }
f(None()) => { 0 }

// desugars to:
f(v) =>{
  v.match().case(x: Some(x) => x + 1).case(_: None() => 0);
}
```

Function-pattern groups are permitted in the application entry file as restricted entry-local dispatch helpers. A bare group cannot capture surrounding runtime variables. A `let` function-pattern group must capture at least one already initialized entry-file runtime binding and is the only entry-file construct that permits such capture. Neither form permits ordinary function declarations, anonymous functions, general closure values, currying, partial application, or escape as a function value.

More about Let and function patterns please refer section [Let and Function Patterns](#)

Single source application file is for testing and getting feel of `fo1ang` Real world applications contain more than single source application file they use concepts of abstraction, encapsulation, inheritance and polymorphism at the core with many other features. Also these applications depend on external libraries, packages to achieve clear boundaries with above features. To develop these kind of applications we need following features at minimum.

1. [package source files](#) under [packages](#)
2. [Entry File](#)
3. [Libraries](#) and [Library surface files](#)
4. [imports](#)

FoLang Supports many features to develop enterprise application with [intent](#) and [FoLang Philosophy — Uniform Object Model](#) are listed below

Complete Feature list:

1. [Packages](#)
2. [UDT](#)
3. [Functions](#)
4. [Units](#)
5. [Imports](#)
6. [Macros](#)
7. [Templates](#)
8. [Annotations and Decorators](#)
9. [Type Classes](#)
10. [Types](#)
11. [Generics](#)
12. [Matchers](#)
13. [Lambdas](#)
14. [Execution Models and Control Abstractions](#)
15. [Extensions](#)
16. [Native code](#)
17. [Indexers](#)
18. [Type Constructors](#)
19. [Dynamic Runtime](#)
20. [Local/Nested Types and Functions](#)
21. [Libraries](#)
22. [Operators](#)
23. [Forward / Extern Declarations](#)
24. [Labels and Named Blocks](#)
25. [Reflections](#)
26. [Comprehensions](#)

In `fo1ang` User defined data types, function, macros, extensions, templates, typeclasses, type constructors, and units must be in its own file and under a package these files are called [package source files](#). , Lets discuss packages before going to UDTs and Functions

Packages

Package Identity

A subfolder containing `.fol` files is a package.

- Dot paths start from subfolders.
- The project root is **not** a package.
- The root folder name never appears in any package dot path.

Examples:

```
/apl/hr/          -> package "hr"
/apl/hr/employee/ -> package "hr.employee"
/apl/auth/        -> package "auth"
```

The project root itself is **not** a package.

Multi-File Packages

Multiple `.fol` files in the same subfolder automatically belong to the same package:

```
hr/employee/
├─ Employee.fol    → hr.employee
├─ EmpService.fol  → hr.employee
└─ EmpValidator.fol → hr.employee
```

Application Project Layout

```
/apl/
├─ app.fol          ← entry file – not a package
├─ hr/              package "hr"
│   └─ employee/    package "hr.employee"
│       ├── Employee.fol
│       └─ EmpService.fol
├─ payroll/         package "hr.payroll"
│   ├── Payroll.fol
│   └─ PayrollCalc.fol
├─ auth/            package "auth"
│   └─ Auth.fol
└─ bindings/        package "bindings"
    └─ CLib.fol
```

Package Aliasing

If there is a folder `/apl/hr/empl` and under that there is a `fol` file called `Employee.fol` then the import statement as we know will be

```
@co.ddap.import(package="hr.empl.Employee" , as="emp")
```

 where `as` is not a mandatory attribute

Now we want to change `empl` to `emp`, simple way is `change the folder name`, but we want to keep the `physical folder name` as is.

For example `/apl/hr/empl` should be named as `hr.emp` instead of `hr.empl`

```
emp co.lang.package;
```

The single line should be put in `package.fol` under `/apl/hr/empl`, and any normal `fol` cgicode must not use this name to the file it is a `restricted name` for a file in `folang`.

The import will be as below,

```
@co.ddap.import(package="hr.emp.Employee", as="emp" )
```

Note: This is a **Planned Feature** not finalized to be part of initial release.

UDT (User defined Data types)

`foLang` provides following constructs to create User Defined Data types, as mentioned UDTs must be in their own source file under package.

1. cstructs
2. structs
3. unionns
4. enums
5. classes
6. Modules
7. interface
8. signature

For more information about UDTs please refer section [Built In Kinds](#) ***

Struct Declaration

```
myStruct co.lang.struct={
    field1 co.lang.int;
    field2 co.lang.string;
    field3 co.lang.bool;
}
```

More about structs please refer section [Structs in detail](#) *** ### C-Struct Declaration

`co.lang.cstruct` is a C-like value type — passed by value, simple memory layout, safe to cross zone boundaries. Unlike `co.lang.struct` which is passed by reference, `co.lang.cstruct` is always copied on pass.

```
Point co.lang.cstruct = {
    x co.lang.int;
    y co.lang.int;
}

Rect co.lang.cstruct = {
    origin Point;
    width co.lang.int;
    height co.lang.int;
}
```

Enum Declaration

```
myEnum co.lang.enum={
    Variant1,
    Variant2,
    Variant3
}
```

Union Declaration

```
myUnion co.lang.union={
    intValue co.lang.int;
    strValue co.lang.string;
}
```

Class Declaration

```
Employee co.lang.class = {
  getEmployeeDetails()->(Employee) = empmodule.getEmployeeDetails;
  // assigning module function to class's method

  getEmployeeInfo()->(Employee) =>> empmodule.getEmployeeDetails();
  // delegating – internally redirecting the call to module function
}

// $1, $2, $3 ... are previous results captured as bind variables
Emp co.lang.class={
  dosomething(a co.lang.int, b co.lang.int)->
  (co.lang.int)=>>somePack.someMethod(a)=>>someOthPack.someOtherMeth($1, b);
}
```

More about classes please refer section [Classes in detail](#) ***

Interface vs Signature

```
// Employee is an ordinary package-level declaration.
MEmployee co.lang.signature = {
  ...
}
```

For more about signatures please refer section [signatures in detail](#)

```
IEmployee co.lang.interface = {
  ...
}
```

For more about interfaces please refer section [interfaces in detail](#)

Module Declaration

```
// Employee.fol – ordinary package-level type
Employee co.lang.struct = {
  ...
}

// EmployeeModule.signature.fol
EmployeeModule co.lang.signature = {
  ...
}

// EmployeeModImpl.fol
@co.dap.module(signature=EmployeeModule)
EmployeeModImpl co.lang.module->(signature=EmployeeModule, matches=EmployeeModule) = {
  ...
}
```

More about modules please refer section [Modules in detail](#) ***

Units

Unit Declaration

A `unit` is a named, non-instantiable container for functions.

Like other named primary declarations, a unit may use an explicit name or `_` for filename-derived naming. For example, `_` `co.lang.unit` in `Math.fol` declares `Math`, while an explicit name such as `Math co.lang.unit` is authoritative regardless of the filename.

Its purpose is structural: it prevents functions from flowing freely at package-file scope and preserves FoLang's rule that every ordinary package source file has one primary top-level declaration. A unit does **not** require its functions to form a domain model, capability, service, or other semantic abstraction.

Functions within a unit may be strongly related by purpose—for example, mathematical, parsing, encoding, or validation functions—or only loosely related. Semantic cohesion is encouraged as a source-design practice but is not enforced by the language.

A unit has two forms:

1. **Standalone unit** — its name does not match a struct in the same package. It acts as an ordinary named container for receiverless functions.
2. **Struct companion unit** — its name matches exactly one `co.lang.struct` in the same package. It provides behaviour associated with that struct while the struct declaration itself remains pure data.

Only `co.lang.struct` can have a companion unit. A same-name unit is not allowed for `co.lang.class`, `co.lang.cstruct`, modules, enums, unions, interfaces, signatures, or other declaration kinds. Classes already contain their own methods and lifecycle behaviour; `cstruct` remains a restricted C-compatible data representation.

```
Text co.lang.unit={  
  
}
```

for more about units please refer section [units in detail](#)

Matchers

Custom Matcher

```
@co.dap.matcher  
Matcher(T) = {  
  matchCase(value T, pattern co.lang.untyped) -> (co.lang.int, co.lang.MatchBindings);  
  // int return: 0 = no match, >0 = match  
}  
  
PositiveEvenMatcher co.lang.Matcher->(for=Matcher, type=co.lang.int) = {  
  matchCase(value co.lang.int, pat co.lang.untyped)->(co.lang.int, co.lang.MatchBindings) = {  
    // user logic  
  }  
}
```

Comprehensions *(planned)*

```
k := (1..10).filter(|x| => x % 2 == 0).map(|x| => x * x);  
  
result := for (x <- List(1,2,3)).yield(x * 2)           // List(2, 4, 6)  
result := for (x <- Set(1,2,3)).yield(x * 2)           // Set(2, 4, 6)  
result := for (x <- Some(5)).yield(x * 2)             // Some(10)  
result := for (x <- fetchData()).yield(x.process())   // Future  
  
ages := {"A":30,"B":40,"c":66,"e":88};  
upper := for ((name, age) <- ages).yield(name.toUpperCase, age)
```

Extensions

```
stringextension co.lang.unit={

  @co.dap.extension(fortype=co.lang.string, what=extends)
  upperCase()->(string)={
    return this.upper()
  }

  @co.dap.extension(fortype=[co.lang.string], what=overrides)
  equals(str string)->(bool)={
    this.return this == str
  }
}
```

Extensions must be **explicitly activated** — they are block-scoped:

```
@co.ddap.use(from="stringextension",extensions=[equals, upperCase])
k.upperCase(); //  explicitly activated
```

Reflections

```
@co.dap.reflection(enable=True, package="co.meta")

x co.lang.int = 10;
x.reflect().getType() → co.lang.int
x.reflect().getValue() → 10
x.reflect().getKind() → value
```

Type Classes

Monads, Applicatives, Functors, Monoids and Transformers

`@co.dap.typeclass(kind=...)` is the single annotation for all typeclass definitions. `kind` specifies the algebraic structure — `Functor`, `Applicative`, `Monad`, `Monoid`, `Transformer`, or any user-defined kind. Instances of any typeclass always use `co.lang.instance`.

Functor

```
@co.dap.typeclass(kind=Functor)
Functor(F) = {
  map(value F(A), f (A)->B) -> (F(B));
}

ListFunctor co.lang.instance->(for=Functor, type=List) = {
  map(value List(A), f (A)->B)->(List(B)) = {
    result = List(B){}
    value.each(_, item).do({ result.append(f(item)) });
    this.return result
  }
}
```

Applicative

```
@co.dap.typeclass(kind=Applicative)
Applicative(F) = {
  pure(x A) -> (F(A));
  apply(fab F(A->B), fa F(A)) -> (F(B));
}

OptionApplicative co.lang.instance->(for=Applicative, type=Option) = {
  pure(x A)->(Option(A)) = { this.return Some(x); }
  apply(fab Option(A->B), fa Option(A))->(Option(B)) = {
    this.return (fab, fa)
      .match
      .case((Some(f), Some(x)) => Some(f(x)))
      .default(None());
  }
}
```

Monad

```
@co.dap.typeclass(kind=Monad)
Monad(F) = {
  pure(x A) -> (F(A));
  flatMap(fa F(A), f (A)->F(B)) -> (F(B));
}

OptionMonad co.lang.instance->(for=Monad, type=Option) = {
  pure(x A)->(Option(A)) = { this.return Some(x); }
  flatMap(fa Option(A), f (A)->Option(B))->(Option(B)) = {
    this.return fa.match().case(Some(x) => f(x)).default(None);
  }
}
```

Monoid

```
@co.dap.typeclass(kind=Monoid)
Monoid(T) = {
  empty() -> (T);
  combine(a T, b T) -> (T);
}

IntMonoid co.lang.instance->(for=Monoid, type=co.lang.int) = {
  empty()->(co.lang.int) = { this.return 0; }
  combine(a co.lang.int, b co.lang.int)->(co.lang.int) = { this.return a + b; }
}
```

Transformer

```
@co.dap.typeclass(kind=Transformer)
Transformer(F(_), G(_)) = {
  map(value F(A), f (A)->B) -> (G(B));
}

ListToSetTransformer co.lang.instance->(for=Transformer, types=[List, Set]) = {
  map(value List(A), f (A)->B)->(Set(B)) = {
    result = Set(B){}
    value.each(_, item).do({ result.insert(f(item)) });
    this.return result;
  }
}
```

Reflection

```
@co.dap.reflection(enable=True, package="co.meta")

x co.lang.int = 10;
x.reflect().getType() → co.lang.int
x.reflect().getValue() → 10
x.reflect().getKind() → value
```

Labels and Named Blocks

```
// Label
outer:{
  // statements
}

//Blocks
{
  // statements
}

// Named Block
labelBlock co.lang.block={

}

labelBlock.expand();
```

imports

FoLang supports three import forms. User packages and libraries must be imported before use. When an import declares `as=`, symbols are accessed through that alias. When `as=` is omitted, the complete imported package or library path is used.

1. Normal Package Import

Use this for ordinary project packages.

```
@co.ddap.import(package="hr.employee", as="emp")

e := emp.getEmployee(1);
co.out.println(e.name);
```

Resolution:

```
package="hr.employee" -> /appl/hr/employee/
```

2. Source Library Import

Use this for same-owner workspace libraries whose source is available.

```
@co.ddap.import(package="com.abc.ffi", src-library=true, expect="ffi", as="ffilib")
```

Resolution:

```
package="com.abc.ffi", src-library=true -> /appl/com/abc/ffi.fo1
```

The resolved file must be a library surface file:

```
@co.dap.library(type="ffi")
ffilib co.lang.library={

}
```

Meaning:

- `package` gives the logical library path
- `src-library=true` means the leaf resolves to a single `.fol` surface file, not a folder
- `expect` is an import-site assertion; the compiler checks it against the actual library type
- only the projected surface API is visible; internal sources are not importable through normal package imports

3. Packaged Library Import

Use this for third-party or prebuilt libraries.

```
@co.ddap.import(library="hrlib", as="hr")
@co.ddap.import(library="paylib", as="pay")
```

Resolution:

```
library="hrlib" -> libs/hrlib.folenc, else libs/hrlib.folib
```

Only the packaged library's projected surface API is visible to the consumer.

Import Directive Fields

Field	Required	Default	Meaning
<code>package</code> or <code>library</code>	one required	—	logical package path or packaged library name
<code>src-library</code>	✗	<code>false</code>	when <code>true</code> , <code>package=</code> resolves to a source library surface file
<code>expect</code>	✗	inferred from library surface	expected library kind such as <code>ffi</code> , <code>system</code> , <code>advanced</code> , <code>dynamicvmrt</code> , or <code>application</code>
<code>as</code>	✗	none — full dot path required when omitted	local alias; valid FoLang identifier
<code>realm</code>	✗	<code>app</code>	isolation domain
<code>parent-realm</code>	✗	—	realm shadowing relationship

Notes:

- `as` is optional — when omitted, no short alias is created and the full imported package path must be used to access symbols
- dots are not allowed in `as`
- `expect` is validation, not the source of truth
- the source of truth is `@co.dap.library(type="...")` on the resolved surface file

Examples:

```
@co.ddap.import(package="hr.employee", as="emp")
emp.Employee

@co.ddap.import(package="hr.employee")
hr.employee.Employee
```

Valid `as` Values

<code>as="hr"</code>	✓
<code>as="v1_hr"</code>	✓
<code>as="v1.hr"</code>	✗
<code>as="123hr"</code>	✗

Realms

Realms provide import isolation, coexistence of versions, and controlled shadowing.

Realm declarations are always syntactically valid — you can always write `realm=` on any import. However realm isolation is **active only when the library marked with `dynamicvmrt`**. Without it, realm declarations are not valid and compiler will throw error.

Default realm:

```
app
```

Hierarchy example:

```
core
└─ app
   └─ app1
      └─ app3
   └─ app2
```

Rules:

- `realm` defaults to `app`
- if `parent-realm` is omitted, parent defaults to `app`
- `parent-realm` is meaningful only when `realm` is explicitly provided and is not `app`
- libraries are always static
- `@co.ddap.dynamicruntime` is allowed only on the application entry file

Version Coexistence

```
@co.ddap.import(package="hr", realm="app", as="hr")
@co.ddap.import(package="v1.hr", realm="x", as="v1_hr")
```

Shadowing

```
@co.ddap.import(package="a", realm="app", as="hr")
@co.ddap.import(package="a", realm="x", parent-realm="app", as="hr")
```

When the alias is the same and the child realm points to the parent realm, the child shadows the parent.

Import Binding Rules

Invalid

Same alias, same realm, different package:

```
@co.ddap.import(package="a", realm="app", as="hr")
@co.ddap.import(package="b", realm="app", as="hr") // error
```

Valid aggregation

```
@co.ddap.import(package="folderx.some", realm="app", as="hr")
@co.ddap.import(package="folderx.some", realm="app", as="hr")
```

Valid coexistence

```
@co.ddap.import(package="a", realm="app", as="hr")
@co.ddap.import(package="a", realm="x", as="v1_hr")
```

Valid shadowing

```
@co.ddap.import(package="a", realm="app", as="hr")
@co.ddap.import(package="a", realm="x", parent-realm="app", as="hr")
```

Cycles

Compiler error if any cycle exists through:

- package imports
- realm parent relationships

Examples:

- `packageA` imports `packageB`, and `packageB` imports `packageA`
- `realm="x", parent-realm="y"` and another import uses `realm="y", parent-realm="x"`

Symbol Resolution

Resolution Order

For symbols without a prefix:

1. current package symbol table
2. parent package chain upward
3. error if not found

For symbols with an alias prefix:

1. current package imported-context map
2. parent package chain imported-context maps
3. error if not found

For imports declared **without** `as`:

1. use the full imported package path directly
2. resolve it through imported-context lookup using that full path
3. error if not found

Example Context Graph

```
pp1 (grandparent package)
├─ SymbolTable
├─ ImportedContexts
└─ ChildCtxs

p1 (parent package)
├─ Parent -> &pp1
├─ SymbolTable
├─ ImportedContexts
└─ ChildCtxs

c1 (current package)
├─ Parent -> &p1
├─ SymbolTable
├─ ImportedContexts
└─ ChildCtxs
```

Resolution Examples

```
lookup "OwnType"
  -> current symbol table

lookup "ParentType"
  -> parent chain

lookup "hr.Employee"
  -> imported-context lookup for alias "hr"

lookup "hr.employee.Employee"
  -> imported-context lookup for full imported package path when no alias exists

lookup "unknown.Type"
  -> imported-context lookup fails -> compiler error
```

Short Summary

- folders define packages
- root is never a package
- each ordinary package source file has exactly one primary top-level declaration
- free functions in package source files must be enclosed in a `co.lang.unit`
- the application entry file is an executable, non-package context with its own restricted declaration rules
- entry-local type aliases, newtypes, opaque types, dependent-type aliases/usages, subtypes, supertypes, bare function-pattern groups, and capturing `let` function-pattern groups are allowed
- ordinary `let` value bindings, ordinary functions, anonymous functions, general closures, classes, structs, enums, cstructs, type constructors, generics, macros, templates, units, and reusable behavioral declarations are forbidden in the entry file
- `co.*` is always available and never imported
- `@co.ddap.alias` optionally shortens a `co.*` path; otherwise the complete path is used
- `@co.ddap.import(package="...")` imports normal packages
- `@co.ddap.import(package="...", src-library=true, ...)` imports same-owner source libraries
- `@co.ddap.import(library="...")` imports packaged external libraries
- `expect="..."` is an import-site assertion, not the source of truth
- `@co.dap.library(type="...")` is the source of truth for library kind
- every library surface exports only boundary `struct` / `cstruct` contracts and public function signatures
- surface function bodies are restricted boundary adapters and are hidden from consumer symbol tables
- application-family boundary structs cross by automatic deep snapshot
- system and FFI boundary cstructs cross by ABI value
- internal packages never depend on surface types; the surface converts between public and internal representations

Let and Function Patterns

Bare Function-Pattern Group

A bare function-pattern group does not capture surrounding runtime bindings:

```
classify(0) => { "zero" }
classify(n).where(n > 0) => { "positive" }
classify(_) => { "negative" }
```

A bare group may use:

- its parameters and names introduced by its patterns
- its own name for recursion
- entry-local type names
- `co.*` APIs
- imported package and library APIs

- compile-time symbols that do not require runtime capture

It may not reference an entry-file runtime variable from the surrounding context:

```
offset := 100;

adjust(n) => { n + offset }
// compiler error: bare function-pattern groups cannot capture `offset`
```

Capturing `let` Function-Pattern Group

The `let` form exists only to declare an entry-local function-pattern group that captures one or more surrounding entry-file runtime bindings:

```
offset := 100;

let adjust(0) = offset
let adjust(n) = n + offset

result := adjust(10);
```

The captured names must resolve to surrounding runtime bindings that are already declared and definitely initialized before the first clause of the group. Built-in names, imported declarations, type names, parameters, and the function's own recursive name are not captures.

A `let` function-pattern group must capture at least one surrounding runtime binding. When no capture is required, the bare form must be used:

```
let fib(0) = 1
let fib(1) = 1
let fib(n) = fib(n - 1) + fib(n - 2)
// compiler error: this group captures nothing; remove `let`

fib(0) => { 1 }
fib(1) => { 1 }
fib(n) => { fib(n - 1) + fib(n - 2) }
```

The `let` marker is therefore not an optional spelling of a bare function-pattern group. It explicitly requests restricted lexical capture.

Similarities

Bare and capturing `let` function-pattern groups both:

- group compatible clauses by function name and arity
- dispatch by parameter patterns and optional `.where(...)` guards
- evaluate clauses in normal pattern-selection order
- may call themselves recursively
- must maintain compatible parameter and result types across all clauses
- follow the normal pattern ordering, reachability, overlap, and exhaustiveness rules
- are private to the entry file
- cannot be imported, exported, or annotated with package visibility
- cannot be converted to, assigned as, passed as, or returned as function values
- cannot be partially applied or curried

Differences

Form	Surrounding runtime capture	Intended use
<code>name(pattern) => body</code>	No	Entry-local pattern dispatch that depends only on its arguments, recursion, built-ins, imports, and compile-time names

<code>let name(pattern) = body</code>	Yes, at least one capture required	Entry-local pattern dispatch that also depends on existing entry-file runtime bindings
---------------------------------------	------------------------------------	--

Clauses of the same name and arity must all use the same form. Mixing bare and `let` clauses in one function-pattern group is a compiler error.

A capturing `let` function-pattern group is still not a general closure facility. It is named, entry-local, non-first-class, and non-escaping. The compiler may internally retain a capture environment, but FoLang does not expose that environment as a closure object.

`let` cannot be used directly in the entry file for value bindings, anonymous functions, general closure expressions, or curried functions:

```
let base = 10;           // compiler error: entry-file let value binding
let result = base + 1;   // compiler error: entry-file let value binding
let add = (a, b) => a + b; // compiler error: anonymous function
let counter = closure { ... }; // compiler error: general closure expression
let add = a => b => a + b; // compiler error: curried function
```

The compiler may lower either function-pattern form to a private entry helper, but neither source construct is a general-purpose function declaration.

Package in detail

Package Identity

A subfolder containing `.fo1` files is a package.

- Dot paths start from subfolders.
- The project root is **not** a package.
- The root folder name never appears in any package dot path.

Examples:

```
/apl/hr/           -> package "hr"
/apl/hr/employee/ -> package "hr.employee"
/apl/auth/         -> package "auth"
```

The project root itself is **not** a package.

It may contain:

- the application entry file which is same as single source application file
- the packaged library surface file when the project itself is a library
- subfolders that define packages or same-owner source libraries

Multi-File Packages

Multiple `.fo1` files in the same subfolder automatically belong to the same package:

```
hr/employee/
├─ Employee.fo1   → hr.employee
├─ EmpService.fo1 → hr.employee
└─ EmpValidator.fo1 → hr.employee
```

Application Project Layout

```
/apl/  
├─ app.fol  
├─ hr/  
│   ├── employee/  
│   │   ├── Employee.fol  
│   │   └─ EmpService.fol  
│   └─ payroll/  
│       ├── Payroll.fol  
│       └─ PayrollCalc.fol  
├─ auth/  
│   └─ Auth.fol  
└─ bindings/  
    └─ CLib.fol
```

← entry file – not a package
package "hr"
package "hr.employee"
package "hr.payroll"
package "auth"
package "bindings"

Package Access Rules

Four access levels control visibility across package boundaries:

Annotation	Same Package	Parent	Sub-Package	Unrelated	Entry / Surface
@co.dap.public	✓	✓	✓	✓	✓
@co.dap.package	✓	✓	✓	✗	✗
@co.dap.protected	✓	✗	✓	✗	✗
@co.dap.private	✓	✗	✗	✗	✗

Meaning:

- @co.dap.public -> visible everywhere
- @co.dap.package -> visible across the package family
- @co.dap.protected -> visible downward into subpackages only
- @co.dap.private -> visible only inside the declaring package

Example:

```
// hr/EmployeeAccess.fol – package "hr"  
  
@co.dap.public  
EmployeeAccess co.lang.unit = {  
  
    @co.dap.public  
    getEmployee(id co.lang.int)->(Employee) = { ... }    // anyone can call  
  
    @co.dap.package  
    validateId(id co.lang.int)->(co.lang.bool) = { ... } // hr package family  
  
    @co.dap.protected  
    baseQuery()->(co.lang.string) = { ... }              // visible to subpackages  
  
    @co.dap.private  
    normalizeId(id co.lang.int)->(co.lang.int) = { ... } // private declaration  
}
```

co.* Paths and Aliases

co.* Is Always Available

All co.* paths are part of the language and are always in scope. They are never imported through @co.ddap.import.


```
co.out.println("hello")
co.in.readLine()
x co.lang.int = 42;
```

Being **in scope** does not automatically mean **permitted in every context**.

A package rule, library kind, or other compiler restriction may still forbid use of a particular `co.*` facility. In such cases the name is still known to the compiler, but its use is rejected at compile time.

`@co.ddap.alias`

Use aliases only to shorten `co.*` paths.

```
@co.ddap.alias(co.out, as="out")
@co.ddap.alias(co.core.list, as="list")
@co.ddap.alias(co.encoding, as="enc")

out.println("hello")
list.of(1, 2, 3)
enc.json.serialize(data)

// full form still works alongside
co.out.println("world")
```

Rules:

- target must be a `co.*` path
- alias must be a valid identifier
- alias scope is file-local
- aliasing user packages is not allowed; use `as=` in `@co.ddap.import`
- duplicate aliases in the same file are compiler errors
- when no alias is declared, the complete `co.*` path is used
- declaring an alias does not disable or hide the complete `co.*` path ***

Package Source Files

A `.fo1` file located inside a package folder contains **exactly one primary top-level declaration**. This gives every package source file a clear structural identity and prevents unrelated declarations or loose functions from being mixed at file scope.

The primary declaration may be a:

- class
- struct
- cstruct
- enum
- union / ADT
- interface
- signature
- module
- type classes/instance
- type constructors
- patterns or objects
- macro
- template
- generics
- annotations
- decorators
- type, type alias, newtype, or opaque type
- unit

File-level import directives, aliases, and annotations may appear before the primary declaration. They do not count as additional primary declarations.

Forbidden directly at package-file scope:

- free-flowing functions
- variables or mutable state
- executable statements
- explicit package declarations
- project metadata
- library metadata
- multiple unrelated primary declarations

The application entry file and library surface files are special source forms and follow their own rules. The single-primary-declaration rule in this section applies to ordinary files inside package folders.

Primary Declaration Names and Filename Inference

The name of a primary declaration may be written explicitly or inferred from the source filename.

When `_` appears in the primary declaration-name position, the compiler derives the declaration name from the filename:

```
// Employee.fo1
_ co.lang.struct = {
  id   co.lang.int;
  name co.lang.string;
}
```

This declares `Employee co.lang.struct`.

In this position, `_` does not declare an anonymous or discarded declaration. It means **use the filename-derived declaration name**.

An explicit declaration name is authoritative and overrides filename inference. The explicit name is not required to match the filename:

```
// employee_model.fo1
Employee co.lang.struct = {
  id   co.lang.int;
  name co.lang.string;
}
```

This still declares `Employee`; `employee_model.fo1` is only the source filename.

Name-selection precedence is therefore:

```
explicit declaration name  -> use the explicit name
_                          -> derive the name from the filename
```

Filename derivation rules:

- for `Name.fo1`, the inferred declaration name is `Name`
- for a kind-qualified filename such as `Name.unit.fo1`, the compiler removes both `.fo1` and the trailing kind suffix when that suffix matches the declaration kind
- the inferred result must be a valid FoLang identifier
- a kind-qualified filename whose suffix conflicts with the declaration kind is a compiler error
- filename inference supplies only the declaration name; generic parameters and all other declaration details remain explicit

Examples:

```
// Status.enum.fol
_ co.lang.enum = {
  active;
  inactive;
}
// inferred name: Status
```

```
// Box.struct.fol
_(T) co.lang.struct = {
  value T;
}
// inferred name: Box; T remains explicit
```

```
// Employee.struct.fol
_ co.lang.class = {
  ...
}
// compiler error: filename kind `struct` conflicts with declaration kind `class`
```

Filename inference may be used by named primary declarations such as classes, structs, cstructs, enums, unions, interfaces, signatures, modules, instances, objects, units, and named type declarations.

Example containing a data declaration:

```
// hr/employee/Employee.fol
_ co.lang.struct = {
  id co.lang.int;
  name co.lang.string;
}
```

When a file needs only ordinary free functions, those functions must be enclosed in a `co.lang.unit`:

```
// hr/employee/EmpService.fol
_ co.lang.unit = {

  getEmployee(id co.lang.int)->(Employee) = {
    this.return Employee{ id: 1, name: "Rao" };
  }
}
```

Application Entry File

The application entry file is a **special executable source form** located at the application root. It is not a package, does not create an importable namespace, and is not subject to the ordinary package-file rule requiring exactly one primary declaration.

The compiler creates a dedicated **entry-file context** for it:

```
ApplicationEntryContext
├─ file directives, imports, and aliases
├─ entry-local non-structural type declarations
├─ non-capturing function-pattern groups
├─ capturing `let` function-pattern groups
└─ executable statements and expressions
```

Everything declared directly in this context is private to the entry file. Entry-local symbols cannot be imported, exported, or referenced by package or library source files.

Allowed Entry-File Constructs

The application entry file uses exactly the same grammar and allowed-construct rules described in [Single Source Application File](#). The earlier section provides the developer-facing overview; this section defines the entry file's formal context, privacy, and dependency direction.

Entry-Local Function Patterns

Function-pattern groups are allowed as a special entry-file construct even though ordinary function declarations are forbidden. FoLang provides two entry-file forms with the same pattern-dispatch model but different capture semantics.

Entry-File Dependency Direction

The entry file may depend on packages and libraries, but packages and libraries may never depend on the entry context:

```
application entry file
  ↓ uses
packages and libraries
```

This allows the entry file to coordinate application startup while preserving package and library independence.

Libraries

Library Project Layout

```
/hrlib/
├─ hrlib.fol           <- library surface – @co.dap.library
├─ emp/                package "emp" – internal, invisible to consumer
│   └─ Employee.fol
│   └─ EmpService.fol
└─ auth/               package "auth" – internal, invisible to consumer
    └─ Auth.fol
    └─ AuthService.fol
```

Consumer only sees what `hrlib.fol` declares. All subfolders are internal.

Library Surface file

FoLang uses surface files in two situations:

1. Packaged library project surfaces
2. Application-workspace source library surfaces

A surface file is a special source form annotated with `@co.dap.library`. It defines one library identity, its public boundary data contracts, and the boundary-adapter functions through which consumers call the library.

```
app.fol    -> application entry
hrlib.fol  -> packaged library surface
ffi.fol    -> source library surface when imported with src-library=true
```

A library surface is not an ordinary package file. It may contain multiple boundary data declarations and public functions inside one `co.lang.library` declaration.

Packaged Library Project Surface

A packaged library project has no application entry file. It has exactly one surface file at the project root.

```
/hrlib/
├─ hrlib.fol           <- library surface
├─ emp/                <- internal package
│   └─ Employee.fol
│   └─ EmployeeService.fol
└─ auth/               <- internal package
    └─ AuthService.fol
```

Rules:

- exactly one `@co.dap.library` surface file exists per packaged library project
- the surface file is located at the project root
- it is compiled into the packaged library artifact, such as `.folib` or `.folenc`

- consumers import it with `@co.ddap.import(library="...")`
- internal package folders are compiled into the library but are not directly visible to consumers

Application-Workspace Source Library Surface

A source library may live inside an application workspace.

```
@co.ddap.import(
  package="com.abc.ffi",
  src-library=true,
  expect="ffi",
  as="ffilib"
)
```

The import resolves to one surface file:

```
/appl/com/abc/ffi.fo1
```

Rules:

- a source-library surface file must be below the application root, never at the application root
- multiple source libraries may exist in one application workspace
- the surface is imported through `package="..."` with `src-library=true`
- only the projected surface API is importable
- internal packages remain hidden even though their source files are physically available
- once a source tree is treated as a library, its subpackages cannot be imported as ordinary packages by consumers
- packaged libraries and source libraries use exactly the same public-surface and API-projection rules

Unified Surface Model

Every library kind uses the same conceptual surface shape:

```
library surface
├─ boundary data contracts
└─ public boundary-adapter functions
```

Library kind changes the permitted boundary representation and transfer semantics, not the conceptual form of the API.

Library kind	Boundary data	Transfer semantics
application	<code>co.lang.struct</code>	automatic deep snapshot
dynamicvmrt	<code>co.lang.struct</code>	automatic deep snapshot
advanced	<code>co.lang.struct</code>	automatic deep snapshot
system	<code>co.lang.cstruct</code>	system ABI value
ffi	<code>co.lang.cstruct</code>	C ABI value

`co.lang.struct` and `co.lang.cstruct` solve different problems:

```
struct -> FoLang semantic data contract
cstruct -> physical ABI-compatible value contract
```

Value transfer must not be confused with C-compatible layout. Application-family libraries therefore use expressive `struct` contracts transferred as deep snapshots, while system and FFI libraries use restricted `cstruct` contracts.

Allowed Surface Declarations

A library surface may contain only:

- file- or library-level imports needed by its adapter implementations
- `co.lang.struct` boundary declarations for `application`, `dynamicvmrt`, and `advanced`
- `co.lang.cstruct` boundary declarations for `system` and `ffi`

- public free-function API declarations with boundary-adapter definitions

The following declaration kinds are forbidden directly in every library surface:

- classes
- modules
- interfaces
- signatures
- units and companion units
- associated functions
- operator functions
- enums, unions, and other ADTs
- type aliases, newtypes, and opaque types
- objects, instances, type classes, and dependent types
- macros, templates, annotations, and decorators
- global variables, pointers, references, addresses, or mutable surface state

Surface `struct` and `cstruct` declarations are data contracts only. They cannot have companion units, associated functions, operators, methods, or other behavior on the surface.

Declarations directly inside the `co.lang.library` body are exported by default. Imports and implementation details are never exported.

Public Signature Type Closure

Every public surface function signature must be closed over the library's exported boundary type set.

For `application`, `dynamicvmrt`, and `advanced` surfaces, parameters and results may use only:

- approved built-in types
- `co.lang.struct` types declared in the same library surface

For `system` and `ffi` surfaces, parameters and results may use only:

- ABI-safe built-in types
- `co.lang.cstruct` types declared in the same library surface

The same closure rule applies recursively to fields of surface boundary types:

- a surface `struct` field may use an approved built-in type or another surface-declared `struct`
- a surface `cstruct` field may use an ABI-safe built-in type or another surface-declared `cstruct`
- an internal package type may never appear in a public function signature or surface boundary-type field
- pointers, references, and addresses may never cross any public library surface

A built-in type is not automatically surface-safe merely because it belongs to `co.lang`. An approved surface built-in must be concrete, fully resolved, and transferable under the library kind's boundary semantics.

The following categories are forbidden in public surface fields and signatures:

- inference-only types such as `co.lang.auto` and `co.lang.infer`
- dynamically typed or unconstrained carriers such as `co.lang.dynamic`, `co.lang.any`, `co.lang.typed`, and `co.lang.untyped`
- function, closure, delegate, loader, realm, AST, reflection, or runtime implementation values
- pointer, reference, address, thunk, and implementation-handle types
- any type whose reachable representation contains a forbidden type

For application-family surfaces, managed built-ins such as `co.lang.string` are permitted when the compiler defines deep-snapshot reconstruction for them. For system and FFI surfaces, only built-ins with a defined ABI representation are permitted; for example, `co.lang.string` is not directly `cstruct`-compatible.

Valid:

```
Employee co.lang.struct = {
  id      co.lang.int;
  name    co.lang.string;
  address Address;
}

Address co.lang.struct = {
  city co.lang.string;
}
```

Invalid:

```
getEmployee(id co.lang.int)->(emp.internal.Employee);
// compiler error: an internal type escapes through the public signature
```

Boundary-Adapter Functions

A public surface function may contain a definition, but its definition is restricted to boundary adaptation.

A boundary-adapter function may:

- read its input parameters and their fields
- declare temporary local values used only for conversion
- construct internal input values
- perform deterministic field-to-field or representation conversion
- invoke exactly one internal implementation operation
- receive that operation's result
- construct and return an allowed surface `struct`, surface `cstruct`, or built-in value
- use a direct delegation expression when no conversion is required

A boundary-adapter function may not:

- implement business rules or domain calculations
- perform business validation or authorization decisions
- orchestrate multiple implementation operations
- call another surface function
- perform persistence, networking, file I/O, retries, caching, or transactions
- start concurrency, scheduling, asynchronous work, processes, or continuations
- contain loops, recursion, workflow branching, or externally observable state mutation
- expose an internal value directly without converting it to an allowed boundary type

The compiler enforces the restricted adapter statement set. The restriction is structural: arbitrary executable logic is not accepted merely because it is placed in a surface file.

A direct delegate is the simplest adapter form:

```
health()->(co.lang.bool)
=> health.internal.Service.health();
```

A converting adapter may map between the public contract and an internal model.

Application Surface Example

```
// hrlib.fol
@co.dap.library(type="application")
hrlib co.lang.library = {

    @co.ddap.import(package="emp", as="emp")

    Employee co.lang.struct = {
        name co.lang.string;
        id   co.lang.int;
    }

    getEmployee(empId co.lang.int)->(Employee) = {
        internalEmployee := emp.EmployeeService.getEmployee(empId);

        this.return Employee{
            name: internalEmployee.name,
            id: internalEmployee.id
        };
    }
}
```

The surface owns conversion from the internal `emp.Employee` representation to the public `hrlib.Employee` contract. The `emp` package owns validation, business rules, persistence, and workflow.

The consumer sees the equivalent API contract:

```
Employee co.lang.struct = {
    name co.lang.string;
    id   co.lang.int;
}

getEmployee(empId co.lang.int)->(Employee);
```

The consumer does not see the body of `getEmployee` or the `emp` package.

System and FFI Surface Example

```
// driver.fol
@co.dap.library(type="system")
driver co.lang.library = {

    @co.ddap.import(package="driver.internal", as="impl")

    Point co.lang.cstruct = {
        x co.lang.int32;
        y co.lang.int32;
    }

    getOrigin()->(Point) = {
        internalPoint := impl.DriverService.getOrigin();

        this.return Point{
            x: internalPoint.x,
            y: internalPoint.y
        };
    }
}
```

Pointers, addresses, native bindings, and hardware state belong in `driver.internal`. They are forbidden in the public surface and cannot appear in the exported signature.

Boundary Transfer Semantics

For `application`, `dynamicvmrt`, and `advanced` libraries:


```
consumer boundary value
  ↓ automatic deep snapshot of the complete reachable value graph
library-local built-in or surface struct
  ↓ surface conversion
internal implementation value
  ↓ surface conversion
library-local built-in or surface struct
  ↓ automatic deep snapshot
consumer-local boundary value
```

The snapshot rule applies to both surface structs and approved built-in arguments/results. The library never receives the consumer's live object identity, and the consumer never receives a live internal-library object.

For `system` libraries, `cstruct` values cross according to the selected backend/platform system ABI.

For `ffi` libraries, `cstruct` values cross according to the declared C ABI, including its layout, alignment, and calling-convention requirements.

Surface-to-Internal Dependency Direction

The source-level dependency is one-way:

```
library surface
  ↓ imports and invokes
internal packages
```

Internal packages:

- do not import the library surface
- do not use surface-declared `struct` or `cstruct` types
- define their own implementation and domain types
- return internal values to the surface adapter
- contain all business logic, validation, workflow, I/O, and state management

This prevents a surface/internal compilation cycle and keeps the public contract independent from the internal domain model.

An internal implementation symbol may be visible to its own library surface without becoming part of the consumer API. Library encapsulation takes precedence over ordinary package visibility: consumers cannot resolve internal package symbols even when those symbols are callable from the surface during library compilation.

Consumer API Projection

The compiler does not expose the complete surface compilation context to a consumer. It creates a projected imported symbol table.

The projected API contains:

- the library identity and kind
- complete public surface `struct` or `cstruct` definitions, including field names and permitted field types
- public function names
- public function parameter names and types
- public function result types
- calling-convention, effect, error, and linkage metadata where applicable

The projected API does not contain:

- function bodies or implementation AST/HIR
- imports used by the surface
- local conversion variables
- delegate or implementation targets
- internal package paths or symbols
- private compiler-generated helpers
- business classes, modules, units, or internal data types

Therefore, a function definition may exist in the surface source and compiled artifact while the consumer's symbol table contains only its signature.

The backend may retain implementation bodies for code generation, linking, or whole-program optimization. Backend possession of implementation code does not make that code semantically visible to the consumer.

Source libraries follow the same rule. Availability of source code does not weaken the projected API boundary.

Library Compilation Order

A library is processed in four logical stages:

1. parse the surface header, boundary data declarations, and public function signatures
2. compile internal packages without depending on surface types
3. type-check and link surface boundary-adapter bodies against internal package symbols
4. emit the compiled implementation and the projected consumer API

This order permits the surface to call internal packages while preventing internal packages from depending on the surface.

Library Kinds

Library kinds are declared on the surface file:

```
@co.dap.library(type="ffi")
@co.dap.library(type="system")
@co.dap.library(type="dynamicvmrt")
@co.dap.library(type="advanced")
@co.dap.library(type="application")
```

Shared Meaning

- `application` -> ordinary safe library implementation
- `dynamicvmrt` -> application features plus full `co.meta` dynamic-runtime support
- `advanced` -> macro, template, concurrency, transformation, and runtime-machinery implementation
- `system` -> unsafe low-level runtime, pointer, process, native, and platform-control implementation
- `ffi` -> foreign-interface implementation

Library kind controls internal capabilities. All kinds still expose only boundary data contracts and public boundary-adapter functions.

`application` Libraries

Internally allowed:

- ordinary safe language features
- classes, modules, interfaces, signatures, units, and ordinary structs
- normal application-level packages and libraries

Internally forbidden:

- pointers, references, and addresses
- native functions annotated with `@co.dap.native`
- macros and templates
- low-level concurrency machinery
- advanced transformation or dynamic-runtime machinery

Public boundary:

- surface `struct` contracts
- approved built-in types
- deep-snapshot transfer

dynamicvmrt Libraries

Internally allowed:

- all `application` capabilities
- full `co.meta` support
- runtime reflection, instrumentation, dynamic loading, patching, and eval-based facilities

Internally forbidden:

- system and FFI capabilities unless reached through an imported public library surface
- raw pointers, addresses, and native functions in its own implementation

Public boundary:

- surface `struct` contracts
- approved built-in types
- deep-snapshot transfer

advanced Libraries

Internally allowed:

- all ordinary safe language features
- macros and templates
- async, parallel, continuation, scheduling, and transformation machinery
- compiler/runtime behavior definitions permitted to advanced libraries

Internally forbidden:

- raw pointers
- references and addresses
- native functions

Public boundary:

- surface `struct` contracts
- approved built-in types
- deep-snapshot transfer

system Libraries

Internally allowed:

- pointers, references, addresses, and word-level types
- native functions
- structs and cstructs
- units containing free functions
- basic value-typed functions
- type aliases
- basic threading and process primitives

Internally forbidden:

- classes
- modules
- interfaces and signatures
- overloading and overriding
- new operator definitions
- generics and templates
- macros
- reflection and dynamic runtime
- dependent types and type classes
- function patterns

- closures and currying
- advanced concurrency machinery such as continuations, defer, lazy evaluation, and thunks
- variadic, named, optional, and default parameters

Public boundary:

- surface `cstruct` contracts
- ABI-safe built-in types
- system ABI value transfer

`ffi` Libraries

Internally allowed:

- native and extern declarations
- pointers, references, addresses, and C ABI types
- cstructs and restricted implementation structs
- units containing free functions
- foreign calling-convention and symbol-linkage metadata

Internally forbidden:

- classes
- modules
- interfaces and signatures
- overloading and overriding
- new operator definitions
- generics and templates
- macros
- reflection and dynamic runtime
- dependent types and type classes
- closures, currying, and advanced concurrency machinery

Public boundary:

- surface `cstruct` contracts
- C-ABI-safe built-in types
- C ABI value transfer

Dependency Direction

Allowed dependency flow is one-way:

```

application
  ↓ through public surface APIs only
dynamicvmrt
  ↓ through public surface APIs only
advanced
  ↓ through public surface APIs only
system
  ↓ through public surface APIs only
ffi
  
```

A library may depend on a library at the same or a lower level only when the dependency does not create a cycle. Reverse dependencies are compiler errors.

Cross-Library Communication

From	To	Allowed?	Notes
<code>application</code>	<code>dynamicvmrt</code>	✓	projected public surface only
<code>application</code>	<code>advanced</code>	✓	projected public surface only

application	system	✓	projected public surface only
application	ffi	✓	projected public surface only
dynamicvmrt	advanced	✓	projected public surface only
dynamicvmrt	system	✓	projected public surface only
dynamicvmrt	ffi	✓	projected public surface only
dynamicvmrt	application	✗	reverse dependency
advanced	system	✓	projected public surface only
advanced	ffi	✓	projected public surface only
advanced	application	✗	reverse dependency
advanced	dynamicvmrt	✗	reverse dependency
system	ffi	✓	projected public surface only
system	application	✗	reverse dependency
system	dynamicvmrt	✗	reverse dependency
system	advanced	✗	reverse dependency
ffi	application	✗	reverse dependency
ffi	dynamicvmrt	✗	reverse dependency
ffi	advanced	✗	reverse dependency
ffi	system	✗	reverse dependency

Units in detail

A `unit` is a named, non-instantiable container for functions.

Like other named primary declarations, a unit may use an explicit name or `_` for filename-derived naming. For example, `_` `co.lang.unit` in `Math.fo1` declares `Math`, while an explicit name such as `Math co.lang.unit` is authoritative regardless of the filename.

Its purpose is structural: it prevents functions from flowing freely at package-file scope and preserves FoLang's rule that every ordinary package source file has one primary top-level declaration. A unit does **not** require its functions to form a domain model, capability, service, or other semantic abstraction.

Functions within a unit may be strongly related by purpose—for example, mathematical, parsing, encoding, or validation functions—or only loosely related. Semantic cohesion is encouraged as a source-design practice but is not enforced by the language.

A unit has two forms:

1. **Standalone unit** — its name does not match a struct in the same package. It acts as an ordinary named container for receiverless functions.
2. **Struct companion unit** — its name matches exactly one `co.lang.struct` in the same package. It provides behaviour associated with that struct while the struct declaration itself remains pure data. For more details please refer section [Struct Companion Units](#)

Only `co.lang.struct` can have a companion unit. A same-name unit is not allowed for `co.lang.class`, `co.lang.cstruct`, modules, enums, unions, interfaces, signatures, or other declaration kinds. Classes already contain their own methods and lifecycle behaviour; `cstruct` remains a restricted C-compatible data representation.

```
Text co.lang.unit = {

    trim(value co.lang.string)->(co.lang.string) = {
        ...
    }

    contains(
        value co.lang.string,
        search co.lang.string
    )->(co.lang.bool) = {
        ...
    }

    repeat(
        value co.lang.string,
        count co.lang.int
    )->(co.lang.string) = {
        ...
    }
}
```

Unit functions are accessed through the unit name:

```
cleaned := Text.trim(input);
found   := Text.contains(input, "FoLang");
```

A unit:

- introduces a named scope for its functions
- contains receiverless functions
- may additionally contain associated and operator functions when it is a struct companion unit
- requires the first declared parameter of every receiverless companion function to have the matching struct type
- uses the explicit receiver, rather than ordinary-parameter matching, to associate an associated function with its struct
- may contain public and private functions
- may use built-in types, user-defined types, or both
- does not introduce a user-defined data type or ADT
- has no fields or unit-level variables
- has no constructors or lifecycle methods
- has no instances, object identity, inheritance, or polymorphic dispatch
- cannot be instantiated or assigned as an object value
- cannot contain nested classes, structs, enums, modules, or other primary type declarations

Physical type or function declarations inside an individual unit function are not permitted. A separately declared class, struct, enum, module, or function may instead be restricted to one or more exact unit functions with `@co.dap.local`, using each function's complete qualified signature in the target set.

```
General co.lang.unit = {

    parseNumber(value co.lang.string)->(co.lang.int) = { ... }

    clamp(
        value co.lang.int,
        min co.lang.int,
        max co.lang.int
    )->(co.lang.int) = { ... }
}
```

The functions inside a unit do not need to use UDTs or ADTs. A unit is especially useful for small operations that consume built-in values, perform a computation, and return built-in values.

A unit may also represent a clearly related family of operations:

```

Math co.lang.unit = {

    abs(value co.lang.int)->(co.lang.int) = { ... }

    max(
        a co.lang.int,
        b co.lang.int
    )->(co.lang.int) = { ... }

    sqrt(value co.lang.float)->(co.lang.float) = { ... }
}

```

The common purpose of these functions makes the source easier to understand, but the compiler does not require or attempt to prove that all functions in a unit are semantically related.

CStructs

`co.lang.cstruct` is a C-like value type — passed by value, simple memory layout, safe to cross zone boundaries. Unlike `co.lang.struct` which is passed by reference, `co.lang.cstruct` is always copied on pass.

```

Point co.lang.cstruct = {
    x co.lang.int;
    y co.lang.int;
}

Rect co.lang.cstruct = {
    origin Point;
    width  co.lang.int;
    height co.lang.int;
}

```

cstruct Rules

always passed by value – never by reference	
simple memory layout – no metadata	
can contain only simple types and other cstructs	
cannot contain <code>co.lang.struct</code>	✗ has metadata
cannot contain <code>co.lang.string</code>	✗ heap allocated
cannot contain <code>co.lang.dynamic</code>	✗ runtime type info
cannot contain classes	✗ vtable, metadata
cannot contain modules	✗
cannot contain any heap allocated type	✗
cannot have methods	
cannot have associated functions	
cannot embed <code>co.lang.struct</code>	
safe to cross direct ABI and zone boundaries	
allowed field types:	
<code>co.lang.int</code> , <code>co.lang.uint</code> , <code>co.lang.float</code>	✓ primitives
<code>co.lang.bool</code> , <code>co.lang.char</code> , <code>co.lang.byte</code>	✓ primitives
<code>co.lang.int->([N])</code>	✓ fixed size arrays
<code>co.lang.cstruct</code>	✓ other cstructs

Packed cstruct — no padding, exact memory layout

Used for hardware registers, binary protocols, exact memory mapped formats:

```

@co.dap.packed
Register co.lang.cstruct = {
    flags  co.lang.uint8;
    status co.lang.uint8;
    data   co.lang.uint16;
}

```

SIMD cstruct — aligned for vector operations

Used for math, graphics, signal processing:

```
@co.dap.simd(align=16)
Vec4 co.lang.cstruct = {
  x co.lang.float;
  y co.lang.float;
  z co.lang.float;
  w co.lang.float;
}
```

Both together

```
@co.dap.packed
@co.dap.simd(align=32)
AVXVec co.lang.cstruct = {
  data co.lang.float;
}
```

`@co.dap.packed` and `@co.dap.simd` are specialisations of `co.lang.cstruct` — same rules, same zone boundary safety. They are not separate types.

Structs

```
myStruct co.lang.struct={
  field1 co.lang.int;
  field2 co.lang.string;
  field3 co.lang.bool;
}
```

Struct Rules

```
structs cannot extend other structs
structs cannot contain methods directly
structs can compose other structs
structs cannot have default values to fields/members
structs can embed other structs (Go lang like)
structs can have a same-package companion unit with the same name
```

The struct declaration remains pure data. All behaviour associated with a struct is declared separately in its matching `co.lang.unit`.

Struct Embedding

Embedding promotes fields of an embedded struct directly into the outer struct — they act as the outer struct's own fields at construction and access sites. This is distinct from composition where the embedded struct is a named field.

```
E co.lang.struct = {
  id co.lang.int;
  name co.lang.string;
}

// ✅ No conflict — id and name promoted as B's own fields
B co.lang.struct = {
  age co.lang.float;
  E; // embedded — id and name promoted
}

b := B{ age: 25.0, id: 1, name: "Rao" }; // all fields at same level
b.id // direct — no b.E.id needed
b.name // direct — no b.E.name needed
b.age // direct
```



```
// ❌ Compiler error — name conflict between B.name and E.name
B co.lang.struct = {
    name co.lang.string; // conflicts with E.name
    E;
    age co.lang.float;
}
// Fix 1 — rename B's conflicting field
// Fix 2 — use explicit composition instead: e E;
```

```
// Explicit composition — no promotion, always qualified access
B co.lang.struct = {
    name co.lang.string;
    e E; // named field — no conflict, no promotion
    age co.lang.float;
}

b.name // B's own name
b.e.id // E's id — always explicit
b.e.name // E's name — always explicit
```

Embedding Rules

Situation	Behavior
Embedded field, no conflict	Promoted — acts as the outer struct's own field
Embedded field, name conflict with outer	❌ Compiler error — rename or use composition
Multiple embeds, no conflict between them	All fields promoted
Multiple embeds, conflict between embedded structs	❌ Compiler error
Explicit composition (<code>e E</code>)	No promotion — always accessed via <code>b.e.field</code>

FoLang does **not** silently shadow conflicting fields. Any name conflict is a compiler error — the programmer must make a conscious decision to rename or switch to explicit composition.

Struct Declaration Relationships

A struct cannot physically declare another struct, enum, class, module, function, signature, interface, or other named declaration inside its body. A struct body remains a pure data declaration containing fields and permitted embeddings only.

Use one of the following instead:

- **composition** through a named field
- **embedding** where the declaration kind's embedding rules permit it
- a separately declared target-local declaration restricted with `@co.dap.local`

```
// EmployeeAddress.fol
@co.dap.local(for=hr.employee.Employee)
EmployeeAddress co.lang.struct = {
    street co.lang.string;
    city co.lang.string;
}
```

```
// Employee.fol
Employee co.lang.struct = {
    id co.lang.int;
    name co.lang.string;
    address EmployeeAddress; // composition
}
```

The following physical nesting is invalid:

```
Employee co.lang.struct = {
  Address co.lang.struct = { // ❌ nested declaration
    city co.lang.string;
  }

  address Address;
}
```

structs cannot declare inner structs	❌	compiler error – only through @co.dap.local
structs can declare inner enums/ADTs	❌	compiler error – only through @co.dap.local or @co.dap.nested
structs cannot declare inner classes	❌	compiler error – struct is pure data
structs cannot declare inner modules	❌	compiler error – struct is pure data

`@co.dap.local` controls declaration visibility only. It does not automatically compose or embed the annotated declaration into the target struct.

Struct Companion Units

When a unit has the same name as a struct in the same package, it is the companion unit of that struct.

The struct and its companion unit are separate primary declarations and therefore normally reside in separate source files within the same package. They are shown together below only to illustrate their relationship.

```
// Vector.fol
Vector co.lang.struct = {
  x co.lang.float;
  y co.lang.float;
}

// Vector.unit.fol
_ co.lang.unit = {

  distance(
    left Vector,
    right Vector
  )->(co.lang.float) = {
    ...
  }

  isZero(value Vector)->(co.lang.bool) = {
    this.return value.x == 0.0 && value.y == 0.0;
  }
}
```

Receiverless functions in a companion unit are accessed through the shared struct/unit name:

```
d := Vector.distance(first, second);
zero := Vector.isZero(value);
```

These functions are analogous to **static functions of a class** in call form, but FoLang additionally requires their **first declared parameter** to have the matching struct type. The first parameter establishes ownership by the companion unit. A matching struct type in a later parameter is insufficient.

```
Vector co.lang.unit = {
  distance(left Vector, right Vector)->(co.lang.float) = { ... } // ✅
  convert(value Vector, radix co.lang.int)->(co.lang.string) = { ... } // ✅

  invalid(scale co.lang.float, value Vector)->(Vector) = { ... }
  // ❌ first parameter is not Vector

  zero()->(Vector) = { ... }
  // ❌ no first parameter identifies Vector ownership
}
```

Therefore, factory-style receiverless functions such as `Vector.create(x, y)` and zero-parameter receiverless functions such as `Vector.zero()` are not valid under this rule because no ordinary parameter establishes ownership. A factory may instead be declared as a type-associated function, described below, or placed in a standalone unit with a different name.

Companion-unit functions do not make the struct a class and do not introduce object identity, inheritance, virtual dispatch, lifecycle methods, or unit-level state.

Associated Functions in a Companion Unit

A companion unit may contain instance-associated and type-associated functions. Both forms use an explicit receiver to establish association with the matching struct, but they differ in whether the receiver is a struct value or the struct type itself.

Instance-Associated Functions

An instance-associated function has an explicit receiver value whose type is the matching struct:

```
Vector co.lang.unit = {  
  (value Vector) magnitude()->(co.lang.float) = {  
    this.return co.math.sqrt(  
      value.x * value.x + value.y * value.y  
    );  
  }  
  
  (value Vector) scale(factor co.lang.float)->(Vector) = {  
    this.return Vector{  
      x: value.x * factor,  
      y: value.y * factor  
    };  
  }  
}
```

Associated functions may be invoked using method-call syntax:

```
v Vector=Vector{  
  length := v.magnitude();  
  scaled := v.scale(2.0);  
}
```

The method-call form is syntactic association. Conceptually:

```
v.magnitude()
```

resolves to the associated function declared in `Vector co.lang.unit` with `v` supplied as its explicit receiver. Associated functions are therefore analogous to **instance methods**, but they remain externally declared functions and do not acquire class semantics.

Instance-associated-function rules:

- the explicit receiver value must have the struct type whose name matches the unit name
- the receiver establishes association; the ordinary parameter list does not need to contain the matching struct type
- the matching struct and unit must be declared in the same package
- the function cannot be declared loose at package scope
- it has no inheritance, overriding, or virtual dispatch
- it receives no special private access beyond normal visibility rules
- an imported struct with the same short name does not create a companion relationship

```
Employee co.lang.struct = {  
  id co.lang.int;  
}  
  
Employee co.lang.unit = {  
  (emp Employee) isValid()->(co.lang.bool) = { ... } //   
  (dept Department) isValid()->(co.lang.bool) = { ... } //  receiver does not match Employee  
}
```

Type-Associated Functions

A type-associated function uses the matching struct type itself as its explicit receiver. It is analogous to a class-level or static factory function in call form, but it does not introduce class lifecycle, inheritance, or object-oriented dispatch.

```
Vector co.lang.unit = {
  (Vector) zero()->(Vector) = {
    this.return Vector{x: 0.0, y: 0.0};
  }

  (Vector) create(
    x co.lang.float,
    y co.lang.float
  )->(Vector) = {
    this.return Vector{x: x, y: y};
  }
}
```

Type-associated functions are invoked through the struct name:

```
origin := Vector.zero();
value  := Vector.create(10.0, 20.0);
```

Type-associated-function rules:

- the explicit receiver must be the struct type whose name matches the unit name
- the type receiver establishes association, so no ordinary parameter is required to have the matching struct type
- zero ordinary parameters are permitted
- the matching struct and unit must be declared in the same package
- the function cannot be declared loose at package scope
- the struct type is not an object instance and cannot be mutated through the type receiver
- type association does not add constructors, lifecycle methods, inheritance, overriding, or virtual dispatch
- an imported struct with the same short name does not create a companion relationship

The three companion-function forms are therefore:

```
receiverless companion function
  -> first ordinary parameter struct
  -> parameter not matching struct (as companion unit name match and must match with struct name)

instance-associated function
  -> explicit receiver is a value of the matching struct

type-associated function
  -> explicit receiver is the matching struct type itself
```

Companion Name Rules

Within one package, FoLang may contain:

```
one Vector co.lang.struct
one Vector co.lang.unit
```

Together they form one struct/companion pair. The following are compiler errors:

```
two units named Vector in the same package
a unit matching a class or cstruct
a companion unit located in a different package from its struct
an instance-associated function whose receiver value is not of the matching struct type
a type-associated function whose type receiver is not the matching struct type
```

Name resolution is determined by context:

```
v Vector;           // type position → Vector struct
v := Vector{x: 1, y: 2}; // construction → Vector struct
z := Vector.isZero(v); // qualified function lookup → Vector companion unit
m := v.magnitude();   // associated-function lookup → Vector companion unit
```

```
x General = General{}; // ❌ compiler error – a unit is not instantiable
```

Companion Function Categories and Operator Functions

```
Employee co.lang.struct = {
  id   co.lang.int;
  name co.lang.string;
}

Employee co.lang.unit = {

  // Receiverless companion function: first parameter establishes ownership.
  compare(
    left  Employee,
    right Employee
  )->(co.lang.int) = {
    ...
  }
  // Receiverless companion function: without matching first parameter with type
  getEmployee(id co.lang.string)->(Employee)={...}

  // Instance-associated function: receiver is an Employee value.
  (emp Employee) fetchEmployee(
    empId co.lang.string
  )->(Employee) = {
    ...
  }

  // Type-associated function: receiver is the Employee type.
  (Employee) getInstance()->(Employee) = {
    ...
  }
}
```

Call forms:

```
order := Employee.compare(emp, other); // receiverless companion function
result := emp.fetchEmployee("E1");    // instance-associated function
emp    := Employee.getInstance();      // type-associated function
emp1   := Employee.getEmployee("E0021"); // receiverless companion function
```

The receiver remains explicit in associated-function declarations even though instance-call or type-call syntax is available at the call site. This does not give the struct class semantics: there is no inheritance, overriding, virtual dispatch, hidden receiver, or lifecycle.

Operator functions associated with a struct must be declared in its companion unit. They may use either an instance receiver or a type receiver:

```

Employee co.lang.unit = {

    @co.dap.operator(symbol="+")
    (emp Employee) add(other Employee)->(Employee) = {
        ...
    }

    @co.dap.operator(symbol=="")
    (Employee) equals(
        left Employee,
        right Employee
    )->(co.lang.bool) = {
        ...
    }
}

```

A receiverless operator function whose first parameter is the matching struct is accepted as syntactic shorthand for the equivalent type-associated operator form. The compiler normalizes both forms to the same companion ownership model; declaring both equivalent forms for the same operator signature is a duplicate-definition error.

Exception from normal associate functions, operator functions cannot have receiverless functions with non matching first parameter to type (struct). otherwise it is a compiler error.

Unions

Unions are untagged ADTs

```

myUnion co.lang.union={
    intValue co.lang.int;
    strValue co.lang.string;
}

```

Enums

```

myEnum co.lang.enum={
    Variant1,
    Variant2,
    Variant3
}

```

classes

```

Employee co.lang.class ={
    getEmployeeDetails()->(Employee) = empmodule.getEmployeeDetails;
    // assigning module function to class's method

    getEmployeeInfo()->(Employee) =>> empmodule.getEmployeeDetails();
    // delegating – internally redirecting the call to module function
}

// $1, $2, $3 ... are previous results captured as bind variables
Emp co.lang.class={
    dosomething(a co.lang.int, b co.lang.int)->
    (co.lang.int)=>>somePack.someMethod(a)=>>someOthPack.someOtherMeth($1, b);
}

```

Class Declaration Relationships

A class cannot physically contain named class, struct, enum, module, function, signature, interface, or other declaration definitions as nested declarations. Class bodies contain fields, lifecycle declarations, and methods permitted by the class model.

Types and helper declarations used only by one class are declared in their ordinary legal source locations and restricted to that class with `@co.dap.local`:

```
// EmployeeAddress.fol
@co.dap.local(for=hr.employee.Employee)
EmployeeAddress co.lang.struct = {
    street co.lang.string;
    city   co.lang.string;
}
```

```
// EmployeeStatus.fol
@co.dap.local(for=hr.employee.Employee)
EmployeeStatus co.lang.enum = {
    Active,
    Inactive,
    Pending
}
```

```
// Employee.fol
Employee co.lang.class = {
    address EmployeeAddress;
    status  EmployeeStatus;

    getAddress()->(EmployeeAddress) = {
        this.return this.address;
    }
}
```

The local declarations are visible while compiling `hr.employee.Employee`, but they do not become nested names such as `Employee.EmployeeAddress` or `Employee.EmployeeStatus`.

The following is invalid:

```
Employee co.lang.class = {
    Address co.lang.struct = { // ❌ physical nested declaration
        city co.lang.string;
    }
}
```

Ordinary visibility annotations do not widen a target-local declaration beyond the declaration or closed target set named by `@co.dap.local` or `@co.dap.nested`.

Method Types

```
Employee co.lang.class = {  
  
  @co.dap.static  
  getEmployee()->(Employee) = {}  
  
  @co.dap.instance  
  getEmployee()->(Employee) = {}  
  
  @co.dap.class  
  getEmployee()->(Employee) = {}  
  
  @co.dap.object  
  getEmployee()->(Employee) = {}  
}  
  
@co.dap.oops(  
  A: { inherit:true, virtual:true },  
  B: { implements:true },  
  C: { inherits:true, abstract=true },  
  D: { inherits:true },  
  E: { uses:true },  
  F: { composes:true },  
  G: { extends:true },  
  H: { with:true },  
  I: { associate:true },  
)  
test co.lang.class = {  
  getTest(id int)->(test) = {}  
}
```

The @@new and @@init Methods

`@@new` and `@@init` are lifecycle methods — compiler-owned, not user-definable outside the class. `@@` signals they are restricted lifecycle symbols, not regular methods.


```

@co.dap.generic(type={T:{typename}, R:{typename}})
Employee co.lang.class = {

    id T
    name R

    // @@new is provided by default even if not overridden.
    // Override only when you genuinely need to change allocation behavior.

    @co.dap.method.class
    @co.dap.private
    @@new()->(co.lang.uninit) = { self.return co.const.none }

    @co.dap.method.class
    @co.dap.public
    @@new(a co.lang.typevalue, b co.lang.typevalue)->(co.lang.uninit) = {
        // Manual type aliasing - @co.dap.generic handles this automatically
        // Override @@new only when you need custom allocation logic
        T co.lang.type = a
        R co.lang.type = b

        // self keyword is allowed only in class methods
        self.parent.new();

        // uninit instance method internally calls new and init
        self.return co.lang.uninit.instance(Employee, self);
    }

    @co.dap.override
    @co.dap.constructor(access=private)
    @@init() = {}

    @co.dap.override
    @co.dap.constructor(access=public)
    @@init(id T, name R) = {
        this.parent.init();
        this.id = id;
        this.name = name;
    }

    getEmployee(id T)->(Employee) = {}
}

```

Anonymous Classes/Types

```

emp := co.lang.class{};

empObj := emp.init();

empobj1 := co.lang.class{
    name string
}.init();

```

Interfaces

```

IEmployee co.lang.interface = {
    storeEmployee(emp Employee)->(Employee);
}

```

Signatures

```
// Employee is an ordinary package-level declaration.  
MEmployee co.lang.signature = {  
  storeEmployee(emp Employee)->(Employee);  
}
```

Structurally they look similar — both are lists of contracts. The difference is **who implements them and how**.

	co.lang.signature	co.lang.interface
Implemented by	module via <code>matches=</code>	class via <code>implements=[]</code>
Number of implementations	Any number of distinct modules may match one signature	Any number of classes may implement one interface
Runtime cardinality of one implementation declaration	One module object per loaded module identity	Any number of independently constructed class objects
State model	One shared state for all references to the same module	Separate per-instance state for each class object
May specify required values	✓	✗ — methods only
May reference package-level types	✓	✓
May declare abstract/fixed type components	✓	✗
May require generic type constructors	✓	✗
May declare physical nested/local types	✗	✗
May use <code>@co.dap.local</code>	✗	✗
Instantiation involved	Module is declared once, not constructed	Class objects are constructed
Reference use	Compatible module references may be used through the signature type without creating another module	Interface references may refer to any implementing object instance
OOP dispatch	✗	✓ virtual/dynamic
Contract style	module values, functions, and type components	behavioral methods on object instances
Practical analogy	singleton component contract	object-instance behavioral contract
Origin	ML/OCaml-inspired modules	Java/C#/Go interfaces

- A `signature` is a **module contract** over values, functions, existing package-level types, and abstract or fixed type components. A type-component specification is a contract slot, not a physical nested type definition. Multiple modules may match one signature, but each module declaration denotes one module object with shared module state.
- An `interface` is a **behavioral contract** tied to class dispatch and polymorphism. It cannot declare module type components or own nested type definitions. A class implementing an interface may create any number of independent runtime objects.
- The approximation `module + signature ≈ singleton object + interface` is useful for understanding cardinality and shared state, but a module is a language-level component rather than a class-based singleton pattern.

Modules

A module is an ML/OCaml-style abstraction governed by an optional signature. A module may use package-level types and may satisfy type components declared by its signature, but it does not physically own or nest arbitrary type declarations. A module should not be introduced merely to prevent functions from appearing loose in a file; use `co.lang.unit` for that simpler structural purpose.

```
// Employee.fol – ordinary package-level type
Employee co.lang.struct = {
  Id   co.lang.int;
  Name co.lang.string;
}

// EmployeeModule.signature.fol
EmployeeModule co.lang.signature = {
  getEmployee(id co.lang.int)->(Employee);
}

// EmployeeModImpl.fol
@co.dap.module(signature=EmployeeModule)
EmployeeModImpl co.lang.module->(signature=EmployeeModule, matches=EmployeeModule) = {

  getEmployee(id co.lang.int)->(Employee) = {
    this.return Employee{
      Id: 10,
      Name: "Rao"
    };
  }
}

mm EmployeeModule = EmployeeModImpl;
v Employee = mm.getEmployee(10);
```

Module Cardinality and Singleton Analogy

A module declaration defines exactly one named module object for that loaded module identity. It is not an instantiable blueprint and does not create a new module object each time its name is referenced.

```
first EmployeeModule = EmployeeModImpl;
second EmployeeModule = EmployeeModImpl;
```

`first`, `second`, and `EmployeeModImpl` refer to the same module object. These bindings copy or retain the module reference; they do not clone or instantiate the module. Consequently, values declared directly by the module represent one shared module state for all references to that module.

A signature does not itself create a module object. It is a contract, and any number of separately declared modules may conform to the same signature:

```
DatabaseBackend signature
├─ PostgreSQLBackend module -> one named module object
├─ MySQLBackend module     -> one named module object
└─ TestDatabaseBackend module -> one named module object
```

Each conforming module declaration contributes its own single module object and its own module state. The signature does not restrict the number of distinct conforming module declarations.

A useful analogy is:

```
signature      ≈ interface contract
conforming module ≈ singleton object implementing that contract
class          ≈ instantiable object type
```

The analogy is intentionally limited. A FoLang module is a compiler-recognized named component, not a class made singleton through a private constructor, static field, or runtime pattern. It cannot be repeatedly constructed. Because module references are first-class in FoLang, they may be bound and used through a compatible signature type, but every reference to the same module declaration still denotes the same module object.

Modules are also broader than ordinary interface implementations. A matching module may provide module values, functions, and abstract, fixed, or generic type-component bindings required by its signature. An interface constrains object behaviour; it does not provide the same module type-component abstraction.

By contrast, a class declaration defines an instantiable type. Every class construction creates a distinct object with independent identity, state, and lifetime:

```
PostgreSQLBackend module
└─ one shared module object and module state

PostgreSQLConnection class
└─ connection1 -> independent object and state
└─ connection2 -> independent object and state
└─ connection3 -> independent object and state
```

Formal mental model: A FoLang module is a single named implementation component that may conform to a signature. It is comparable to a singleton object implementing an interface, but it is not instantiated from a class. Multiple distinct modules may conform to the same signature, while each module declaration denotes one module object. Unlike an ordinary singleton-interface implementation, a module may also satisfy abstract, fixed, and generic type components required by its signature.

Module instantiation A FoLang class or struct declaration introduces an instantiable type but does not create an instance. A FoLang module declaration introduces one named module component directly into its package. The module name acts as the binding for that component, so no separate construction expression is required. The module's runtime state is initialized once according to the language's module-initialization rules.

A module declaration is a singleton component declaration and binding, rather than merely an object definition.

Module Signature Contents

A `co.lang.signature` is a declarative contract for a module. It may specify required module values, functions, and type components. A signature does not allocate storage, initialize variables, execute statements, or provide function bodies.

A signature may contain:

- value specifications
- function signatures
- references to already existing accessible types
- abstract type-component specifications
- fixed or manifest type-component specifications
- abstract generic type-constructor specifications

A signature may not contain:

- value initializers
- executable statements
- function bodies
- concrete class, struct, enum, module, interface, or signature definitions
- arbitrary nested or target-local declarations

Type-component specifications are part of module conformance. They are not Java-, C++-, or C#-style inner types and do not participate in `@co.dap.local`.

Value Specifications

A declaration such as:

```
Counter co.lang.signature = {
    count co.lang.int;
    increment(amount co.lang.int)->();
}
```

requires a matching module to provide a value named `count` of type `co.lang.int` and a compatible `increment` function. The signature does not initialize `count` and does not define `co.lang.int`; the built-in type already exists.

```
CounterImpl co.lang.module->(
  signature=Counter,
  matches=Counter
) = {
  count co.lang.int = 0;

  increment(amount co.lang.int)->() = {
    count.value = count + amount;
  }
}
```

The same rule applies when a value or function specification uses an existing accessible package type:

```
EmployeeRepository co.lang.signature = {
  current hr.employee.Employee;
  find(id co.lang.int)->(hr.employee.Employee);
}
```

The matching module must provide `current` and `find`. It does not redefine `hr.employee.Employee`.

Abstract Type Components

An abstract type component declares that every matching module must supply a type binding for that component:

```
Repository co.lang.signature = {
  Entity co.lang.type;

  current Entity;
  find(id co.lang.int)->(Entity);
}
```

`Entity co.lang.type;` does not define the representation of `Entity`. It defines a required module type component. A matching module binds it to a compatible existing type:

```
EmployeeRepositoryImpl co.lang.module->(
  signature=Repository,
  matches=Repository
) = {
  Entity co.lang.type = hr.employee.Employee;

  current Entity = ...;
  find(id co.lang.int)->(Entity) = { ... }
}
```

Within a matching module, `Entity co.lang.type = ...` is a **signature-component binding**, not an arbitrary nested type declaration. Its name must correspond to an abstract type component declared by the matched signature. A module cannot use this form to introduce unrelated module-local types.

An abstract type component differs from `forward` and `extern` declarations:

```
abstract signature type component
  -> each matching module supplies its own compatible type binding

forward declaration
  -> one specific declaration is completed later

extern declaration
  -> one specific declaration is defined in another linkage or compilation unit
```

Fixed or Manifest Type Components

A signature may fix a type component to an already known type:

```
IntegerRepository co.lang.signature = {
  Id co.lang.type = co.lang.int;

  find(id Id)->(co.lang.bool);
}
```

Here `Id` is predetermined as `co.lang.int`. A matching module uses that type component but does not choose or redefine it.

```
Entity co.lang.type;           -> abstract; implementor supplies the binding
Id co.lang.type = co.lang.int; -> fixed; signature supplies the type equality
```

Abstract Generic Type Constructors

A signature may require a generic type constructor without defining its representation:

```
StackSignature co.lang.signature = {
  Stack(T) co.lang.type;

  empty(T)->(Stack(T));
  push(value T, stack Stack(T))->(Stack(T));
  pop(stack Stack(T))->(T, Stack(T));
}
```

`Stack(T) co.lang.type;` declares an **abstract generic type component**, also described as an abstract type constructor of arity one. The signature specifies that `Stack` accepts one type argument, but it does not define what `Stack(T)` is.

A matching module must provide a compatible type-constructor binding with the same name, arity, and declared constraints:

```
ListStackModule co.lang.module->(
  signature=StackSignature,
  matches=StackSignature
) = {
  Stack(T) co.lang.type = co.core.list(T);

  empty(T)->(Stack(T)) = { ... }
  push(value T, stack Stack(T))->(Stack(T)) = { ... }
  pop(stack Stack(T))->(T, Stack(T)) = { ... }
}
```

Another matching module may choose another representation:

```
ArrayStackModule co.lang.module->(
  signature=StackSignature,
  matches=StackSignature
) = {
  Stack(T) co.lang.type = collections.ArrayStack(T);
  ...
}
```

Therefore:

```
StackSignature
  -> requires a generic type constructor Stack(T)

ListStackModule
  -> binds Stack(T) to co.core.list(T)

ArrayStackModule
  -> binds Stack(T) to collections.ArrayStack(T)
```

A signature-component binding does not permit physical type nesting. When an implementation needs a new concrete struct, class, or enum representation, that declaration remains an ordinary package declaration and may be restricted to the implementing module with `@co.dap.local`; the module then binds the signature component to that declaration.

Signature Conformance Rules for Type Components

For every type component in a matched signature:

- an abstract component must receive exactly one compatible module binding
- a fixed component must retain the type equality declared by the signature
- a generic component binding must preserve generic arity, parameter kinds, bounds, variance, and other declared constraints
- component names must be unique within the signature
- component bindings cannot contain executable code
- extra module-local type bindings that do not correspond to signature components are compiler errors
- types referenced by value and function specifications must resolve after applying the module's type-component bindings

Module Declaration Relationships

A module cannot physically declare nested structs, enums, classes, modules, signatures, interfaces, or other arbitrary named declarations. It references ordinary package-level declarations through its functions and signature. The only type-like declarations permitted directly in a matching module are signature-component bindings that satisfy abstract type components declared by its matched signature; such bindings do not create independent nested declarations.

A declaration intended only for one module may be restricted with `@co.dap.local`:

```
// EmployeeModuleConfig.fol
@co.dap.local(for=hr.employee.EmployeeModImpl)
EmployeeModuleConfig co.lang.struct = {
  timeout co.lang.int;
  retries co.lang.int;
}
```

```
// EmployeeModImpl.fol
EmployeeModImpl co.lang.module = {
  connect(cfg EmployeeModuleConfig)->(co.lang.bool) = {
    ...
  }
}
```

The following remains invalid:

```
EmployeeModImpl co.lang.module = {
  Config co.lang.struct = { // ❌ physical nested declaration
    timeout co.lang.int;
  }
}
```

A target-local declaration does not automatically become a module member name and is not projected through the module's signature. It becomes part of the signature view only when an explicit signature type component is bound to it or a signature value/function specification references it through an allowed type component.

Structs vs Classes vs Modules vs Units vs Packages

	Struct	CStruct	Class	Module	Unit	Package
Purpose	Pure data shape	C-like value type	Behaviour + data	Signature-backed ML-style abstraction	Named function container; optionally a struct companion	Folder- based grouping
Fields	✅	✅ simple only	✅ per instance	❌	❌	❌
Module-level values	❌	❌	❌	✅ when declared directly or required by a signature	❌	❌

Functions / methods	Optional static-like functions whose first parameter is the struct, plus receiver-based associated functions, through a matching companion unit	✗	✓ methods	✓ module functions	✓ receiverless functions; associated functions only when matching a struct	✗
Lifecycle (<code>@@new</code> / <code>@@init</code>)	✗	✗	✓	✗	✗	✗
<code>this</code> / <code>self</code>	✗	✗	✓	✗	✗	✗
Value/literal construction	✓	✓	✗	✗	✗	✗
Lifecycle instantiation (<code>new</code> / <code>init</code>)	✗	✗	✓ multiple objects	✗ — one module object per declaration	✗	✗
Runtime state cardinality	Per bound struct object	Per value	Per class object	One shared state for the module declaration	—	—
First class	✓	✓	✓	✓ module reference; referencing does not instantiate	✗	✗
Pass by	Reference	Value	Reference	Reference to the same module object	—	—
Contract	—	—	<code>interface</code> via <code>implements=[]</code>	<code>signature</code> via <code>matches=</code>	none	—
OOP / inheritance	✗	✗	✓	✗	✗	✗
Physically nested declarations	✗	✗	✗	✗	✗	✓ separate declarations across package files
May be an <code>@co.dap.local</code> target	✓	✗	✓	✓	✗	✗
Pattern matching	✓	✓	✓	✗	✗	✗
Direct ABI / zone boundary safe	✗ — library boundaries require snapshots	✓	✗	✗	✗	✗
Associated functions	✓ through matching companion unit	✗	—	—	✓ only in a struct companion unit	✗
Embedding	✓	✗	—	—	✗	✗
Declared with	<code>co.lang.struct</code>	<code>co.lang.cstruct</code>	<code>co.lang.class</code>	<code>co.lang.module</code>	<code>co.lang.unit</code>	folder path
C++ backend analogy	struct without methods	plain C struct	class	struct/class abstraction	namespace or static function scope	namespace

Closest mental model	Rust struct	C struct	Java/C# class	singleton implementation component with ML-style type members	named function scope; optional struct companion	filesystem namespace
----------------------	-------------	----------	---------------	---	---	----------------------

Mental model:

```

reach for struct    → pure data; use a same-name companion unit for behaviour
reach for cstruct   → physical ABI-compatible value data crossing direct zone or native boundaries
reach for class     → behaviour, lifecycle, multiple instances
reach for module    → one named implementation component with shared state, governed by an optional signature and capable of satisfying type components
reach for unit      → named function container; same-name struct unit acts as companion
reach for package   → folder-based grouping only, not a value

```

Declaration scoping rule: FoLang does not permit physical nested or function-local named declarations. Classes, structs, enums, modules, and functions use ordinary package declarations and may restrict a supported declaration to one or more exact targets with `@co.dap.local`. The annotation accepts either one declaration reference or a non-empty closed target list. The local declaration and every target must belong to the same exact folder-derived package; parent and subpackages are different packages. Non-function targets are identified by complete qualified name; overloaded function targets are identified by complete qualified signature. Visibility is the union of the explicitly listed target scopes and is neither inherited nor transitive. Signatures and interfaces cannot own or target local declarations. A signature may nevertheless declare abstract, fixed, and generic type components as module-conformance requirements; these are contract slots rather than physical nested declarations. A matching module may bind only those declared components. Units contain functions only, and a struct companion unit remains a separate declaration from its struct. ***

Local and/or Nested types and functions

FoLang does not provide Java-, C++-, or C#-style physical nesting of named declarations inside another type, module, function, or executable block. Instead, FoLang keeps declarations in their ordinary legal source locations and may restrict one declaration to one or more explicitly identified target declarations.

```
@co.dap.local(for=<declaration-reference>)
```

or:

```

@co.dap.local(
  for=[
    <declaration-reference>,
    <declaration-reference>
  ]
)

```

The annotated declaration is called a **target-local declaration**. The declarations named by `for` form its **local target set**.

Supported Declaration and Target Kinds

`@co.dap.local` may be applied only to these declaration kinds:

- `co.lang.class`
- `co.lang.struct`
- `co.lang.enum`
- `co.lang.module`
- a named function declared in a context that normally permits functions

Every entry in the local target set must resolve to exactly one:

- class
- struct
- enum
- module

- function overload

A target list may contain any combination of supported target kinds.

Signatures and interfaces do not support target-local or physically nested declarations. A `co.lang.signature` or `co.lang.interface` cannot be annotated with `@co.dap.local` and cannot appear in a local target set. A signature may declare abstract, fixed, or generic **type-component specifications** as part of its module contract; these are contract requirements rather than local or nested type definitions. Interfaces do not declare signature type components.

Other declaration kinds are not target-local unless a later section explicitly permits them.

Target-Set Syntax

The `for` argument accepts either:

1. one declaration reference; or
2. a non-empty list of declaration references.

The scalar form is equivalent to a singleton target list:

```
@co.dap.local(for=hr.employee.Employee)
```

```
@co.dap.local(for=[hr.employee.Employee])
```

The scalar form is canonical when only one target is required. Use the list form when two or more declarations require access:

```
@co.dap.local(
  for=[
    hr.employee.Employee,
    hr.employee.EmployeeService,
    hr.employee.EmployeeValidator
  ]
)
EmployeeState co.lang.enum = {
  Active,
  Inactive
}
```

Rules:

- the target list must contain at least one declaration reference
- each reference must resolve independently and unambiguously
- duplicate references to the same resolved declaration are a compiler error
- list order does not affect visibility or declaration identity
- the target list is a closed set; access is not granted to declarations omitted from it
- repeated `@co.dap.local` annotations are not used to accumulate targets; use one annotation with one complete target list

Invalid:

```
@co.dap.local(for=[])
State co.lang.struct = { ... }
// ❌ empty target list
```

```
@co.dap.local(
  for=[
    hr.employee.Employee,
    hr.employee.Employee
  ]
)
State co.lang.struct = { ... }
// ❌ duplicate resolved target
```

Declaration-Reference Syntax

Each `for` entry is a compiler-resolved declaration reference, not a string, runtime expression, function call, or `co.meta.symbol` value.

For a non-function target, use only its complete qualified declaration name:

```
@co.dap.local(for=hr.employee.Employee)
@co.dap.local(for=hr.employee.EmployeeStatus)
@co.dap.local(for=hr.employee.EmployeeRules)
```

For a function target, use its complete qualified function signature because FoLang permits overloads:

```
@co.dap.local(
  for=hr.employee.Employee.calculate(co.lang.decimal)->()
)
```

Function references in a target list follow the same rule:

```
@co.dap.local(
  for=[
    hr.employee.Employee.calculate(co.lang.decimal)->(),
    hr.employee.Employee.calculate(co.lang.int)->(),
    hr.employee.Employee.validate()->(co.lang.bool)
  ]
)
CalculationState co.lang.struct = { ... }
```

Parameter names are not part of the reference:

```
// ✅ canonical
hr.employee.Employee.calculate(co.lang.decimal)->()

// ❌ parameter names are not declaration identity
hr.employee.Employee.calculate(amount co.lang.decimal)->()
```

The parameter and return types must match one exact overload. An abbreviated function name, unresolved target, or ambiguous overload is a compiler error.

```
@co.dap.local(for=hr.employee.Employee.calculate)
// ❌ ambiguous when calculate is overloaded
```

Each listed overload is an independent target. Listing one overload does not grant access to sibling overloads with the same function name.

Same-Package Requirement

The target-local declaration and **every declaration in its local target set** must belong to the same exact package. The compiler compares their complete folder-derived package identities; matching only a parent package, package family, import alias, realm, library, or source root is not sufficient.

For a function target, the target package is the package that owns the function's enclosing class, struct companion unit, module, unit, or other legal function container. A target-local function is checked in the same way: the package containing its legal function-owning declaration must match the package of every listed target.

The invariant is:

```
package(target-local declaration)
== package(target 1)
== package(target 2)
== ...
```

```
// hr/employee/Employee.fol
Employee co.lang.class = { ... }

// hr/employee/EmployeeService.fol
EmployeeService co.lang.class = { ... }

// hr/employee/EmployeeState.fol
@co.dap.local(
  for=[
    hr.employee.Employee,
    hr.employee.EmployeeService
  ]
)
EmployeeState co.lang.enum = { Active, Inactive }
// ✅ all declarations belong to package hr.employee
```

A declaration in another package cannot participate in the local target set, even when ordinary package visibility, imports, aliases, parent/subpackage relationships, or library membership would otherwise make the declaration resolvable.

```
// EmployeeState is declared in package hr.internal
@co.dap.local(
  for=[
    hr.employee.Employee,
    hr.internal.EmployeeService
  ]
)
EmployeeState co.lang.enum = { Active, Inactive }
// ❌ local declaration and every target must have the same package
```

```
// CalculationState is declared in package hr.employee.internal
@co.dap.local(
  for=hr.employee.Employee.calculate(co.lang.decimal)->()
)
CalculationState co.lang.struct = { ... }
// ❌ subpackages are distinct packages; both sides must be exactly hr.employee
```

The same-package rule prevents `@co.dap.local` from becoming a cross-package friendship or visibility-bypass mechanism. A local declaration is a selectively shared implementation detail inside one package, not an imported helper attached from elsewhere.

Visibility Domain

A target-local declaration may be resolved only from:

1. each exact declaration in its local target set; and
2. lexical scopes contained within each listed target's implementation.

For targets `A`, `B`, and `C`, the visibility domain is the union of their implementation scopes:

```
visibility(local declaration)
= scope(A) ∪ scope(B) ∪ scope(C)
```

It is not an intersection, and code does not need to be simultaneously inside every target.

Consequences:

- a declaration local to a class is available to that class body and its methods
- a declaration local to a module is available to that module body and its functions
- a declaration local to a struct is available while resolving that struct declaration
- a declaration local to an enum is available while resolving that enum declaration
- a declaration local to a function is available only in the body of the exact targeted overload and its nested statement scopes
- when several targets are listed, each listed target receives access independently
- sibling declarations and sibling overloads not listed cannot resolve it
- subclasses, companion units, extensions, callees, callers, and related declarations do not gain access automatically

- visibility is not inherited, transitive, or propagated through calls, composition, embedding, imports, or other local declarations
- it cannot be imported, exported, or projected through a library surface

```
@co.dap.local(
  for=[
    hr.employee.Employee,
    hr.employee.EmployeeService
  ]
)
EmployeeState co.lang.enum = {
  Active,
  Inactive
}

Employee co.lang.class = {
  state EmployeeState; // ✅ listed target
}

EmployeeService co.lang.class = {
  state EmployeeState; // ✅ listed target
}

Payroll co.lang.class = {
  state EmployeeState; // ❌ not listed
}
```

Interaction with Ordinary Visibility

`@co.dap.local` establishes the final visibility boundary. Ordinary visibility annotations such as `@co.dap.public`, `@co.dap.package`, `@co.dap.protected`, or `@co.dap.private` cannot widen the declaration beyond its closed local target set.

```
@co.dap.public
@co.dap.local(
  for=[
    hr.employee.Employee,
    hr.employee.EmployeeService
  ]
)
EmployeeState co.lang.enum = { Active, Inactive }
```

`EmployeeState` is still visible only to the listed targets. The ordinary visibility annotation is redundant for external resolution and may produce a compiler warning, but it never overrides `@co.dap.local`.

When most or all declarations in a package require access, ordinary package visibility should be used instead of maintaining a large target list.

Source Placement and Identity

A target-local declaration remains in the source position normally required for its declaration kind:

- a class, struct, enum, or module remains a normal primary declaration and follows the one-primary-declaration-per-package-file rule
- a function remains inside a unit, class, module, or another declaration context that normally permits functions
- `@co.dap.local` does not permit a free function to appear loose at package-file scope

The declaration retains a stable package-owned compiler identity. Its package identity must be exactly the same as the package identity of every target. It does not become physically nested and is not addressed as `Target.LocalName`.

```
hr.employee.EmployeeState
  local-for [
    hr.employee.Employee,
    hr.employee.EmployeeService
  ]
```

The normalized local target set is declaration metadata. Target-list order does not create a different declaration identity.

No Implicit Relationship or Privilege

`@co.dap.local` changes visibility only. It does not imply:

- composition
- embedding
- inheritance
- lifecycle ownership
- memory ownership
- friendship or privileged private-member access
- automatic membership in any listed target
- shared runtime state among the targets

Composition must still be written explicitly through a field or another supported relation. Embedding remains a separate facility: only struct and enum declarations are eligible for embedding, and each use must follow the embedding rules defined for that declaration kind.

Escape Restrictions

A target-local type must not leak outside its complete visibility domain.

It cannot appear in:

- a public, package, or protected API visible outside the local target set
- a library surface
- the parameter or return types of a function to which it is local
- a field or method signature that exposes it beyond the local target set
- an exported generic specialization or type alias

A value whose static type is target-local must be converted to an externally visible type before leaving the union of the listed targets' visibility domains.

Invalid Physical Nesting

```
Employee co.lang.class = {  
  Address co.lang.struct = { ... } // ✗  
}  
  
EmployeeModule co.lang.module = {  
  Config co.lang.struct = { ... } // ✗  
}  
  
process()->() = {  
  State co.lang.enum = { Ready, Done } // ✗  
}  
  
EmployeeContract co.lang.signature = {  
  Employee co.lang.struct; // ✗  
}  
  
EmployeeApi co.lang.interface = {  
  Result co.lang.struct; // ✗  
}
```

Use separately declared package declarations, composition, embedding where allowed, and `@co.dap.local` when a closed set of declarations requires selective access.

There is another annotation `@co.dap.nested` which is similar to local but captures target state.

`@co.dap.nested(target=hr.emp.Employee)`

Instead of `for` we use `target` and `target` is always a single fully qualified type/function.

`@co.dap.nested` behaves exactly like nested or inner classes or functions

Comparison Table

Annotation	attribute	multiple targets	capture the target state
@co.dap.local	for	✓ as list for single target can mention without list syntax	✗
@co.dap.nested	target	✗	✓

Statements

A statement is a complete executable or declarative instruction. It may contain one or more expressions and may change program state, control execution, introduce declarations, or produce observable effects.

Common statement categories in `FoLang`

1. Declaration Statement
2. Initialization Statement
3. Expression Statement
4. Conditional Statement
5. Loop Statement etc.,

Expressions

An expression is a construct that evaluates to a value, produces an effect, or both, and may be contained within a larger expression or statement.

Expression Evaluation Order

FoLang defines a deterministic evaluation order for expressions.

Expression structure is determined first by:

1. explicit grouping with parentheses;
2. operator precedence;
3. operator associativity.

After the expression structure has been determined, operands and subexpressions are evaluated from left to right in their source-code order, except where a construct explicitly defines conditional, lazy, asynchronous, concurrent, or otherwise specialized evaluation behavior.

Ordinary Expressions

For an ordinary expression:

```
result = first() + second();
```

the evaluation order is:

1. evaluate `first()`;
2. evaluate `second()`;
3. apply the `+` operator;
4. assign the resulting value to `result`.

Precedence and Evaluation Order

Left-to-right evaluation does not override operator precedence.

For example:

```
result = first() + second() * third();
```

operator precedence determines the expression structure as:

```
result = first() + (second() * third());
```

The function calls are evaluated in source order:

1. evaluate `first()`;
2. evaluate `second()`;
3. evaluate `third()`;
4. multiply the results of `second()` and `third()`;
5. add the multiplication result to the result of `first()`;
6. assign the final result to `result`.

Therefore, precedence determines how values are combined, while evaluation order determines when each operand or subexpression is evaluated.

Function Calls

For a function call, the callable expression is evaluated first, followed by the arguments from left to right.

```
result = calculate(first(), second(), third());
```

The evaluation order is:

1. evaluate the callable expression `calculate`;
2. evaluate `first()`;
3. evaluate `second()`;
4. evaluate `third()`;
5. invoke `calculate` with the resulting argument values;
6. assign the returned value to `result`.

Name resolution performed at compile time does not constitute runtime evaluation.

Method Calls

For a method call, the receiver expression is evaluated first, followed by the arguments from left to right.

```
result = getEmployee().calculate(first(), second());
```

The evaluation order is:

1. evaluate `getEmployee()`;
2. resolve the resulting object as the method receiver;
3. evaluate `first()`;
4. evaluate `second()`;
5. invoke `calculate`;
6. assign the returned value to `result`.

The receiver expression must be evaluated exactly once.

Member Access

For member access, the expression producing the containing object is evaluated before the member is accessed.

```
value = getEmployee().address.city;
```

The evaluation order is:

1. evaluate `getEmployee()`;
2. access `address` on the resulting object;
3. access `city` on the resulting address;
4. assign the resulting value to `value`.

Each intermediate receiver is evaluated only once.

Indexing

For an indexing expression, the indexed object is evaluated first, followed by index expressions from left to right.

```
value = getMatrix()[row()][column()];
```

The evaluation order is:

1. evaluate `getMatrix()`;
2. evaluate `row()`;
3. perform the first indexing operation;
4. evaluate `column()`;
5. perform the second indexing operation;
6. assign the resulting value to `value`.

Assignment

For assignment, FoLang evaluates the assignment target first, then evaluates the right-hand-side expression, and finally performs the assignment.

```
array[index()] = calculate();
```

The evaluation order is:

1. evaluate `array`;
2. evaluate `index()`;
3. determine the destination location;
4. evaluate `calculate()`;
5. assign the resulting value to the destination.

Determining the assignment target does not read the previous value stored at that location unless the assignment operation explicitly requires it.

Simple Variable Assignment

For assignment to a simple variable:

```
result = first() + second();
```

the evaluation order is:

1. determine the binding represented by `result`;
2. evaluate `first()`;
3. evaluate `second()`;
4. apply the `+` operator;
5. assign the resulting value to `result`.

Compound Assignment

A compound assignment evaluates its target exactly once.

```
array[index()] += calculate();
```

The evaluation order is:

1. evaluate `array`;
2. evaluate `index()`;
3. determine the destination location;
4. read the current value from that location;
5. evaluate `calculate()`;
6. apply the `+` operator;
7. assign the resulting value to the same destination.

The expression above must not behave as though it were expanded into a form that evaluates `array` or `index()` more than once.

Multiple Assignment

When an assignment contains multiple right-hand-side expressions, those expressions are evaluated completely from left to right before values are assigned to their targets.

```
a, b = first(), second();
```

The evaluation order is:

1. determine the target for `a`;
2. determine the target for `b`;
3. evaluate `first()`;
4. evaluate `second()`;
5. assign the first resulting value to `a`;
6. assign the second resulting value to `b`.

Right-hand-side evaluation must complete before any target receives its new value.

This permits value exchange without an explicit temporary variable:

```
a, b = b, a;
```

Operator Expressions

Built-in and user-defined operators follow the same operand-evaluation rules.

For a binary operator:

```
left() + right()
```

FoLang evaluates `left()` before `right()`.

For a prefix unary operator:

```
-operand()
```

FoLang evaluates `operand()` before applying the operator.

For a postfix unary operator:

```
operand()!
```

FoLang evaluates `operand()` before applying the operator.

Operator overloading must not change the specified evaluation order of operands.

Short-Circuit Boolean Operations

Short-circuit Boolean operators evaluate the left operand first.

The right operand is evaluated only when it is required to determine the result.

For logical AND:

```
left() && right()
```

the evaluation order is:

1. evaluate `left()`;
2. when the result is false, return false without evaluating `right()`;
3. otherwise, evaluate `right()` and use its Boolean result.

For logical OR:

```
left() || right()
```

the evaluation order is:

1. evaluate `left()`;
2. when the result is true, return true without evaluating `right()`;
3. otherwise, evaluate `right()` and use its Boolean result.

A skipped operand produces no value, mutation, side effect, or error.

Conditional Expressions

A conditional expression evaluates its condition first and then evaluates exactly one selected result expression.

```
result = condition()  
    .return(whenTrue())  
    .otherwise.return(whenFalse());
```

The evaluation order is:

1. evaluate `condition()`;
2. when the condition is true, evaluate `whenTrue()` only;
3. otherwise, evaluate `whenFalse()` only;
4. assign the selected result to `result`.

The unselected expression must not be evaluated.

Conditions and Branches

For a condition-and-branch construct, conditions are evaluated sequentially from left to right.

```
firstCondition().do({  
    firstBranch();  
}).otherwise(secondCondition()).do({  
    secondBranch();  
}).otherwise.do({  
    finalBranch();  
});
```

The evaluation order is:

1. evaluate `firstCondition()`;
2. when true, execute `firstBranch()` and skip the remaining conditions and branches;
3. otherwise, evaluate `secondCondition()`;
4. when true, execute `secondBranch()` and skip the final branch;
5. otherwise, execute `finalBranch()`.

Only the selected branch is evaluated.

Pattern Matching

A pattern-matching subject is evaluated exactly once.

Cases are examined in source order unless a particular matcher explicitly defines another ordering rule.

```
getValue().match  
    .case(firstPattern => firstResult())  
    .case(secondPattern => secondResult())  
    .default(defaultResult());
```

The evaluation order is:

1. evaluate `getValue()` exactly once;

2. test `firstPattern`;
3. when it matches, evaluate `firstResult()` and stop;
4. otherwise, test `secondPattern`;
5. when it matches, evaluate `secondResult()` and stop;
6. otherwise, evaluate `defaultResult()`.

Results belonging to unmatched cases are not evaluated.

Pattern guards are evaluated only after their corresponding structural pattern has matched.

Collection Literals

Elements of an array, list, tuple, set, map, or other collection literal are evaluated from left to right as they appear in the source.

```
values = [first(), second(), third()];
```

The evaluation order is:

1. evaluate `first()`;
2. evaluate `second()`;
3. evaluate `third()`;
4. construct the collection from the resulting values.

For a map entry, the key expression is evaluated before its corresponding value expression.

```
values = {  
    firstKey(): firstValue(),  
    secondKey(): secondValue()  
};
```

The evaluation order is:

1. evaluate `firstKey()`;
2. evaluate `firstValue()`;
3. evaluate `secondKey()`;
4. evaluate `secondValue()`;
5. construct the map.

Range Expressions

For a bounded range, the lower-bound expression is evaluated before the upper-bound expression.

```
range = lower()..upper();
```

The evaluation order is:

1. evaluate `lower()`;
2. evaluate `upper()`;
3. construct the range.

An omitted bound is not evaluated and does not produce an implicit function call or side effect.

Comprehensions and Iteration

A comprehension or iteration construct follows its own defined iteration semantics.

Within each iteration:

- source expressions are evaluated in the order declared;
- filter conditions are evaluated before result expressions;
- a result expression is evaluated only when its filters succeed;
- individual operand evaluation remains left to right.

The language does not implicitly evaluate comprehension iterations concurrently unless concurrency is explicitly requested.

Lazy Expressions

An expression declared lazy is not evaluated when the lazy binding is created.

Its evaluation occurs only when demanded according to the rules of the corresponding lazy construct.

Once evaluation begins, the expression itself follows the normal FoLang evaluation-order rules unless the lazy construct specifies otherwise.

The specification for each lazy construct must also state whether its result is:

- evaluated once and cached;
- evaluated again for every demand;
- safe for concurrent demand;
- permitted to propagate errors more than once.

Asynchronous and Concurrent Expressions

Left-to-right evaluation determines the order in which asynchronous or concurrent operations are created, invoked, or submitted.

It does not necessarily determine the order in which independently executing operations complete.

```
submit(first());  
submit(second());
```

FoLang evaluates and submits `first()` before `second()`. However, completion order depends on the execution-model rules unless explicit ordering is requested.

Concurrent execution never arises implicitly from an ordinary expression. It must be introduced by an explicit FoLang concurrency, parallelism, asynchronous-execution, scheduling, or execution-model construct.

Errors During Evaluation

When evaluation of a subexpression produces an error that prevents further evaluation, later subexpressions are not evaluated.

```
result = first() + failing() + third();
```

When `failing()` terminates evaluation with an error:

1. `first()` has already been evaluated;
2. `failing()` produces the error;
3. `third()` is not evaluated;
4. no final value is assigned to `result`.

The applicable error-handling rules determine whether the error is propagated, matched, converted, recovered from, or terminates execution.

Side Effects

All observable side effects occur according to the defined evaluation order.

Observable effects include:

- object mutation;
- variable rebinding;
- input and output;
- file-system operations;
- network operations;
- synchronization;
- exception or error generation;
- interaction with foreign code;
- interaction with the runtime environment.

An implementation must not reorder operations when doing so could change any observable effect.

Compiler Optimizations

A compiler may optimize, combine, eliminate, or internally reorder operations only when the transformation preserves all behavior required by this specification.

In particular, an optimization must not change:

- the resulting value;
- the order or presence of observable side effects;
- the identity or aliasing behavior of objects;
- the timing or presence of specified errors;
- the selected conditional or pattern-matching branch;
- the number of times an effectful expression is evaluated;
- synchronization or concurrency guarantees.

Pure internal operations may be reordered when no conforming FoLang program can observe the difference.

Implementation Requirement

Every conforming FoLang implementation must preserve the evaluation order defined in this section.

An implementation may use any parser, intermediate representation, optimizer, runtime, or backend, but those implementation choices must not alter the externally observable behavior required by these rules.

Operators

Custom Operator Definition & Overloading

Operator functions are functions and cannot be declared loose at package scope. An operator function for a struct must be declared inside that struct's same-package companion unit.

The **first declared operand parameter** must have the matching struct type. A matching struct type in a later operand is insufficient. For a unary operator, the sole operand must have the matching struct type.

For infix operators, this makes the first parameter the ownership or left-operand type used for companion-unit lookup. For example, an operator in `Vector co.lang.unit` may define `Vector * scalar`, but it may not define `scalar * Vector` unless that operation is owned and provided by the scalar type's applicable implementation.

```
Employee co.lang.struct = {
    salary co.lang.int;
}

Employee co.lang.unit = {

    @co.dap.operator(symbol='+', mode=overload)
    add(a Employee, b Employee)->(Employee) = {
        this.return Employee{
            salary: a.salary + b.salary
        };
    }
}
```

The expression:

```
combined := first + second;
```

is resolved through the `Employee` companion unit.

```

Vector co.lang.struct = {
  x co.lang.float;
  y co.lang.float;
}

Vector co.lang.unit = {

  @co.dap.operator(
    symbol='U',
    mode=define,
    fixity=infix,
    precedence=60,
    associativity=left,
    arity=binary,
    commutative=true,
    idempotent=true,
    identity="∅",
    foldable=true,
    vectorizable=false,
    distributes_over=['n'],
    desugar="intrinsic:set_union"
  )
  union(left Vector, right Vector)->(Vector) = {
    ...
  }
}

```

Invalid placement and ownership:

```

@co.dap.operator(symbol='+', mode=overload)
add(a Employee, b Employee)->(Employee) = { ... }
// ✗ operator function cannot appear at package scope

Math co.lang.unit = {
  @co.dap.operator(symbol='+', mode=overload)
  add(a Employee, b Employee)->(Employee) = { ... }
  // ✗ Math is not the companion unit of Employee
}

Employee co.lang.unit = {
  @co.dap.operator(symbol='+', mode=overload)
  add(a co.lang.int, b Employee)->(Employee) = { ... }
  // ✗ first operand is not Employee; a later matching operand is insufficient
}

```

`mode=override` is not supported in the foreseeable future; the compiler reports an error.

fixity values: `infix`, `postfix`, `prefix`, `circumfix`, `postcircumfix`, `prescircumfix`, `mixfix`, `ternary`, `distfix`

Forward / Extern Declarations

Variables extern declaration

```

@co.dap.declare(extern)
someBool co.lang.bool;

```

Functions forward declaration

```

@co.dap.declare(forward)
getEmployee(id co.lang.int)->(somepack.Employee);

// or - @co.dap.declare is optional for functions
getEmployee(id co.lang.int)->(somepack.Employee);

```

Types external declaration

```
@co.dap.declare(extern)
Employee co.lang.struct;

// or - @co.dap.declare is optional for types
Employee co.lang.struct;
```

For functions and types `@co.dap.declare` is optional. For variables it is required.

Functions

`foLang` doesn't allow free flowing functions as UDTs they must be in package source file apart from that they must be enclosed in a kind of container call Unit `co.lang.unit`

Normal

```
General co.lang.unit = {

    fun1(k co.lang.int, b co.lang.char)->(co.lang.int, co.lang.char) = {
        // function body
    }
}
```

`foLang` function can return multiple values

Default Parameters

```
fun1 (k co.lang.int, b co.lang.char = 10)->(co.lang.int, co.lang.char)={
}
```

Variadic Functions

Curried functions are not allowed to be variadic, and vice versa.

```
fun1 (k co.lang.int, ...b co.lang.char)->(co.lang.int, co.lang.char)={
}
```

Optional Parameters

```
fun1(k? co.lang.int)->()={
    if k.omitted{

    }else{

    }
}
```

Named Parameters

```
fun1(~k co.lang.int)->()={
}
```

Named Returns

```
doManythings(a co.lang.int, b co.lang.int->(&, meta={type=out}))->(r co.lang.int, e co.lang.exception)={}
```

Function Delegates

```
@co.dap.delegate someDelegate co.lang.delegate = (a co.lang.int, b co.lang.int) -> (co.lang.int, co.lang.int);
```


Function Chaining

```
fetchEmployee(empId co.lang.string)->(Employee)=>>empMod.getEmployee(this, empId);
```

Anonymous Functions

```
add := (a int, b int) -> (int) {  
    this.return a + b;  
};  
  
res := (a int, b int) -> (int) {  
    this.return a * b;  
}(10, 20);
```

for more details on functions please refer section [Functions in Details](#)

Functions in detail

Inline

```
Math co.lang.unit = {  
    @co.dap.inline  
    add(a co.lang.int, b co.lang.int)->(co.lang.int) = {  
        this.return a + b;  
    }  
}
```

Anonymous Functions

```
add := (a int, b int) -> (int) {  
    this.return a + b;  
};  
  
res := (a int, b int) -> (int) {  
    this.return a * b;  
})(10, 20);
```

Lambda

Only allowed as an inline callback argument to collection operations (e.g. `map`, `filter`, `reduce`, `forEach`, `sortBy`, `groupBy`). Using `|...|` anywhere else is a syntax/lint error.

```
// Syntax  
|x, y| => x + y;  
  
// Collection use – allowed  
nums.map(|x| => x*x)  
words.filter(|s| => s.len() > 3)  
pairs.reduce(|acc, e| => acc + e, 0)  
dict.map(|k, v| => v * 10)  
list.sortBy(|a, b| => a.score - b.score)
```

Inner Function

```
myfun(a co.lang.int, b co.lang.int)->(co.lang.int)={
  p co.lang.int = 10;
  someother()->()={
    co.out.println(p);
  }
  someother();
  p = 20;
  someother();
}
```

As we have already seen the inner function capabilities achieved through either `@co.dap.local` or `@co.dap.nested`

Those two forms allow separation of inner function logic from main function so that improves readability. The above form is useful when inner functions are small and compact.

Curried

```
add(first co.lang.int)(second co.lang.int)->(co.lang.int)={
  this.return first + second
}
```

Closure

```
adder() -> ((co.lang.int) -> co.lang.int) = {
  sum co.lang.int = 0
  this.return (x co.lang.int) -> (co.lang.int) = {
    sum += x
    this.return sum
  }
}
```

Functions Taking and Returning Functions

Syntax 1 — Inline signature

```
someFunction (r (co.lang.int, co.lang.int)->(co.lang.int))->((co.lang.int)->(co.lang.int))={}
```

Syntax 2 — Named type alias

```
someFArg co.lang.type = (co.lang.int, co.lang.int)->(co.lang.int)
someFRet co.lang.type = (co.lang.int)->(co.lang.int)

someFunction (r someFArg)->(someFRet)={}
```

Syntax 3 — Function objects

```
someFArg co.lang.function = (a co.lang.int, b co.lang.int) -> (co.lang.int)={
  this.return a + b;
}

someFRet co.lang.function = (a co.lang.int) -> (co.lang.int)={
  this.return a * 2;
}
```

Other ways to declare closures/function objects and types/ curried functions

```
myobj co.lang.function = (a co.lang.int, b co.lang.int)->(co.lang.int)={
  this.return a + b;
}

add (a co.lang.int, b co.lang.int)->(co.lang.int){ this.return a + b; }
oObj co.lang.function = add;

funtype co.lang.type = (a co.lang.int, b co.lang.int)->(co.lang.int);

closure(factor int) => (x int) = x * factor;

curry(factor int)(val int) = factory * val;
```

Associated Functions

For a user-defined struct, associated functions must be declared inside the same-package companion unit whose name matches the struct. For more details on associated function please refer section [Associated Functions in a Companion Unit](#)

Some Restrictions on Special Functions

1. Special functions

- Curried functions
- Functions with named arguments
- Functions with optional arguments
- Functions with default arguments
- Variadic functions
- Functions that take or return functions or function types
- Dynamically scoped and mixed-scoped functions

2. Restrictions

- They cannot be overloaded.
- They cannot be used as callbacks.
- They cannot participate in [Execution Models and Control Abstractions](#).

Scoping Rules for Functions

All functions in FoLang have a defined scope — the set of variables a function can access.

Default — Lexical / Static Scope

All of the following are **always** lexically scoped — this cannot be changed:

```
methods
inner methods
free functions
inner functions
target-local named functions
closures
lambdas
Generic Functions
Anonymous functions
Curried functions
```

Lexical scope means a function resolves names from its **declaration site**, not from the scope of its caller. Unit-level variables are forbidden, so a unit-scoped free function receives runtime values through parameters or introduces them locally. Inner functions may capture variables from the enclosing function scope.

```
ScopeExample co.lang.unit = {

  foo()->() = {
    x co.lang.int = 10;

    printX()->() = {
      co.out.println(x); // ✅ x from the enclosing lexical scope
    }

    printX();
  }
}
```

Associated Functions — Additional Scope Options

Only associated functions support non-lexical scoping via annotations:

The examples below are members of the same-package `Employee` companion unit.

`@co.dap.lexicalscope` — default, explicit declaration

```
Employee co.lang.unit = {
  @co.dap.lexicalscope
  (emp Employee) process()->() = {
    co.out.println(emp.name); // ✅ declaration scope
  }
}
```

`@co.dap.dynamicscope` — accesses caller's scope

```
Employee co.lang.unit = {
  @co.dap.dynamicscope
  (emp Employee) process()->() = {
    co.out.println(name); // name comes from caller's scope
  }
}
```

`@co.dap.mixedscope` — accesses both scopes

```
Employee co.lang.unit = {
  // caller scope takes priority – shadows declaration scope on conflict
  @co.dap.mixedscope
  (emp Employee) process()->() = {
    co.out.println(name); // caller scope takes priority
    co.out.println(emp.id); // falls back to declaration scope
  }
}
```

Callback Scope Inside Dynamically Scoped Associated Functions

A callback block or lambda does not independently select lexical, dynamic, or mixed scope. The associated function that executes the callback determines how the callback's free runtime names are resolved.

Callback parameters and declarations made inside the callback always belong to the callback's local scope. Names that are not callback parameters or callback-local declarations follow the executing associated function's scope policy.

```
nums.reduce(|acc, e| => {
  total.value += e;
  acc + e
}, 0)
```

For this example:

```
acc, e          -> callback parameters
temporary locals -> callback-local context
total           -> reduce's dynamic caller context
co.* and imports -> normal built-in/import resolution
```

The callback syntax remains uniform. `reduce` is dynamically scoped, so unresolved runtime names in the callback are resolved through the active caller-context chain. A lexically scoped associated function would instead resolve free runtime names through its declaration context.

Why Dynamic Scope Exists — `.do`, `.loop`, `.each`, and Collections

FoLang's control-flow model is built on dynamically scoped associated functions. The executing function supplies the scope policy, so blocks do not require separate capturing and non-capturing forms.

```
x      co.lang.int = 10;
total  co.lang.int = 0;
arr     co.lang.int->([5]) = [1, 2, 3, 4, 5];

// .do reads and modifies the caller's x
(x > 5).do({
  x.value = 20;
  co.out.println(x);
})

// .loop modifies the caller's x
(x > 0).loop({
  x.value--;
})

// .each modifies the caller's total
arr.each(_, val).do({
  total.value += val;
})

// .filter, .map, and .reduce use the same dynamic caller context
nums.filter(|x| => x > 10)
nums.reduce(|acc, e| => {
  total.value += e;
  acc + e
}, 0)
```

The compiler does not create a separate capture description for each control block. It resolves names according to the scope mode of the associated function executing that block.

FoLang Control Flow Uses Dynamic Scope

```
no if/else keywords  -> .do / .otherwise  - dynamic scope
no for/while keywords -> .loop           - dynamic scope
no foreach keywords  -> .each           - dynamic scope
no in keyword         -> .contains       - dynamic scope
no map/filter keywords -> .map / .filter  - dynamic scope

all control flow is implemented as associated functions
control associated functions are dynamically scoped
caller variables are accessible and mutable under the dynamic lookup rules
block syntax requires no separate capture mode
```

Dynamic and Mixed Lookup Order

For `@co.dap.dynamicscope`, runtime variable lookup follows this order:

1. function/callback parameters and local declarations
2. immediate caller activation context
3. caller activation contexts outward through the dynamic call chain
4. built-in ``co.*`` names and imported declarations visible to the source file
5. compiler error if no compatible declaration is available

For `@co.dap.mixedscope`, lookup follows this order:

1. function/callback parameters and local declarations
2. immediate caller activation context
3. caller activation contexts outward through the dynamic call chain
4. lexical declaration context
5. built-in ``co.*`` names and imported declarations visible to the source file
6. compiler error if no compatible declaration is available

Local declarations shadow caller declarations. Caller declarations shadow declarations from the lexical declaration context in mixed scope.

Types, annotations, imports, and other compile-time declarations continue to use ordinary static resolution. Dynamic lookup applies to runtime bindings, not to the identity of types or imported APIs.

Call-Site Validation of Dynamic Requirements

A dynamically or mixed-scoped associated function may reference a caller-provided runtime name that is not declared in its lexical context:

```
@co.dap.dynamicscope
(Employee) addToTotal(value co.lang.int)->>() = {
    total.value += value;
}
```

At every statically known call site, the compiler validates that the active caller-context chain provides a definitely initialized binding named `total` with a compatible type and permitted mutability. A call is rejected when the required binding cannot be proven to exist.

Because non-lexically scoped functions are non-first-class and non-escaping, their call sites remain statically identifiable. This avoids runtime string-based name lookup and allows the compiler to represent dynamic requirements using context and symbol-table references.

Scope Rules Summary

Function Type	Lexical	Dynamic	Mixed
methods	✓ only	✗	✗
inner methods	✓ only	✗	✗
free functions	✓ only	✗	✗
inner functions	✓ only	✗	✗
target-local named functions	✓ only	✗	✗
anonymous functions	✓ only	✗	✗
closures	✓ only	✗	✗
lambdas/callback blocks	determined by executing associated function	determined by executing associated function	determined by executing associated function
associated functions	✓ default	✓ opt-in	✓ opt-in
<code>.do</code> / <code>.loop</code> / <code>.each</code>	✗	✓ built-in	✗
<code>.map</code> / <code>.filter</code> / <code>.reduce</code>	✗	✓ built-in	✗

Additional Restrictions

- dynamically or mixed-scoped functions cannot be returned

- dynamically or mixed-scoped functions cannot be stored as values
- dynamically or mixed-scoped functions cannot be passed as ordinary callbacks
- dynamic or mixed caller contexts cannot cross thread or process boundaries
- dynamic or mixed execution cannot continue after the caller frame ends
- dynamic or mixed-scoped functions cannot participate in [Execution Models and Control Abstractions \(library type=advanced\)](#)
- dynamically or mixed-scoped functions cannot be curried
- named, optional, variadic, and default-parameter forms follow the same non-escaping and call-site-validation rules

Types

The three axes — each adds one new power:

Axis 1: Polymorphism (terms depend on types)

```
// "Give me a type, I'll give you a value"

// Without: write separate functions
identityInt(x int) → int
identityStr(x string) → string

// With: one function works for all types
identity(T)(x T) → T

// This is generics / parametric polymorphism
// System F, Java generics, your @co.dap.generic
```

Axis 2: Type operators (types depend on types)

```
// "Give me a type, I'll give you a type"

List(Int)    → List of Int      // type → type
Map(String, Int) → Map          // type → type → type
Option(T)    → Some(T) | None   // type constructor

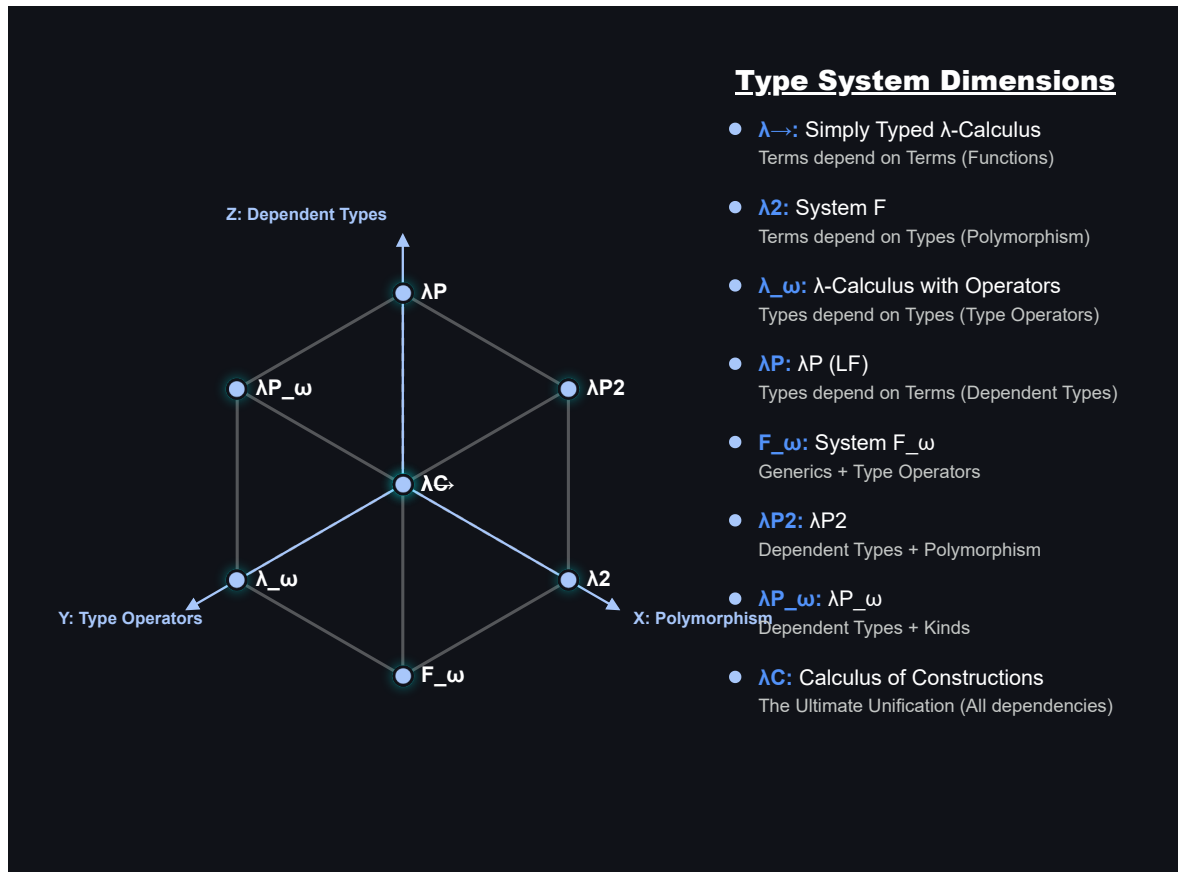
// This is kinds / higher-kinded types
// Your folang: Option(T) co.lang.type = Some(T) | None()
```

Axis 3: Dependent types (types depend on values)

```
// "Give me a value, I'll give you a type"

Vector(3)    → array of exactly 3 elements
Matrix(2, 3) → 2x3 matrix
NonZero(n)   → proof that n ≠ 0

// The type changes based on a runtime value
divide(a int, b NonZero(int)) → int
// compiler PROVES b can't be zero
```



Lambda Cube

Dependent Types

Type Constructors — Functions That Return Types

A type constructor is a function that takes a value or type and returns a type. The returned type depends on the input value — this is what makes it a dependent type.

```
// Vector – type constructor function
// takes → co.lang.int (size)
// returns → co.lang.dependentType (a type)
Vector(n co.lang.int)→(co.lang.dependentType) =
    co.lang.int→([n]);

// calling Vector(3) returns a TYPE at compile time
// that type is co.lang.int→([3])
v3 Vector(3) = [1, 2, 3]; // type is Vector(3)
v4 Vector(4) = [1, 2, 3, 4]; // type is Vector(4) – different type!

// Vector(3) ≠ Vector(4) – completely different types
// size is part of the type – compiler knows at compile time
```


More About Type

```
Name(T) co.lang.data = variants;  
  → concrete ADT type-constructor definition  
  → right-hand-side definition is mandatory  
  
Name(T) co.lang.type;  
  → abstract type-constructor requirement  
  → permitted only inside a signature  
  
Name(T) co.lang.type = ExistingType(T);  
  → concrete type alias or signature-component binding  
  → permitted in modules and ordinary type declarations
```

Type Constructor Is A Function

```
Vector      → function (type constructor)  
Vector(3)   → function call → returns type co.lang.int->([3])  
Vector(4)   → function call → returns type co.lang.int->([4])  
  
just like:  
  add(1, 2) → returns a value  (3)  
  Vector(3) → returns a type   (int[3])
```

Compiler Enforced Size Safety

```
// dot product – only valid for same size vectors  
// compiler enforces this via dependent types  
dotProduct(a Vector(n), b Vector(n))->(co.lang.int) = {  
  // n is same for both – compiler verified  
}  
  
v3 Vector(3) = [1, 2, 3];  
v4 Vector(4) = [1, 2, 3, 4];  
  
dotProduct(v3, v3) // ✓ same type Vector(3)  
dotProduct(v3, v4) // ✗ compiler error – Vector(3) ≠ Vector(4)
```

Matrix — Two Parameter Type Constructor

```
// Matrix – takes rows and cols, returns dependent type  
Matrix(r co.lang.int, c co.lang.int)->(co.lang.dependentType) =  
  co.lang.int->([r, c]);  
  
m34 Matrix(3, 4) = [[1,2,3,4],[5,6,7,8],[9,10,11,12]];  
m45 Matrix(4, 5) = ...;  
  
// matrix multiply – cols of A must equal rows of B  
// n must match – compiler verified  
multiply(a Matrix(r, n), b Matrix(n, c))->(Matrix(r, c)) = {  
  // compiler ensures dimensions are compatible  
}  
  
multiply(m34, m45) // ✓ Matrix(3,4) × Matrix(4,5) = Matrix(3,5)  
multiply(m34, m34) // ✗ compiler error – 4 ≠ 3
```

Stack — Value and Type Parameter

```
// Stack – takes size and element type
Stack(n co.lang.int, T co.lang.type)->(co.lang.dependentType) =
  T->([n]);

s Stack(10, co.lang.int)    = ...; // stack of max 10 ints
t Stack(5, co.lang.string) = ...; // stack of max 5 strings
```

Type Is Value + Kind Combined

```
Vector(3):
  kind  = Vector    (what it is)
  value = 3         (how many)
  type  = Vector(3) (both together – the dependent type)

Vector(3) ≠ Vector(4)  ← different types entirely
Vector(3) = Vector(3)  ← same type
```

Connects to Type Constructor in Spec

```
// Option – type constructor for ADT
@co.dap.hokrt
Option(T) co.lang.data = Some(T) | None();

// Vector – type constructor for dependent type
Vector(n co.lang.int)->(co.lang.dependentType) =
  co.lang.int->([n]);

// same concept:
// Option(T) takes a type → returns ADT type
// Vector(n) takes a value → returns dependent type
```

Simple Dependent Type

```
identity(x co.lang.int)->(x.type) = x
```

Compile-Time and Runtime Values in Type-Related Computation

FoLang distinguishes value-indexed dependent types, compile-time type computation, runtime type descriptors, and runtime values whose concrete types may differ. These mechanisms are related, but they are not interchangeable.

1. Value-Indexed Dependent Types

A dependent type may contain a value as part of its type identity. The value index may be a compile-time constant, a symbolic type-level value, or a runtime function parameter whose relationship to the result is tracked by the compiler.

```
readVector(n co.lang.int)->(Vector(n)) = {
  ...
}
```

The compiler does not need to know the concrete value of `n` while compiling `readVector`. It records the relationship:

```
input index = n
result type = Vector(n)
```

Likewise:

```
dotProduct(
  left Vector(n),
  right Vector(n)
)->(co.lang.int) = {
  ...
}
```

requires both vectors to have the same value index. Dependent typing means that a type contains or is constrained by a value; it does not mean that an arbitrary runtime branch can silently change the static type of an already compiled variable.

2. Compile-Time Type Computation

A function may compute and return a type when it is guaranteed to execute during compilation.

```
@co.dap.comptime
@co.dap.eager
chooseType(value co.lang.int)->(co.lang.type) = {
  (value < 100)
    .return(co.lang.string)
    .otherwise.return(co.lang.bool);
}
```

The arguments must be compile-time evaluable when the result is used in a static type position:

```
Selected co.lang.type = chooseType(10);
value Selected = "Hello";
```

Invalid:

```
input := co.in.readInt();
Selected co.lang.type = chooseType(input);
// compiler error: `input` is not compile-time evaluable
```

Conceptually:

```
compile-time value
  -> compiler executes the type function
  -> one concrete static type is available before ordinary type checking completes
```

3. Built in compile type computation

A function may compute and return a type when it is guaranteed to execute during compilation.

`decltype` built in method

The arguments must be compile-time evaluable when the result is used in a static type position:

```
someIntVar co.lang.int ;
someVar co.hokrlt.type.decltype(someIntVar) = 200;
```

4. Runtime Type Descriptors

An ordinary function returning `co.lang.type` produces a runtime type descriptor when it is not executed at compile time.

```
selectType(value co.lang.int)->(co.lang.type) = {
  (value < 100)
    .return(co.lang.string)
    .otherwise.return(co.lang.bool);
}

selectedType co.lang.type = selectType(input);
```

Here, `selectedType` is a runtime object that describes a type. It may be used for reflection, runtime validation, dynamic loading, metadata inspection, or dynamic object creation where those capabilities are permitted.

A runtime type descriptor cannot ordinarily be used as the static type of a declaration:

```
value selectType(input);  
// compiler error: runtime type descriptor used in a static type position
```

Conceptually:

```
runtime value  
  -> runtime type-selection function  
  -> co.lang.type descriptor object
```

A value that represents a type is not automatically a compile-time-resolved static type. ***

5. `@co.dap.typefromvalue`

`@co.dap.typefromvalue` derives a type from the type of a returned compile-time value. In the initial FoLang specification, it is permitted only for compile-time evaluation.

```
@co.dap.compiletime  
@co.dap.typefromvalue  
inferType(value co.lang.int)->(co.lang.type) = {  
  (value < 100)  
    .return("Hello")  
    .otherwise.return(co.const.true);  
}
```

The compiler derives:

```
"Hello"      -> co.lang.string  
co.const.true -> co.lang.bool
```

Every argument must be compile-time evaluable when the result is used in a type position. Runtime value-based type selection must instead use an explicit runtime representation.

6. Runtime Values with Different Concrete Types

When a runtime branch may produce unrelated concrete value types, the function must return one stable outer type. FoLang may use a tagged value, an ADT, `co.lang.dynamic` where permitted, or another explicitly packaged representation.

Using `co.lang.tag`:

```
selectValue(value co.lang.int)->(co.lang.tag) = {  
  (value < 100)  
    .return(co.lang.tag(co.lang.string, "Hello"))  
    .otherwise.return(  
      co.lang.tag(co.lang.bool, co.const.true)  
    );  
}
```

The outer static type is always `co.lang.tag`:

```
co.lang.tag  
├─ runtime type descriptor  
└─ value compatible with that descriptor
```

An ADT provides a more strongly typed alternative:

```

SelectedValue co.lang.data =
  StringValue(co.lang.string)
| BoolValue(co.lang.bool);

selectValue(value co.lang.int)->(SelectedValue) = {
  (value < 100)
    .return(StringValue("Hello"))
    .otherwise.return(BoolValue(co.const.true));
}

```

Summary

```

Vector(n)
  -> value-indexed dependent type

@co.dap.comptime function returning co.lang.type
  -> compile-time type computation

ordinary runtime function returning co.lang.type
  -> runtime type descriptor

runtime branch returning unrelated value types
  -> tag, ADT, dynamic value, or another explicit package

```

A runtime type descriptor is a value that represents a type. It must not be confused with a statically resolved dependent type.

Indexer

Indexer functions for a struct are associated functions and must be declared inside the matching companion unit.

```

MyList co.lang.struct = {
  eles co.lang.int->([...]);
}

MyList co.lang.unit = {

  @co.dap.indexer(symbol="[]")
  (g MyList) get(index co.lang.int)->(co.lang.int) = {
    this.return g.eles[index]
  }

  @co.dap.indexer(symbol="[]=")
  (g MyList) set(index co.lang.int, value co.lang.int)->() = {
    g.eles[index] = value
  }
}

lst MyList;
co.out.println(lst[0]);
lst[1] = 22;

```

Generics

```

@co.dap.generic(
  at=runtime,
  type={
    T: {variance:invariant, bound=Number, kind:param, impredicative:false},
    R: {variance:invariant, bound=Number, kind:return}
  }
)
add(a T, b T)->(R) = { this.return a + b; }

```

Generic annotation fields:

- `at` — `runtime` | `compiletime` (acts like C++ templates)
- `refied` — `true` | `false`
- `where` — `usesite` | `callsite`

typename/type attributes:

Attribute	Values
variance	<code>covariant</code> , <code>invariant</code> , <code>contravariant</code>
bound	type to bind
kind	<code>param</code> , <code>result</code> , <code>var</code> , <code>arg</code>
default	default type
nullable	bool
inclusive	bool
impredicative	bool — when <code>true</code> , allows <code>T</code> to be instantiated with a <code>forall</code> type (v2)
typekind	<code>type</code> , <code>class</code> , <code>function</code> , <code>module</code> , <code>unit</code> , <code>package</code>
types	list of allowed types for constraints

Generic Functions — Parameters and Return Values

Rank-1: Outer function is generic; parameter uses the same type variable

`T` is fixed at the call site before the function parameter is used. The passed function is already monomorphic inside the body.

Syntax 1 — Inline signature

```
@co.dap.generic(type={T:{variance:invariant}})
someFunction(f (T, T)->(T), a T)->(T) = {}
```

Syntax 2 — Named type alias

```
@co.dap.generic(type={T:{variance:invariant}})
someFArg co.lang.type = (T, T)->(T)

someFunction(f someFArg, a T, b T)->(T) = {}
```

Rank-2: The function parameter is itself polymorphic (higher-rank)

The passed function stays generic **inside the callee**. The callee decides what `T` is. Uses existing `forall`.

Syntax 1 — Inline signature

```
someFunction(f forall(T).(T, T)->(T))->(co.lang.int) = {
  this.return f(1, 2);
}
```

Syntax 2 — Named type alias

```
someFArg co.lang.type = forall(T).(T, T)->(T)

someFunction(f someFArg)->(co.lang.int) = {}
```

```
// Correct — Syntax 2 with co.lang.type
someFArg co.lang.type = forall(T).(T, T)->(T)

someFunction(f someFArg)->(co.lang.int) = {}
```

Returning Generic Functions

Rank-1 return

```
@co.dap.generic(type={T:{variance:invariant}})
makeAdder(a T)->((T)->(T)) = {
  this.return (b T)->(T) = { this.return a + b; };
}
```

Rank-2 return — returning a polymorphic function

```
makeIdentity()->( forall(T).(T)->(T) ) = {
  this.return forall(T).(x T)->(T) = { this.return x; };
}
```

Rank-3: A Parameter is Itself a Rank-2 Function

Rank-3 works naturally in FoLang via `forall` nesting. No new constructs needed.

Syntax 1 — Inline

```
// f takes a Rank-2 function as its argument — that is Rank-3
applyRank2(
  f (forall(T).(T, T)->(T)) -> (co.lang.int)
) -> (co.lang.int) = {
  this.return f(1, 1);
}
```

Syntax 2 — Named type aliases (cleaner)

```
rank2FnType co.lang.type = forall(T).(T, T)->(T)
rank3ArgType co.lang.type = (rank2FnType) -> (co.lang.int)

applyRank2(f rank3ArgType) -> (co.lang.int) = {
  this.return f(1, 1);
}
```

Rank-3 return

```
makeRank2Consumer() -> ((forall(T).(T)->(T)) -> (co.lang.int)) = {
  this.return (f forall(T).(T)->(T)) -> (co.lang.int) = {
    this.return f(42);
  };
}
```

Impredicativity — Instantiating `T` with a `forall` Type

Impredicativity is when a type variable `T` in a generic is itself instantiated with a `forall` type. Example of what this means:

```
@co.dap.generic(type={T:{variance:invariant}})
box(x T) -> (Box(T)) = {}

// Impredicative call — T being set to forall(U).(U)->(U)
result := box(forall(U).(U)->(U)); // ❌ not legal without explicit opt-in
```

Most type systems reject this by default. FoLang takes an opt-in approach.

v1 Workaround — Option C: Wrapping with `co.lang.type`

Not true impredicativity but solves 90% of practical cases:

```
polyId co.lang.type = forall(U).(U)->(U)

// box takes co.lang.type – no impredicative unification needed
box(x co.lang.type) -> (Box(co.lang.type)) = {}

result := box(polyId);    // ✔ works – x is co.lang.type, not a forall type
```

v2 — Option A: `impredicative:true` in `@co.dap.generic`

Explicit opt-in via the existing annotation. The compiler only permits `forall` instantiation where declared:

```
@co.dap.generic(
  type={T:{variance:invariant, impredicative:true}}
)
box(x T) -> (Box(T)) = {}

polyId co.lang.type = forall(U).(U)->(U)
result := box(polyId);    // ✔ legal – impredicative:true explicitly opts in
```

Generic Function Rank Support Matrix

Scenario	Allow?	Notes
Rank-1 generic param (Syntax 1, 2, 3)	✔ Yes	Natural extension, no new concepts
Rank-1 generic return (Syntax 1, 2, 3)	✔ Yes	Same as above
Rank-2 param via <code>forall</code> (Syntax 1, 2)	✔ Yes	<code>co.lang.type</code> naturally holds polymorphic types; no new kind needed
Rank-2 param via Syntax 3 <code>co.lang.function</code>	✗ Compiler error	Function objects are concrete values; use <code>co.lang.type = forall(T).(T)->(T)</code> instead
Rank-2 return via <code>forall</code> (Syntax 1, 2)	✔ Yes	Same reasoning as param
Rank-3 via <code>forall</code> nesting (Syntax 1, 2)	✔ Yes	No new constructs — <code>forall</code> nesting works naturally
Rank-3 return	✔ Yes	Same reasoning as Rank-3 param
Rank-3 via Syntax 3 <code>co.lang.function</code>	✗ Compiler error	Same rule as Rank-2; function objects are concrete
Impredicative — v1 workaround (Option C)	✔ Yes	Wrap <code>forall</code> type in <code>co.lang.type</code> ; solves 90% of real cases
Impredicative — true opt-in (Option A)	➡ SOON v2	<code>impredicative:true</code> in <code>@co.dap.generic</code> ; explicit opt-in

Generic Types

```
@co.dap.generic(typename=T)
LinkedList co.lang.struct={
    value T
    next  LinkedList
    prev  LinkedList
}

k := LinkedList.new(co.lang.int); // when we call new it returns an object of type co.lang.uninit
actualList := k.init(); // this is what create a fully formed object of type class

@co.dap.generic(type={T:{typename}, R:{typename}})
Employee co.lang.class = {
    id  T
    name R

    @co.dap.override
    @co.dap.constructor(access=private)
    @@init() = {}

    @co.dap.override
    @co.dap.constructor(access=public)
    @@init(id T, name R) = {
        this.parent.init();
        this.id  = id;
        this.name = name;
    }

    getEmployee(id T)->(Employee)={}
}

a := Employee.new(co.lang.int, co.lang.string);
b := a.init(1, "Rao");

Normally we need not use @@new and @@init it is special case only applicable for Generics
Normal conditions to create/instantiate object of class we just call init which internally call new
In specific cases as above we need to do two calls or use call chain like below

c:= Employee.new(co.lang.int,co.lang.string).init(1,"Rao");
```

Generics Inheritances and Types

This is in conceptual stage not supported.

- A) Abstract vs concrete type members
- B) Path-dependent types
 1. Type-level projection
 2. Path-dependent In folang how it would be

forall

What `forall` Is — and Is Not

`forall` is **not** a general-purpose generic declaration keyword. It is a **type-level expression only**, restricted to contexts where a polymorphic type must appear as an anonymous value inline — specifically Rank-2 and Rank-3 parameter and return positions, and `co.lang.type` aliases.

`@co.dap.generic` is the **one and only** way to declare generic functions, structs, classes, and other named things. `forall` at declaration level is a **compiler error**.

Where forall Is Allowed — Type Expression Form Only

forall(T). followed by an anonymous type body. The . is the syntactic signal that what follows is a type body, not a declaration name.

Pattern:

```
forall(T). <anonymous type body>

// co.lang.type alias - naming a polymorphic type for reuse
someFArg co.lang.type = forall(T).(T, T)->(T)

// Rank-2 inline parameter - callee decides what T is
someFunction(f forall(T).(T)->(T)) -> (co.lang.int) = {}

// Rank-2 return type - returning a polymorphic function
makeIdentity() -> (forall(T).(T)->(T)) = {}

// Rank-3 inline parameter - f takes a Rank-2 function
applyRank2(f (forall(T).(T, T)->(T)) -> (co.lang.int)) -> (co.lang.int) = {}
```

Where forall Is Banned — Use @co.dap.generic Instead

```
// ❌ compiler error - forall at declaration level
forall(T) identity(x T)->(T) = {}

// ✅ correct
@co.dap.generic(type={T:{variance:invariant}})
identity(x T)->(T) = {}

// ❌ compiler error
forall(T) LinkedList co.lang.struct = { value T; next LinkedList; }

// ✅ correct
@co.dap.generic(typename=T)
LinkedList co.lang.struct = { value T; next LinkedList; }

// ❌ compiler error - Rank-1 generics belong to @co.dap.generic
forall(T) someFunction(f (T,T)->(T), a T)->(T) = {}

// ✅ correct
@co.dap.generic(type={T:{variance:invariant}})
someFunction(f (T,T)->(T), a T)->(T) = {}
```

Quick Reference

Form	Status	Context
forall(T) name ...	❌ Compiler error	Declaration level — use @co.dap.generic instead
forall(T).(T)->(T)	✅ Allowed	Type level only — Rank-2/3 param, return, co.lang.type alias

The rule in one sentence: forall(T). is a type constructor for anonymous polymorphic types; it is never a declaration keyword.

Templates

Typed

```
@co.dap.template
add(a co.lang.int, b co.lang.int)->(co.lang.int) ={
    this.return a + b;
}
```

Untyped

```
@co.dap.template
add(a, b)->(co.lang.untyped) ={
    this.return a + b;
}
```

Annotations and Decorators

```
// Annotation – static object, can carry data

myAnnotation co.lang.object->(for=annotation) = {
    value    co.lang.string;
    enabled  co.lang.bool;
}

// Decorator – function, transforms target, returns
@co.dap.decorator
myDecorator(target co.lang.function)->(co.lang.function) = { }
```

Note Directives and Pragmas are not allowed to create as they are language internals

Macros

```
// a. Basic macro
@co.dap.macro
say()->()={ this.return co.macro.quote({ println("Line 1") println("Line 2") }); }

// b. Escape assign
@co.dap.macro
yes_esc_assign()->(co.lang.untyped)={
  this.return co.macro.quote({
    co.macro.esc(y) = 42
    println("Inside macro: y = ", y)
  });
}

// c. Debug macro with gensym
@co.dap.macro
debug(expr)->(co.lang.untyped)={
  let tmp = co.macro.gensym(co.lang.var, "tmp")
  this.return co.macro.quote({
    tmp = co.macro.esc(expr)
    println("Result: ", tmp)
    tmp
  });
}

// d. if/else condition macro
@co.dap.macro(
  group = {items:["if","else"], chain:true},
  sugarform={forms:["if expr block"]},
  bind={vars:["x"]},
  isolate={vars:["temp", "index"]},
  gensym={prefix:"tmp_"},
  hygienic=true,
  argtransform={param:"body", wrap:"lambda", whentype:"block"},
  desugar={exprs:["if($cond) { $block }" => "if($cond,$block)"]},
  mode="inject"
)
if(condition expr, body block)->()={

  blockormacro co.lang.Kind = block | macro

  @co.dap.macro(
    group={items:["if","else"], chain:true},
    sugarform={forms:["else block","else if"]},
    chainswith={macro:"if", position:"immediate", required:true},
    argtransform={param:"body", wrap:"lambda", whentype:"block"},
    standalone=false,
    desugar={exprs:[
      "else if($cond) { $block }" => "else(if($cond, $block))",
      "else { $elseblock }" => "else($elseblock)"
    ]},
  )
  else(body blockormacro)->()={}
```

Other macro utilities: 1. `@co.dap.compose(using=["base_if", "blockify"])` 2. `@co.dap.guard(expr="is_bool_expr(expr)")` 3. Quasiquote macros use `co.macro.quote` and `co.macro.unquote`

Execution Models and Control Abstractions (library type=advanced)

FoLang executes code sequentially by default. It also provides a uniform execution model for concurrency, parallelism, asynchronous execution, coroutines, continuations, scheduling, and structured task execution.

Developers express the intended execution semantics by applying annotations such as `@co.dap.thread`, `@co.dap.task`, or `@co.dap.process` to a method. When the method is submitted through facilities such as `co.cpa.submitToPool`, `co.cpa.submitThread`, or `co.cpa.submitToEventLoop`, the FoLang runtime selects and manages the appropriate execution mechanism.

Depending on the annotation, submission operation, runtime environment, and execution policy, FoLang may use a thread pool, virtual or green threads, an event loop, a dedicated operating-system thread, or a separate process. Communication operations such as sending and receiving values are also handled through the `co.cpc` package. Developers therefore describe the required execution behavior without directly managing the underlying threads, processes, pools, or event loops.

The `@co.dap.continuation` annotation enables continuation support for a function. An annotated function can use constructs provided by the `co.cpc` package to suspend execution, yield control or a value, preserve its execution state, and later resume from the suspension point.

Native Code (Library type system/ffi)

The `@co.dap.native` annotation enables access to the `co.native` package. Through this package, developers can write assembly and machine-level code using facilities such as `co.native.asm` and `co.native.inline`, providing low-level capabilities similar to those available in C++.

Native Functions

```
@co.dap.native
nativeMethod(a co.lang.int, b co.lang.int)-(co.lang.int) ={
    // native implementation
}
```

Dynamic Runtime (library type=dynamicvmrt)

The `@co.ddap.dynamicruntime` annotation enables full access to the `co.meta` package. Through this package, developers can use dynamic class and type loading, monkey patching, runtime reflection, instrumentation, eval-based code execution, and other advanced metaprogramming capabilities.

Variable Kinds Support

Kind	Where
Normal	All
Pointers	Library of type <code>system</code> and <code>ffi</code>
Arrays	All
References Heap, Lvalue, Rvalue	Library of type <code>system</code> and <code>ffi</code>
Addresses	Library of type <code>system</code> and <code>ffi</code>
Thunks	Library <code>application</code> or <code>system</code> or <code>ffi</code> or <code>advanced</code> or <code>dynamicvmrt</code>
Ranges	All
Slices	All

Builtin Data Types

Type	Kind
<code>co.lang.string</code>	
<code>co.lang.int</code>	
<code>co.lang.bit</code>	
<code>co.lang.double</code>	
<code>co.lang.float</code>	
<code>co.lang.long</code>	

co.lang.byte	
co.lang.char	
co.lang.any	
co.lang.dynamic	
co.lang.auto	
co.lang.infer	
co.lang.bool	
co.lang.void	
co.lang.data	
co.lang.value	
co.lang.typed	
co.lang.untyped	
co.mem.region	
co.lang.nothing	
co.lang.word	
co.lang.MatchBindings	
co.lang.tag	
co.lang.typevalue	
co.lang.uninit	
co.lang.Literal	
co.lang.pointer	
co.lang.address	
co.lang.reference	
co.lang.thunk	
co.lang.array	
co.lang.slice	
co.lang.range	

Builtin Directives

Built-in Directives

Kind		
PRAGMA	"@co.pdap.compiler", "@co.pdap.scale"	
DIRECTIVE	"@co.ddap.movetotop", "@co.ddap.import", "@co.ddap.dynamicruntime", "@co.ddap.use", "@co.ddap.alias"	

ANNOTATION	"@co.dap.template", "@co.dap.macro", "@co.dap.operator", "@co.dap.annotation", "@co.dap.library", "@co.dap.module", "@co.dap.pragma", "@co.dap.directive", "@co.dap.native", "@co.dap.class", "@co.dap.static", "@co.dap.instance", "@co.dap.object", "@co.dap.inline", "@co.dap.ctfe", "@co.dap.friend", "@co.dap.sealed", "@co.dap.extension", "@co.dap.override", "@co.dap.virtual", "@co.dap.abstract", "@co.dap.delegate", "@co.dap.dynamicscope", "@co.dap.lexicalscope", "@co.dap.staticscope", "@co.dap.mixedscope", "@co.dap.typeclass", "@co.dap.matcher", "@co.dap.constructor", "@co.dap.oops", "@co.dap.hokrt", "@co.dap.hokrlt", "@co.dap.indexer", "@co.dap.generic", "@co.dap.comptime", "@co.dap.typefromvalue", "@co.dap.local", "@co.dap.private", "@co.dap.public", "@co.dap.package", "@co.dap.protected", "@co.dap.internal", "@co.dap.export", "@co.dap.eager", "@co.dap.lazy", "@co.dap.packed", "@co.dap.declare", "@co.dap.simd", "@co.dap.reflection", "@co.dap.mop", "@co.dap.nested", "@co.dap.inner", "@co.dap.declare"	//mop => meta object programming
DECORATOR	"@co.dap.before", "@co.dap.after", "@co.dap.around", "@co.fx.onErrExcept", "@co.fx.InvokeAlways", "@co.fx.HandleEffect", "@co.dap.callback", "@co.dap.defer", "@co.dap.continuation", "@co.dap.event", "@co.dap.scale", "@co.dap.distributed", "@co.dap.concurrent", "@co.dap.parallel", "@co.dap.subroutine", "@co.dap.generator", "@co.dap.goroutine", "@co.dap.coroutine", "@co.dap.async", "@co.dap.promise", "@co.dap.future", "@co.dap.thread", "@co.dap.task", "@co.dap.fiber", "@co.dap.process", "@co.dap.spawn", "@co.dap.exec", "@co.dap.fork", "@co.dap.csp", "@co.dap.actor", "@co.dap.synthetic", "@co.dap.bridge", "@co.dap.greenlet", "@co.dap.channel", "@co.dap.callable", "@co.dap.iterator"	

Builtin Kinds

Kind	Purpose
co.lang.type	
co.lang.struct	
co.lang.cstruct	
co.lang.realm	similar to loader where symbols reside
co.lang.loader	class, type, functions loader where objects reside loader takes realm as parameter
co.lang.class	
co.lang.interface	
co.lang.union	
co.lang.role	
co.lang.record	
co.lang.property	
co.lang.indexer	
co.lang.object	
co.lang.instance	
co.lang.matcher	
co.lang.trait	
co.lang.mixin	
co.lang.extension	
co.lang.delegate	
co.lang.typeclass	

co.lang.concept	
co.lang.typealias	
co.lang.module	
co.lang.unit	named, non-instantiable function container; same-name struct unit acts as companion
co.lang.macro	
co.lang.template	
co.lang.lambda	
co.lang.block	
co.lang.behavior	
co.lang.package	
co.lang.signature	
co.lang.function	
co.lang.method	
co.lang.operator	
co.lang.namespace	
co.lang.stex	
co.lang.kind	
co.lang.level	
co.lang.order	
co.lang.rank	
co.lang.newtype	
co.lang.opaquetype	
co.lang.subtype	
co.lang.supertype	
co.lang.dependentType	
co.lang.refinementType	
co.lang.associatedtype	
co.lang.hokr1t	higer rank order kind level types
co.lang.data	
co.lang.enum	
co.lang.typetype	
co.lang.typekind	
co.lang.alias	
co.lang.value	
co.lang.just	
co.lang.nothing	

Builtin Operators

Arithmetic operators

`+`, `-`, `*`, `/`, `%`, `**`, `++`, `--`

Logical operators

`&&`, `||`, `!`, `&`, `|`

Comparison operators

`==`, `!=`, `<`, `>`, `<=`, `>=`

Other operators

`@`, `#`, `!`, `~`, `$`, `^`, `(`, `)`, `_`, ```, `?`, `{`, `[`, `]`, `}`, `\`, `:`, `;`, `"`, `'`, `=`, `.`, `?=`, `:=`, `::=`, `,`, `..`, `...`, `<..`, `..<`, `<..<`, `=>>`, `=>`, `->`, `<-`, `->>`, `<->`, `@@`

Special Operators (Reserved for future)

`λ`, `(o)`, `â`, `ĩ`, `∇`, `∃`, `o`, `ö`, `U`, `Š`, `Š`, `ṁ`, `Π`, `⇒`, `f`, `ℱ`, `∇`, `ℱ`, `↓`, `λ`, `ð`, `⊥`, `↓`, `↓`

Reserved words

`co`, `let`, `this`, `self` (contextual keyword), `for`, `forall`, `fo` (reserved word)

Difference between `this` and `self`

- `this` is for instances and objects
- `self` is for classes
- `static` — no shortcut; can be on variable or classname
- Both `self` and `this` can access member variables

Builtin Methods

method	Responsibilit
<code>to_str</code>	
<code>to_int</code>	
<code>to_float</code>	
<code>to_double</code>	
<code>classof</code>	
<code>typeof</code>	
<code>new</code>	
<code>prototype</code>	
<code>proto</code>	
<code>make</code>	
<code>objectof</code>	
<code>instanceof</code>	
<code>is</code>	
<code>as</code>	
<code>iskindof</code>	
<code>has</code>	
<code>hasown</code>	
<code>uses</code>	

match	
matchall	
matchany	
matchnone	
matchtype	
case	
with	
print	
println	
printsp	
echo	
contains	
cast	
to	
dummy	
clone	
of	
for	
when	
where	
then	
callback	
getAttr	
inject	
isinstance	
cast_to	
cast_from	
do	
map	
flatMap	
orElse	
filter	
fold	
recover	
peek	
loop	
istrue	
isfalse	
if	
elif	
else	

return	
otherwise	
each	
containsVal	
in	
decltype	deduce the type at compile time
replace	
send	
receive	
submitToPool	
submitToEventLoop	

Builtin Packages

`co` — root (reserved word)

The only package provided by default.

Sub-package	Responsibility
<code>co.lang</code>	All data types and kinds
<code>co.sys</code>	file, concurrent, parallel, goto, invoke, bind, call, apply, setTimeout, setInterval, scheduler, cron, event
<code>co.os</code>	signal, cmd, execute, run, env, getenv, setenv, sleep, exit, cwd, chdir, fork, wait, pipe, dup, dup2, close, readfd, writefd, random
<code>co.meta</code>	ast, instrument, transform, augment, reflect, introspect, patch, inject, create, runtime(eval,proto,prototype,etc), realm
<code>co.core</code>	list, set, map, tree, trie, sort, search, array, pointer, ref, address, ptr, matrix, word
<code>co.native</code>	load, register, asm, inline, emit, ffi, spawnnon[gpu,cpu,npu,apu,fpga,asic,tpu,mki,mcu],arch[x86,x86-64,risc,arm,vliw]
<code>co.in</code>	read, readln
<code>co.out</code>	println, print
<code>co.regex</code>	stex, pattern, match, search
<code>co.crypto</code>	rsa, aes, hash, md5, rand, uuid, ssl, tls
<code>co.dap</code>	built-in decorators, annotations
<code>co.ddap</code>	built-in directives
<code>co.pdap</code>	built-in pragmas
<code>co.net</code>	tcp, udp, http
<code>co.const</code>	<code>true</code> , <code>false</code> , <code>none</code>
<code>co.encoding</code>	base64Encode, base64Decode, json, yaml, bson
<code>co.utils</code>	makeImmutable, makeShared, copyOnWrite, toSnapshot — object behaviour policies
<code>co.dynamic</code>	dynamic capabilities
<code>co.runtime</code>	
<code>co.compiletime</code>	

<code>co.macro</code>	
<code>co.pattern</code>	
<code>co.control</code>	
<code>co.cpa</code>	concurrent, async, await, defer, lazy, parallel, process, thread, fiber, task, coroutine, continuation, cps, pool, channel ...
<code>co.hokrt1</code>	
<code>co.hokrt</code>	

FoLang Philosophy — Uniform Object Model

Core Principle

Everything in **FoLang** is an object. That is the reason for the name **FO — Functional Objects**.

FoLang gives the programmer one uniform object model across all types instead of forcing them to switch between separate type families for:

- ordinary values
- immutable values
- shared/concurrent values
- copy-on-write values
- literal values

The programmer writes against a single conceptual model and opts into the required object behaviour only when needed.

Uniform Object Principle

In FoLang, **everything is an object**.

Unless explicitly stated otherwise, the reference and mutation rules in this section apply to **managed FoLang objects**.

`co.lang.cstruct` remains an object in the language model, but it is an explicitly value-semantic ABI representation: assignment and parameter passing copy its value rather than a managed object reference.

This includes:

- scalar objects
- collection objects
- user-defined objects
- ADT objects
- function objects

Functions are objects in the same sense as all other values.

This is true for both **named functions** and **anonymous functions**.

FoLang does not introduce a separate semantic model for functions, scalars, UDTs, or ADTs.

They all follow the same core object principles:

- assignment of managed objects copies references
- assignment of `co.lang.cstruct` copies its value
- `==` compares values
- mutation is applied through the object
- behaviour policies such as immutability, shared, copy-on-write, and literal conversion apply uniformly where meaningful

So in FoLang, the programmer does not need one mental model for data objects and another for function values.

The language treats them under one consistent object model.

1. Default Object Model

Every value in FoLang is an object and is **mutable by default**.

This includes:

```
co.lang.int, co.lang.float, co.lang.string → built-in scalar types
user-defined types (structs, classes, ADTs) → user-defined objects
co.core.list, co.core.map, co.core.array → built-in collections
```

Example:

```
positive_int co.lang.int = 10;
```

`positive_int` denotes an object.

Its current value is `10`.

By default, it is mutable.

2. Assignment, Aliasing, and Mutation

FoLang distinguishes three related ideas:

- **assignment**
- **aliasing**
- **mutation**

2.1 Assignment copies references

When one object variable is assigned to another, FoLang copies the **reference**, not the internal contents.

```
b := a
```

After this, `a` and `b` refer to the same object.

2.2 Mutation changes object contents

If two names refer to the same object, mutating through one name is visible through the other.

```
Employee co.lang.class = {
  Name co.lang.string
}

a Employee = { Name = "Kamesh" };
b := a;
c Employee = { Name = "Kamesh" };

a == b; // true
a == c; // true
b == c; // true

b.Name = "Ramesh"; // changes a.Name too
c.Name = "Ramesh"; // does not change a.Name
```

Why:

- `a` and `b` are the **same object**
- `c` is a **different object** with the same earlier value

2.3 `==` means value equality

In FoLang, `==` compares **values**, not object identity.

That rule is uniform across all object kinds, including built-in types and user-defined types.

So:

```
a == b
```

means:

- compare contents deeply
- return `true` when the values match
- even if `a` and `b` are not the same object

2.4 Mutation accessors

The accessor used for mutation depends on the object type:

```
scalar (int, float, bool, string, char, byte) → .value
struct field                                → .fieldname
class member                               → .membername
map entry                                  → .key
array / list element                       → .index
```

Examples:

```
count.value = 30
emp.name = "Rao"
marks.math = 95
nums.0 = 42 or nums[0] = 42 //both forms supported
```

2.5 Function-call behaviour

Function parameters introduce a **new local binding**.

So:

- rebinding a parameter does **not** affect the caller
- mutating the object through the parameter **does** affect the caller if both still refer to the same object

```
checkPositive(a co.lang.int)->(co.lang.bool) = {
  a = 20;           // local rebinding only – caller unchanged
  a.value = 30;     // object mutation – caller sees 30
}
```

If `positive_int` is passed to `checkPositive(a)`:

```
a = 20           → local rebinding only
a.value = 30     → mutates the passed object
```

3. Literal Objects

All literals in FoLang create **Literal Objects**.

```
"Kumar" → Literal Object – string
42       → Literal Object – int
true     → Literal Object – bool
```

A Literal Object is an **anonymous object created from a literal expression**.

Literal Objects participate in value equality just like every other object in FoLang:

```
10 == 10 // true
```

In FoLang, `==` compares **values**, not object identity.

So two literal expressions with the same value compare equal, but that does not mean they are the same object.

Binding a Literal Object

When a literal is assigned to a named object, the name refers to the object created from that literal expression.

```
a co.lang.int = 10;
```

Here, `a` refers to the anonymous integer object created from literal `10`.

So:

```
a.value = 30;
```

mutates that same object, and `a` now has value `30`.

The important points are:

- literal expressions create anonymous objects
- literal-created objects are still ordinary objects unless another policy is applied
- `==` compares values, not identity
- once bound to a name, a literal-created object behaves like any other normal mutable object
- immutability applies only when explicitly requested, for example through `makeImmutable(...)`

Why Literal Objects Cannot Be Mutated Directly

Mutation can occur in two ways:

1. Through rebinding

```
a co.lang.int = 10; a = 20;
```

The above statement is valid because `a` is valid identifier and named handle to `co.lang.int` type.

```
10 = 20; // ❌ invalid
```

The above statement is invalid because `10` is not a valid identifier and if `10` in literal object type there is no named handle.

The rebinding happens through named handles which are valid identifiers of `foLang`

2. Through a property or method

An object may also be mutated through one of its mutable properties or methods. `a co.lang.int = 10; a.value = 20;`

A literal object cannot be mutated directly because it does not provide properties/methods that can be accessed.

```
10.value = 20; // ❌ invalid
```

```
a co.lang.int = 10;  
a.value = 30;
```

is valid after binding, a bare literal such as `10` cannot be mutated directly because there is no name or handle through which to perform mutation.

Not the Same as `toSnapshot(...)`

A literal object created by a literal expression is **not** the same thing as the internal snapshot representation produced by `co.utils.toSnapshot(...)`.

- literal expression → anonymous object
- `toSnapshot(x)` → internal snapshot representation used to reconstruct a fresh local object later

These are related concepts, but they are not the same mechanism.

Summary

- literal objects are real objects
- literal expressions create anonymous objects
- identical literals compare equal by value
- identical literals are not automatically the same object
- literal-created objects are mutable by default once bound to a handle
- only `makeImmutable(...)` makes an object immutable

4. Object Behaviour Policies

Any object can be given a behaviour policy using `co.utils.*`.

All four policy calls are **in-place transformations**:

- the object itself changes behaviour kind
- there is no wrapper object
- there is no alternate binding to capture
- the original name now refers to the transformed object

All policies are **deep by default**.

They flow through nested structs, members, collection elements, and all reachable objects in the graph unless the specification later states otherwise.

4.1 Immutable

```
co.utils.makeImmutable(positive_int)
```

After this call the object itself is immutable.

This is an in-place transformation, not a wrapper.

Any attempt to mutate the object or any reachable object beneath it is a **compiler error** where detectable statically, and a **runtime error** otherwise.

```
positvie_int co.lang.int = 10;
positive_int.value = 30      // ✖ compiler error
positive_int = 30           // ✖ compiler error
```

For nested objects:

```
Employee co.lang.struct = { address Address; }
Address co.lang.Address = {city co.lang.string; state co.lang.string; lane co.lang.string;pin co.lang.string;}

emp Employee = Employee{
    address = Address{
        city: "Pune";
    }
}
co.utils.makeImmutable(emp);

emp = Employee{
    address=Address{
        city: "ABC";
    }
} // ✖ compiler error

emp.address = newAddress          // ✖ compiler error
emp.address.city.value = "Mumbai" // ✖ compiler error
```

Immutability is deep and total.

4.2 Value Immutable

```
positive_int co.lang.int = 20;

co.utils.makeValueImmutable(positive_int);

positive_int.value = 30; // ❌ current object is immutable
positive_int = 30;      // ✅ binding may reference a new object

co.utils.makeValueImmutable(emp);

emp.address.city = "ABC"; // ❌
emp.address.city.value = "ABC"; // ❌

emp = Employee{
  address= Address{
    city: "ABC";
  }
}; // ✅
```

Difference between Immutable and Immutable Value

```
makeValueImmutable(x)
└─ current object graph cannot change

makeImmutable(x)
└─ current object graph cannot change
   └─ binding cannot be reassigned
```

Table

Operation	Binding	Current value/object graph
<code>makeValueImmutable(x)</code>	Mutable	Immutable
<code>makeImmutable(x)</code>	Immutable	Immutable

4.3 Shared

```
co.utils.makeShared(positive_int)
```

A Shared Object is safe for concurrent and multi-threaded use.

The programmer continues to use the same object model — same accessors, same syntax — while FoLang guarantees the required safety properties internally.

Shared behaviour is deep:

- all reachable objects within the shared object are also shared

Note on Analogies

Comparisons to things like Java's `AtomicInteger` or `ConcurrentHashMap` are **analogies only**.

They may help explain the concept, but they are **not part of the formal language contract**.

FoLang specifies the behavioural guarantee, not the exact runtime implementation.

A good explanatory statement is:

A shared `co.lang.int` may be thought of similarly to an atomic integer, and a shared map may be thought of similarly to a concurrent map. These comparisons are explanatory only. FoLang does not require any particular internal runtime representation.

4.4 CopyOnWrite

```
co.utils.copyOnWrite(positive_int)
```

A CopyOnWrite Object passes by reference like a normal object.

The copy is deferred.

It is created only when mutation is attempted in a context that must not affect the original source object.

```
process(a co.lang.int)->() = {
  a.value = 99;  // deep copy is made here – caller's object unchanged
}

co.utils.copyOnWrite(positive_int)
process(positive_int)

// positive_int still holds its old value
```

Copy-on-write is deep.

When a copy is made, the entire reachable object graph is copied.

Cyclic references must be handled by the runtime/compiler's structural clone logic with proper identity tracking.

4.5 toSnapshot

```
co.utils.toSnapshot(positive_int)
```

`toSnapshot` converts an object into a **snapshot representation** — a value descriptor, not a live object. The snapshot representation itself cannot be mutated.

When passed to a function, the compiler/runtime uses the snapshot representation to construct a fresh independent local variable bound to the parameter name. That local is a normal mutable object with no shared identity with the original. All of this happens automatically — the developer writes only `co.utils.toSnapshot(positive_int)`.

```
process(a co.lang.int)->() = {
  a.value = 99;  // mutates the fresh local – positive_int completely unaffected
}

process(co.utils.toSnapshot(positive_int))

// positive_int unchanged
```

The flow:

```
positive_int
  ↓ co.utils.toSnapshot(positive_int)
  snapshot representation ← value descriptor – immutable by nature, not by policy
  ↓ passed to function
  compiler constructs fresh local 'a' from the snapshot representation
  'a' is a new independent normal object – mutations never reach positive_int
```

`toSnapshot` is deep — the snapshot representation covers the entire reachable object graph.

5. Policy Summary

Object Kind	Mutation allowed	Caller sees mutation	Thread safe	Copy on write	Deep
Literal expression object	✗ directly in source — no handle	—	value-like use	—	✓
Normal (default)	✓ via accessor	✓	✗	✗	—
Immutable	✗	—	✓	—	✓

Shared	✓ via accessor	✓	✓	✗	✓
CopyOnWrite	✓ on own copy	✗	✓	✓	✓
toSnapshot result	✓ on reconstructed local object	✗	independent snapshot	—	✓

6. No Type Fragmentation

FoLang deliberately avoids special types for mutability or concurrency concerns.

There is no need for separate public type families such as:

```
ImmutableInt
SharedMap
CopyOnWriteList
AtomicInt
ConcurrentMap
```

Instead, FoLang keeps the type model uniform and applies behaviour policies through `co.utils.*`.

In many ecosystems, a programmer must constantly choose between:

```
normal integer    vs atomic integer
normal map        vs concurrent map
mutable value     vs immutable wrapper
raw structure     vs copy-on-write structure
```

FoLang instead aims to provide:

- one object model
- one programming style
- one mental model

while still allowing the programmer to opt into immutability, sharing, copy-on-write, or literal conversion when needed.

7. What Still Needs Precision

The philosophy is sound, but the formal specification still needs to define these precisely:

```
aliasing behaviour      → when two names refer to the same object
rebinding vs mutation interaction → edge-case coverage
deep policy propagation rules → exactly which objects are reachable
cyclic reference handling in COW → identity-tracking specification
visibility of mutation across calls → formal function-call model
identity vs value equality → what operators or built-ins expose identity
policy stacking         → can an object be both shared and COW?
                        → can an object be both shared and immutable?
```

8. Formal Philosophy Statement

All managed FoLang objects use reference semantics by default. `co.lang.cstruct` is an explicitly value-semantic ABI representation and is an exception to managed-object reference semantics.

In FoLang, everything is an object and managed objects are mutable by default.

Assignment of managed objects copies references, `co.lang.cstruct` assignment copies values, and `==` compares values deeply.

Developers may opt into immutability, shared behaviour, copy-on-write behaviour, or literal conversion depending on their needs.

All behaviour policies are deep — they apply to the entire reachable object graph unless stated otherwise by the formal specification.

This policy model is uniform across managed types, so programmers do not need separate type families for ordinary, immutable, concurrent, or snapshot-oriented use.

Familiar analogies such as atomic integers or concurrent maps may help explain the design, but they are not part of the formal implementation contract.

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- type-system definitions and constraints;
- execution-model and control-abstraction descriptions;
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- explanatory and instructional material.

This includes the documented expression and presentation of FoLang-specific constructs such as:

```
(booleanExpression).do({  
  ...  
}).otherwise.do({  
  ...  
});
```

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Example:

Appendix A - Complete FoLang EBNF Grammar

The following grammar is the normative lexical and syntactic grammar for FoLang.

(*

FoLang consolidated EBNF – decision-complete draft, revision 5
Derived from language-ref.md.

Revision 5 closes twelve constructs that revision 4 could not parse, removes ten unreferenced productions, resolves three ambiguous ones, and records the reference divergences that revision 4 left unlabelled. Every new choice is labelled DECISION-* below.

Notation:

= defines a production
; ends a production
| alternative
[...] optional
{ ... } zero or more
(...) grouping
"..." terminal text
? ... ? a precisely described lexical or context-sensitive terminal

STATUS

The language reference remains authoritative. Where that document did not select a lexical or parsing rule, this revision makes an explicit design decision based on FoLang requirements and C++ backend compatibility. Every such decision is labelled DECISION-* in this grammar and in the companion decision register so it can be copied into language-ref.md.

PRINCIPAL DECISIONS

DECISION-SYN-001: A semicolon is mandatory after every simple statement and every standalone built-in directive. Newlines never terminate statements and there is no semicolon insertion. A block-bodied declaration is not followed by a semicolon.

DECISION-OP-001: Built-in operators use the precedence table encoded in section 11. Assignment has the lowest built-in precedence.

DECISION-OP-002: Runtime assignment operators are right-associative. Therefore `a = b = c` parses as `a = (b = c)`. An assignment expression yields the assigned value. FoLang's separately specified target-first, left-to-right evaluation order is retained.

DECISION-OP-003: `:=` and `?` are statement-level definition operators, not general expression operators; they cannot be chained. `::=` remains reserved and is not accepted by this grammar.

DECISION-LEX-001: Source files are UTF-8. FoLang identifiers are the ASCII subset `[A-Za-z][A-Za-z0-9_]*`, but may not contain consecutive underscores or end in an underscore. A lone `_` is a dedicated contextual token, never an identifier.

DECISION-LEX-002: `//` and non-nesting `/* ... */` comments are supported. Line breaks are ordinary whitespace outside literals.

DECISION-LEX-003: The lexer applies maximal munch. Reserved multi-character operators are chosen before their shorter prefixes.

DECISION-BACKEND-001: Each resolved user-defined FoLang identifier is lowered to C++ by appending the suffix `_fo`. Built-in names, keywords, and compiler-internal symbols use their own compiler-defined lowering rules.

DECISION-LIT-000: FoLang accepts C++-compatible built-in literal spellings for the configured C++ backend dialect, preserves the complete source lexeme, and emits it unchanged in C++ IR. The C++ pointer literal `nullptr` is not introduced.

DECISION-LIT-004: FoLang user-defined literals use FoLang syntax and semantics. They do not use C++ literal-operator lookup. The backend lowers them to an equivalent C++ literal, constructor, aggregate initializer, factory, generated helper, or ADT/variant construction.

DECISION-LIT-001: Integer literals use C++ binary, leading-zero octal, decimal, and hexadecimal forms, apostrophe digit separators, and standard integer suffixes.

DECISION-LIT-002: Floating literals use C++ decimal and hexadecimal forms, exponent rules, apostrophe separators, and standard or backend-supported extended floating suffixes.

DECISION-LIT-003: Character and string literals use C++ encoding prefixes, escapes, universal character names, and raw strings. Adjacent string literals form one string-literal sequence.

DECISION-COL-001: Commas separate enum variants, map entries, annotation-map entries, and object initializers; a trailing comma is allowed. Object and annotation-map fields use colon.

DECISION-BLK-001: A block may end in one unterminated tail expression. That expression is the block value and is not a statement.

DECISION-EXT-001: User-defined operators are parsed by the compiler's registered precedence table. Overloads of built-in symbols retain built-in precedence. New symbols require declared fixity, numeric precedence, and associativity.

REVISION 5 DECISIONS

DECISION-ANN-001: An annotation argument and an annotation-map entry both accept "=" or ":" as the key/value binder, and a bare key with no value is a flag. This admits the reference forms `@co.dap.oops(A: { inherit:true })` and `@co.dap.generic(type={T:{typename}}, R:{variance:invariant, bound=Number})`, which mix both binders freely.

DECISION-TYP-001: Every type derivation, not only a pointer derivation, may carry a trailing attribute list. This admits the reference form `co.lang.int->(&, meta={type=out})`.

DECISION-TYP-002: A type-constructor body binds a type-expression, so `Vector(n co.lang.int->(co.lang.dependentType) = co.lang.int->([n]);` is well formed. Where a token sequence satisfies both type-expression and expression, the type-expression reading is selected.

DECISION-TYP-003: An array dimension may be elided in any position, so `->([,])` and `->([])` are both well formed.

DECISION-GEN-001: A generic parameter may declare its arity, so a higher-kinded parameter is written `Transformer(F(_), G(_))`. An arity slot is "_" or a named placeholder.

DECISION-KIND-001: Any built-in kind that has no dedicated production is parsed by general-kind-declaration, which accepts a block body, a type-expression body, or a forward form. This stops a declaration such as `blockormacro co.lang.kind = block | macro;` from silently parsing as a variable. Kind names that also appear in the built-in data-type table (`co.lang.value`, `co.lang.nothing`, `co.lang.just`) are deliberately excluded; in a declarator they read as types.

DECISION-SYN-003: A statement-level local function declaration requires both a return-type clause and a block body. This admits the reference inner-function form `someother()->()={...}` while keeping `foo()`; an expression statement rather than a forward declaration.

DECISION-SYN-004: Annotations may prefix an expression statement, admitting `@co.dap.lazy applied to x = add(1, 2);`.

DECISION-SYN-005: A block is a statement in its own right and takes no trailing semicolon.

DECISION-FUN-001: The "=" before a function block body is optional, so both `add(a T)->(T) = { ... }` and `add(a T)->(T) { ... }` parse.

DECISION-FUN-002: Curried and arrow closure declarations are recognized. The curried form requires two or more parameter lists, so an ordinary assignment to a call result stays unambiguous.

DECISION-OP-005: `::=`, `->>`, `<->`, the backtick, and the backslash are reserved operator tokens. The lexer recognizes them and the parser rejects them, so they cannot be silently reused by a user-defined operator before the language assigns them meaning. The reserved-for-future glyph set is reserved the same way outside literals and declared operator symbols.

DECISION-LIT-005: FoLang boolean literals are `co.const.true` and `co.const.false` and the null literal is `co.const.none`. `true` and `false` are NOT reserved words and NOT literals. Inside an annotation argument, a bare `true`, `false`, or `True` is an ordinary annotation-value name.

DECISION-LEX-004: An identifier may begin with a single underscore when at least one alphanumeric character follows, so `_x` and `_1` are

identifiers. Consecutive underscores and a trailing underscore remain errors, and a lone `_` is still the dedicated contextual token.

DECISION-BACKEND-002: When a FoLang identifier begins with an underscore, the C++ lowering also prepends `fo`, giving `fo_x_fo`, so the emitted name never matches a C++ reserved identifier.

CONFORMANCE NOTE ON DECISION-SYN-001

Several examples in `language-ref.md` omit a required semicolon and must be corrected in that document: generic class and struct field lists, ordinary `let` value bindings, function-pattern clauses, and the module binding `mm EmployeeModule = EmployeeModImpl`. This grammar does not relax the rule.

Context-sensitive rules such as source-file kind, declaration legality, type checking, visibility, operator ownership, capture, and definite initialization remain semantic constraints and are documented separately.

REVISION 5 CHANGE LOG

Constructs that revision 4 could not parse, now accepted:

1	inner/nested functions in a block	local-function-declaration
2	type-constructor bodies	type-constructor-binding
3	@co.dap.oops(A: {...})	annotation-binder
4	bare flag keys such as {typename}	annotation-map-entry
5	"=" inside an annotation map	annotation-binder
6	co.lang.int->(&, meta={type=out})	attribute tails on every derivation form
7	co.lang.int->([,])	array-dimension-content
8	Transformer(F(_), G(_))	generic-parameter, generic-arity-clause
9	@co.dap.lazy over x = add(1, 2);	expression-statement
10	add(a T)->(T) { ... } without "="	function-definition
11	closure(f int) => (x int) = x * f;	arrow-closure-declaration
	curry(f int)(v int) = f * v;	curried-closure-declaration
12	a bare block used as a statement	statement

Silent misparse removed:

~28 built-in kinds had no declaration production and were absorbed by `variable-declaration`. They are now parsed by `general-kind-declaration`.

Token inventory completed:

`::=` `->>` `<->` backtick backslash are reserved-operator; the reserved glyph set is reserved-future-operator.

Reference divergences corrected or labelled:

booleans are `co.const.true/false`, `co.const.none` added, `true` and `false` removed from `hard-reserved-word`; a leading underscore is admitted in identifiers with a matching C++ lowering rule; the semicolon rule is retained with an explicit list of reference examples to amend.

Hygiene:

removed 10 unreferenced productions (`co-path`, `contextual-keyword`, `declaration-prefix`, `external-variable-declaration`, `hard-reserved-word`, `index-expression`, `member-access-expression`, `octal-digit-sequence`, `result-binding`, `white-space`) by wiring in the ones that carried meaning and deleting the redundant ones; removed the ambiguous `word-type-specification`; narrowed `array-dimension` and `type-or-value-argument` to drop subsumed alternatives.

Verified: 317 productions, no undefined nonterminal, no dead production.

*)

```
(* ===== *)
(* 1. Compilation units                               *)
(* ===== *)
```

```
compilation-unit = package-source-file
                  | application-entry-file
                  | library-surface-file ;
```

```

package-source-file = file-preamble, primary-declaration ;

application-entry-file = file-preamble, { entry-item } ;

library-surface-file = file-preamble, library-declaration ;

file-preamble = { file-directive } ;

file-directive = import-directive
                | alias-directive
                | use-directive
                | dynamic-runtime-directive
                | pragma-directive
                | generic-directive ;

entry-item = file-directive
            | entry-type-declaration
            | bare-function-pattern-clause
            | capturing-function-pattern-clause
            | statement ;

primary-declaration = struct-declaration
                    | cstruct-declaration
                    | enum-declaration
                    | union-declaration
                    | data-declaration
                    | class-declaration
                    | interface-declaration
                    | signature-declaration
                    | module-declaration
                    | unit-declaration
                    | type-declaration
                    | object-declaration
                    | instance-declaration
                    | matcher-instance-declaration
                    | function-object-declaration
                    | delegate-declaration
                    | named-block-declaration
                    | annotated-contract-declaration
                    | annotated-function-primary
                    | type-constructor-primary
                    | forward-type-declaration
                    | general-kind-declaration
                    | package-alias-declaration ;

(* ===== *)
(* 2. Directives, annotations, and metadata *)
(* ===== *)

annotations = { annotation } ;

one-or-more-annotations = annotation, { annotation } ;

annotation = "@", qualified-name,
            [ "(", [ annotation-argument-list ], ")" ] ;

annotation-argument-list = annotation-argument,
                          { ",", annotation-argument }, [ ",", ] ;

(*
  DECISION-ANN-001: "=" and ":" are interchangeable binders.
  The optional group is written as a unit so a recursive-descent or PEG
  parser backtracks it cleanly: for a bare value such as co.lang.int the
  group tries "co" then finds no binder, abandons the group as a whole, and
  the value is matched by annotation-value.
*)
annotation-argument = [ annotation-key, annotation-binder ], annotation-value ;

annotation-binder = "=" | ":" ;

```

```

annotation-key = identifier, { "-", identifier } ;

annotation-value = literal
    | type-expression
    | qualified-name
    | declaration-reference
    | annotation-list
    | annotation-map
    | annotation-arrow-pair ;

annotation-list = "[", [ annotation-value,
    { ",", annotation-value }, [ ",", ] ], "]" ;

annotation-map = "{", [ annotation-map-entry,
    { ",", annotation-map-entry }, [ ",", ] ], "}" ;

(*
  DECISION-COL-001 / DECISION-ANN-001:
  A map entry uses either binder, and a bare key is a flag whose value is the
  boolean true. This admits {T:{typename}} and {variance:invariant,
  bound=Number} and {type=out}.
*)
annotation-map-entry = annotation-key, annotation-binder, annotation-value
    | annotation-key ;

annotation-arrow-pair = string-literal, "=>", string-literal ;

(* DECISION-SYN-001: standalone directives are terminated by semicolons. *)
import-directive = "@co.ddap.import", "(", import-field,
    { ",", import-field }, [ ",", ], ")", statement-end ;

import-field = ( "package" | "library" | "src-library" | "expect"
    | "as" | "realm" | "parent-realm" ), "=", annotation-value ;

(* The alias target must be a co.* path, so co-path is used directly. *)
alias-directive = "@co.ddap.alias", "(", co-path, ",",
    "as", "=", string-literal, [ ",", ], ")",
    statement-end ;

use-directive = "@co.ddap.use", "(", annotation-argument-list, ")",
    statement-end ;

dynamic-runtime-directive = "@co.ddap.dynamicruntime",
    [ "(", [ annotation-argument-list ], ")" ],
    statement-end ;

pragma-directive = ( "@co.pdap.compiler" | "@co.pdap.scale" ),
    [ "(", [ annotation-argument-list ], ")" ],
    statement-end ;

generic-directive = "@co.ddap.", identifier,
    [ "(", [ annotation-argument-list ], ")" ],
    statement-end ;

(* ===== *)
(* 3. Names and references *)
(* ===== *)

declaration-name = identifier | "_" ;

qualified-name = ( identifier | "co" ), { ".", identifier } ;

co-path = "co", ".", identifier, { ".", identifier } ;

declaration-reference = qualified-function-reference | qualified-name ;

qualified-function-reference = qualified-name, "(", [ type-list ], ")",
    return-type-clause ;

```

```

lifecycle-name = "@@", identifier ;

special-binding = result-binding | self-binding ;

(* $ alone is the self-referential let binding. *)
self-binding = "$" ;

(* $1, $2, ... capture the previous result in a =>> delegation chain. *)
result-binding = "$", digit, { digit } ;

wildcard = "_" ;

(* ===== *)
(* 4. Type syntax *)
(* ===== *)

type-expression = forall-type | union-type-expression ;

forall-type = "forall", "(", type-parameter-list, ")", ".", type-expression ;

type-parameter-list = identifier, { ",", identifier } ;

union-type-expression = arrow-type-expression,
    { "|", arrow-type-expression } ;

arrow-type-expression = type-postfix-expression,
    [ ">", arrow-type-tail ] ;

arrow-type-tail = type-derivation
    | parenthesized-type-list
    | type-expression ;

type-postfix-expression = type-atom, { type-argument-list } ;

type-atom = qualified-name
    | "(", type-expression, ")"
    | "(", type-list, ")" ;

type-argument-list = "(", [ type-or-value-argument,
    { ",", type-or-value-argument } ], ")" ;

(* identifier is already reachable through type-expression. *)
type-or-value-argument = type-expression | constant-expression ;

type-list = type-expression, { ",", type-expression } ;

parenthesized-type-list = "(", [ type-list ], ")" ;

type-derivation = "(", derivation-specification, ")" ;

(*
    DECISION-TYP-001:
    Every derivation form may carry a trailing attribute list. The bare
    derivation-attribute-list alternative covers the repr/sign/region word and
    address forms, so the former word-type-specification production, which was
    textually identical to it, has been removed.
*)
derivation-specification = pointer-specification
    | array-specification
    | reference-specification
    | range-type-specification
    | slice-type-specification
    | thunk-type-specification
    | address-type-specification
    | derivation-attribute-list ;

pointer-specification = pointer-stars,
    [ ",", derivation-attribute-list ] ;

pointer-stars = "**", { "*" } ;

```

```

reference-specification = ( "&" | "&&" | "~" ),
    [ ",", derivation-attribute-list ] ;

address-type-specification = "@", [ ",", derivation-attribute-list ] ;

thunk-type-specification = "^", [ ",", derivation-attribute-list ] ;

slice-type-specification = ":", [ ",", derivation-attribute-list ] ;

range-type-specification = "..", [ ",", derivation-attribute-list ] ;

array-specification = array-dimension-group, { array-dimension-group },
    [ ",", derivation-attribute-list ] ;

array-dimension-group = "[", array-dimension-content, "]" ;

(* DECISION-TYP-003: any dimension may be elided, including the first. *)
array-dimension-content = [ array-dimension ], { ",", [ array-dimension ] } ;

(* integer-literal and identifier are already reachable through expression. *)
array-dimension = "... " | ". " | expression ;

derivation-attribute-list = derivation-attribute,
    { ",", derivation-attribute } ;

derivation-attribute = annotation-key, "=", annotation-value ;

return-type-clause = "->", "(", [ return-item-list ], ")" ;

return-item-list = return-item, { ",", return-item } ;

return-item = [ identifier ], type-expression ;

(* ===== *)
(* 5. Common declaration components *)
(* ===== *)

(*
    DECISION-GEN-001:
    A generic parameter may declare arity, giving higher-kinded parameters such
    as Transformer(F(_), G(_)).
*)
generic-parameter-clause = "(", generic-parameter,
    { ",", generic-parameter }, ")" ;

generic-parameter = identifier, [ generic-arity-clause ] ;

generic-arity-clause = "(", generic-arity-slot,
    { ",", generic-arity-slot }, ")" ;

generic-arity-slot = "_" | identifier ;

kind-options = "->", "(", [ annotation-argument-list ], ")" ;

field-declaration = annotations, identifier, type-expression,
    [ "=", expression ], statement-end ;

embedded-field-declaration = annotations, type-expression, statement-end ;

value-specification = annotations, identifier, type-expression, statement-end ;

(* DECISION-SYN-002: comma-separated variable declarations are one statement. *)
variable-declaration = annotations, typed-variable-declarator,
    { ",", typed-variable-declarator }, statement-end ;

typed-variable-declarator = identifier, type-expression,
    [ "=", expression ] ;

inferred-variable-declaration = annotations, inferred-variable-declarator,

```

```

        { ",", inferred-variable-declarator },
        statement-end ;

inferred-variable-declarator = identifier,
        ( ":@" | "?@" ), expression ;

(* ===== *)
(* 6. Data and type declarations *)
(* ===== *)

struct-declaration = annotations, declaration-name,
        [ generic-parameter-clause ], "co.lang.struct", "=",
        struct-body ;

struct-body = "{", { struct-member }, "}" ;

struct-member = field-declaration | embedded-field-declaration ;

cstruct-declaration = annotations, declaration-name,
        [ generic-parameter-clause ], "co.lang.cstruct", "=",
        cstruct-body ;

cstruct-body = "{", { field-declaration }, "}" ;

enum-declaration = annotations, declaration-name,
        [ generic-parameter-clause ], "co.lang.enum", "=",
        enum-body ;

enum-body = "{", [ enum-variant,
        { enum-separator, enum-variant }, [ enum-separator ] ], "}" ;

(* DECISION-COL-001: enum variants are comma-separated. *)
enum-separator = "," ;

enum-variant = annotations, identifier,
        [ "(", [ type-list ], ")" ],
        [ "=", constant-expression ] ;

union-declaration = annotations, declaration-name,
        [ generic-parameter-clause ], "co.lang.union", "=",
        union-body ;

union-body = "{", { field-declaration }, "}" ;

data-declaration = annotations, declaration-name,
        [ generic-parameter-clause ], "co.lang.data", "=",
        data-variant, { "|", data-variant }, statement-end ;

data-variant = qualified-name,
        [ "(", [ type-list ], ")" ] ;

type-declaration = annotations, declaration-name,
        [ generic-parameter-clause ], type-declaration-kind,
        [ "=", type-expression ], statement-end ;

type-declaration-kind = "co.lang.type"
        | "co.lang.typealias"
        | "co.lang.newtype"
        | "co.lang.opaque"
        | "co.lang.subtype"
        | "co.lang.supertype"
        | "co.lang.associatedtype"
        | "co.lang.refinementType" ;

entry-type-declaration = type-declaration ;

forward-type-declaration = annotations, declaration-name,
        [ generic-parameter-clause ],
        forward-declarable-kind, [ kind-options ],
        statement-end ;

```

```

forward-declarable-kind = "co.lang.struct"
                        | "co.lang.cstruct"
                        | "co.lang.class"
                        | "co.lang.interface"
                        | "co.lang.signature"
                        | "co.lang.module"
                        | "co.lang.enum"
                        | "co.lang.union"
                        | "co.lang.data"
                        | "co.lang.object"
                        | "co.lang.instance"
                        | "co.lang.function" ;

package-alias-declaration = declaration-name, "co.lang.package", statement-end ;

(*)
  DECISION-KIND-001:
  A built-in kind with no dedicated production is parsed here rather than
  falling through to variable-declaration. Ordered choice matters: a specific
  declaration in section 6 or 7 is tried first, and in statement position
  variable-declaration is preferred, so an ordinary declarator is unaffected.
*)
general-kind-declaration = annotations, declaration-name,
                        [ generic-parameter-clause ],
                        general-declarable-kind, [ kind-options ],
                        general-kind-binding ;

general-kind-binding = "=", general-kind-block
                    | "=", type-expression, statement-end
                    | "=", expression, statement-end
                    | statement-end ;

general-kind-block = "{", { general-kind-member }, "}" ;

general-kind-member = field-declaration
                    | embedded-field-declaration
                    | signature-type-component
                    | function-declaration
                    | function-specification ;

general-declarable-kind = "co.lang.realm"
                        | "co.lang.loader"
                        | "co.lang.role"
                        | "co.lang.record"
                        | "co.lang.property"
                        | "co.lang.indexer"
                        | "co.lang.trait"
                        | "co.lang.mixin"
                        | "co.lang.extension"
                        | "co.lang.typeclass"
                        | "co.lang.concept"
                        | "co.lang.macro"
                        | "co.lang.template"
                        | "co.lang.lambda"
                        | "co.lang.behavior"
                        | "co.lang.method"
                        | "co.lang.operator"
                        | "co.lang.namespace"
                        | "co.lang.stex"
                        | "co.lang.kind"
                        | "co.lang.level"
                        | "co.lang.order"
                        | "co.lang.rank"
                        | "co.lang.dependentType"
                        | "co.lang.hokrIt"
                        | "co.lang.typeType"
                        | "co.lang.typeKind"
                        | "co.lang.alias" ;

```

```

(* ===== *)
(* 7. Containers and behavioral declarations *)
(* ===== *)

unit-declaration = annotations, declaration-name, "co.lang.unit", "=",
                    unit-body ;

unit-body = "{", { function-declaration }, "}" ;

class-declaration = annotations, declaration-name,
                    [ generic-parameter-clause ], "co.lang.class",
                    [ kind-options ], "=", class-body ;

class-body = "{", { class-member }, "}" ;

class-member = field-declaration
              | function-declaration
              | lifecycle-method-declaration ;

lifecycle-method-declaration = annotations, lifecycle-name,
                               parameter-list, [ return-type-clause ],
                               function-definition ;

interface-declaration = annotations, declaration-name,
                        [ generic-parameter-clause ], "co.lang.interface", "=",
                        interface-body ;

interface-body = "{", { function-specification }, "}" ;

signature-declaration = annotations, declaration-name,
                        [ generic-parameter-clause ], "co.lang.signature", "=",
                        signature-body ;

signature-body = "{", { signature-member }, "}" ;

signature-member = value-specification
                  | function-specification
                  | signature-type-component ;

signature-type-component = annotations, declaration-name,
                           [ generic-parameter-clause ], "co.lang.type",
                           [ "=", type-expression ], statement-end ;

module-declaration = annotations, declaration-name, "co.lang.module",
                     [ kind-options ], "=", module-body ;

module-body = "{", { module-member }, "}" ;

module-member = variable-declaration
               | inferred-variable-declaration
               | function-declaration
               | signature-type-component ;

library-declaration = annotations, declaration-name, "co.lang.library", "=",
                      library-body ;

library-body = "{", { library-member }, "}" ;

library-member = import-directive
                | struct-declaration
                | cstruct-declaration
                | function-declaration ;

object-declaration = annotations, declaration-name, "co.lang.object",
                     [ kind-options ], "=", object-body ;

object-body = "{", { field-declaration | function-declaration }, "}" ;

instance-declaration = annotations, declaration-name, "co.lang.instance",
                       [ kind-options ], "=", instance-body ;

```



```

instance-body = "{", { function-declaration | variable-declaration }, "}" ;

matcher-instance-declaration = annotations, declaration-name,
                                ( "co.lang.Matcher" | "co.lang.matcher" ),
                                [ kind-options ], "=", instance-body ;

annotated-contract-declaration = one-or-more-annotations, declaration-name,
                                [ generic-parameter-clause ], "=",
                                contract-body ;

contract-body = "{", { function-specification | value-specification }, "}" ;

named-block-declaration = annotations, declaration-name, "co.lang.block", "=",
                           block ;

delegate-declaration = annotations, declaration-name, "co.lang.delegate", "=",
                       function-type, statement-end ;

function-object-declaration = annotations, declaration-name,
                              "co.lang.function", "=",
                              anonymous-function-expression,
                              statement-end ;

annotated-function-primary = one-or-more-annotations, function-declaration ;

(*
  DECISION-TYP-002:
  A type constructor returns a type, so its body binds a type-expression. The
  type-expression alternative precedes the expression alternative, so
  Vector(n co.lang.int)->(co.lang.dependentType) = co.lang.int->([n]);
  is parsed as a type, not as an unparseable expression.
*)
type-constructor-primary = annotations, function-name, parameter-list,
                           { parameter-list }, return-type-clause,
                           type-constructor-binding ;

type-constructor-binding = function-definition
                          | function-delegation
                          | "=", type-expression, statement-end
                          | "=", expression, statement-end
                          | statement-end ;

(* ===== *)
(* 8. Functions *)
(* ===== *)

function-declaration = annotations, [ receiver-clause ], function-name,
                       parameter-list, { parameter-list },
                       [ return-type-clause ], function-binding ;

function-name = identifier | lifecycle-name ;

receiver-clause = "(", ( type-expression
                        | identifier, type-expression ), ")" ;

parameter-list = "(", [ parameter,
                      { ",", parameter }, [ ",", ] ], ")" ;

parameter = [ "..." ], [ "~" ], identifier, [ "?" ],
            [ type-expression ], [ "=", expression ] ;

function-binding = function-definition
                  | function-delegation
                  | function-alias-binding
                  | statement-end ;

(* DECISION-FUN-001: the "=" before a block body is optional. *)
function-definition = [ "=" ], block ;

```

```

function-delegation = ( ">" | ">>" ), expression,
                      { ">>", expression }, statement-end ;

function-alias-binding = "=", expression, statement-end ;

function-specification = annotations, [ receiver-clause ], function-name,
                          parameter-list, { parameter-list },
                          [ return-type-clause ], statement-end ;

function-type = "(", [ type-list ], ")", return-type-clause ;

anonymous-function-expression = [ "forall", "(", type-parameter-list, ")", "." ],
                                parameter-list, return-type-clause,
                                [ "=" ], block ;

lambda-expression = "|", [ lambda-parameter,
                          { ",", lambda-parameter } ], "|", ">",
                          ( expression | block ) ;

lambda-parameter = identifier, [ type-expression ] ;

(*)
  DECISION-FUN-002:
  Two abbreviated closure declaration forms from the reference. The curried
  form requires two or more parameter lists so that an ordinary assignment to
  a call result is never captured by this production.
*)
closure-declaration = curried-closure-declaration
                    | arrow-closure-declaration ;

curried-closure-declaration = annotations, identifier, parameter-list,
                              parameter-list, { parameter-list },
                              [ return-type-clause ], "=", expression,
                              statement-end ;

arrow-closure-declaration = annotations, identifier, parameter-list,
                           ">", parameter-list, "=", expression,
                           statement-end ;

(*)
  DECISION-SYN-003:
  A function declared inside a block requires a return-type clause and a
  block body, which keeps foo(); an expression statement.
*)
local-function-declaration = annotations, function-name, parameter-list,
                              { parameter-list }, return-type-clause,
                              function-definition ;

(*) ===== *)
(*) 9. Function-pattern groups and patterns (*)
(*) ===== *)

bare-function-pattern-clause = identifier, pattern-parameter-list,
                              [ where-clause ], ">", pattern-result ;

capturing-function-pattern-clause = "let", identifier,
                                   pattern-parameter-list,
                                   [ where-clause ], "=", pattern-result ;

pattern-parameter-list = "(", [ pattern,
                              { ",", pattern } ], ")" ;

where-clause = ".where", "(", expression, ")" ;

pattern-result = expression | block ;

pattern = wildcard
        | literal-pattern
        | binding-pattern
        | constructor-pattern

```

```

| record-pattern
| tuple-pattern
| qualified-name ;

literal-pattern = literal ;

binding-pattern = identifier ;

constructor-pattern = qualified-name, "(", [ pattern,
                                     { ",", pattern } ], ")" ;

record-pattern = qualified-name, "{", [ record-pattern-field,
                                     { ",", record-pattern-field } ], "}" ;

record-pattern-field = identifier, [ ":", pattern ] ;

tuple-pattern = "(", pattern, ",", pattern,
               { ",", pattern }, ")" ;

match-case = ".case", "(", match-case-body, ")" ;

match-case-body = pattern, [ ":", expression ], "=>",
                      ( expression | block ) ;

match-default = ".default", "(", ( expression | block ), ")" ;

(* ===== *)
(* 10. Statements and blocks *)
(* ===== *)

(*
  DECISION-SYN-001:
  - Every simple statement ends with ";".
  - A newline is whitespace and never terminates a statement.
  - A block statement and a block-bodied declaration do not take a trailing
    semicolon merely because their final token is "}".

  DECISION-BLK-001:
  A final expression without a semicolon is a block tail expression, not an
  expression statement. It supplies the block's value.
*)
block = "{", { block-item }, [ block-tail-expression ], "}" ;

block-item = statement ;

block-tail-expression = expression ;

statement = variable-declaration
| inferred-variable-declaration
| grouped-variable-declaration
| let-value-declaration
| local-function-declaration
| closure-declaration
| multiple-assignment-statement
| return-statement
| expression-statement
| labeled-block
| block
| empty-statement ;

grouped-variable-declaration = "(", typed-variable-declarator,
                              { ",", typed-variable-declarator }, ")",
                              statement-end ;

let-value-declaration = "let", identifier, [ type-expression ], "=",
                        expression, statement-end ;

(*
  DECISION-OP-003:
  := and ?= occur only in inferred-variable-declaration. They are not

```

```

assignment-expression operators and cannot participate in a = b = c-style
chain. ::= remains reserved for a future feature and is rejected.
*)

(* Multiple assignment is a statement because it has multiple destinations. *)
multiple-assignment-statement = assignment-target, ",", assignment-target,
                                { ",", assignment-target }, "=",
                                expression-list, statement-end ;

assignment-target = postfix-expression
                  | tuple-assignment-target ;

tuple-assignment-target = "(", assignment-target, ",", assignment-target,
                          { ",", assignment-target }, ")" ;

return-statement = ( "this" | "self" ), ".return",
                   [ expression-list ], statement-end ;

(* DECISION-SYN-004: an expression statement may carry annotations. *)
expression-statement = annotations, expression, statement-end ;

labeled-block = identifier, ":", block ;

empty-statement = ";" ;

expression-list = expression, { ",", expression } ;

statement-end = ";" ;

(* ===== *)
(* 11. Expressions and built-in operator precedence *)
(* ===== *)

(*
  DECISION-OP-001 – built-in precedence, highest to lowest:

100 postfix: calls, indexing, member access, postfix !, ++, --   left
  90  exponentiation: **                                           right
  80  prefix: +, -, !, ~, @, #, ^, ++, --                         right
  70  multiplicative: *, /, %                                       left
  60  additive: +, -                                               left
  55  ranges: .., <.., <..<, <..<                                  none
  50  relational: <, <=, >, >=                                       left
  45  equality: ==, !=                                              left
  40  bitwise AND: &                                                left
  38  bitwise XOR: ^                                               left
  36  bitwise OR: |                                                 left
  30  logical AND: &&                                               left
  20  logical OR: ||                                               left
  10  assignment: =, +=, -=, *=, /=, %=, **=, &=, ^=, |=         right

Operands are still evaluated according to FoLang's normative left-to-right
and target-first evaluation rules. Associativity determines grouping, not
the order in which operand subexpressions are evaluated.

DECISION-EXT-001:
A use of at least one newly defined custom operator is parsed by the
registered operator table. The contextual production extended-operator-
expression denotes that compiler-generated precedence grammar. Overloads of
built-in symbols use the built-in table above and do not alter precedence.
*)
expression = assignment-expression
            | extended-operator-expression ;

(* DECISION-OP-002: right recursion makes assignment right-associative. *)
assignment-expression = logical-or-expression,
                       [ runtime-assignment-operator,
                         assignment-expression ] ;

runtime-assignment-operator = "="

```

```

        | compound-assignment-operator ;

compound-assignment-operator = "+=" | "-=" | "*=" | "/=" | "%="
        | "**=" | "&=" | "^=" | "|=" ;

constant-expression = logical-or-expression ;

logical-or-expression = logical-and-expression,
        { "||", logical-and-expression } ;

logical-and-expression = bitwise-or-expression,
        { "&&", bitwise-or-expression } ;

bitwise-or-expression = bitwise-xor-expression,
        { "|", bitwise-xor-expression } ;

bitwise-xor-expression = bitwise-and-expression,
        { "^", bitwise-and-expression } ;

bitwise-and-expression = equality-expression,
        { "&", equality-expression } ;

equality-expression = relational-expression,
        { equality-operator, relational-expression } ;

equality-operator = "==" | "!=" ;

relational-expression = range-expression,
        { relational-operator, range-expression } ;

relational-operator = "<" | "<=" | ">" | ">=" ;

(* A range expression contains at most one range operator. *)
range-expression = additive-expression,
        [ range-operator, [ additive-expression ] ]
        | range-operator, additive-expression ;

range-operator = ".." | "<.." | "..<" | "<..<>" ;

additive-expression = multiplicative-expression,
        { additive-operator, multiplicative-expression } ;

additive-operator = "+" | "-" ;

multiplicative-expression = unary-expression,
        { multiplicative-operator, unary-expression } ;

multiplicative-operator = "*" | "/" | "%" ;

unary-expression = { prefix-operator }, power-expression ;

(* DECISION-OP-004: ++ and -- support both prefix and postfix forms. *)
prefix-operator = "+" | "-" | "!" | "~" | "@" | "#" | "^"
        | "++" | "--" ;

(* Right recursion makes exponentiation right-associative. *)
power-expression = postfix-expression,
        [ "**", unary-expression ] ;

postfix-expression = primary-expression,
        { postfix-suffix | postfix-operator } ;

postfix-operator = "!" | "++" | "--" ;

postfix-suffix = call-suffix
        | index-suffix
        | member-suffix
        | match-suffix ;

call-suffix = "(", [ argument-list ], ")" ;

```

```

argument-list = argument, { ",", argument }, [ ",", " ] ;

argument = ( [ identifier, "=" ], expression )
           | block
           | lambda-expression ;

index-suffix = "[", [ expression-list ], "]" ;

member-suffix = ".", ( identifier | lifecycle-name ) ;

(*
   Member access and indexing are already expressed by postfix-expression with
   member-suffix and index-suffix; the former duplicate productions were
   removed in revision 5.
*)

primary-expression = literal
                   | special-binding
                   | "this"
                   | "self"
                   | qualified-name
                   | wildcard
                   | grouped-expression
                   | tuple-expression
                   | array-literal
                   | map-literal
                   | object-construction
                   | anonymous-class-expression
                   | block
                   | anonymous-function-expression
                   | lambda-expression
                   | let-expression
                   | comprehension-expression ;

grouped-expression = "(", expression, ")" ;

tuple-expression = "(", expression, ",", expression,
                  { ",", expression }, [ ",", " ], ")" ;

array-literal = "[", [ expression,
                      { ",", expression }, [ ",", " ] ], "]" ;

map-literal = "{", [ map-entry,
                    { ",", map-entry }, [ ",", " ] ], "}" ;

map-entry = expression, ":", expression ;

(* DECISION-COL-001: object fields use colon and comma. *)
object-construction = type-postfix-expression, "{",
                      [ object-field-initializer,
                        { ",", object-field-initializer }, [ ",", " ] ], "}" ;

object-field-initializer = identifier, ":", expression ;

anonymous-class-expression = "co.lang.class", "{",
                             { class-member }, "}" ;

let-expression = "let", "(", "{", let-binding,
                 { ",", let-binding }, "}", ")",
                 ".in", "(", "{", expression, "}", ")" ;

let-binding = ( identifier | special-binding ), "=", expression ;

comprehension-expression = "for", "(", comprehension-binding, ")",
                          ".yield", "(", expression-list, ")" ;

comprehension-binding = pattern, "<-", expression ;

(*

```

At least one case is required semantically. The grammar accepts zero so a malformed chain produces a semantic diagnostic rather than a parse error.

```

*)
match-suffix = ".match", [ "(", [ expression ], ")" ],
               { match-case }, [ match-default ] ;

extended-operator-expression =
  ? expression containing a registered non-built-in operator, parsed by
    precedence climbing from its declared fixity, precedence, associativity,
    and arity; all built-in subexpressions obey the table above ? ;

(* ===== *)
(* 11a. Control-flow chain shapes (informative) *)
(* ===== *)

(*
  FoLang has no if, else, for, while, or foreach keyword. All control flow is
  an associated-function chain and therefore already parses as
  postfix-expression with member-suffix and call-suffix, where argument
  admits a block. The productions below are INFORMATIVE. They document the
  canonical shapes the semantic analyzer enforces; they are not a second
  parse path and must not be entered from expression.

  Enforcing these shapes in the parser instead would require unbounded
  lookahead to distinguish a control chain from any other method chain, so
  the check belongs to semantic analysis.
*)

informative-condition-chain =
  "(", expression, ")", ".do", "(", block, ")",
  { ".otherwise", "(", expression, ")", ".do", "(", block, ")" },
  [ ".otherwise", ".do", "(", block, ")" ] ;

informative-loop-chain =
  "(", expression, ")", ".loop", "(", block, ")",
  { ".otherwise", "(", expression, ")", ".loop", "(", block, ")" },
  [ ".otherwise", ".loop", "(", block, ")" ] ;

informative-mixed-chain =
  "(", expression, ")", informative-branch-verb, "(", block, ")",
  { ".otherwise", "(", expression, ")",
    informative-branch-verb, "(", block, ")" },
  [ ".otherwise", informative-branch-verb, "(", block, ")" ] ;

informative-branch-verb = ".do" | ".loop" ;

informative-ternary-chain =
  "(", expression, ")", ".return", "(", expression, ")",
  { ".otherwise", "(", expression, ")", ".return", "(", expression, ")" },
  ".otherwise", ".return", "(", expression, ")" ;

informative-each-chain =
  postfix-expression, ".each", "(", ( identifier | "_" ), ",", identifier,
  ")", ".do", "(", block, ")" ;

informative-contains-chain =
  postfix-expression, ".contains", "(", expression, ")",
  ".do", "(", block, ")",
  [ ".otherwise", ".do", "(", block, ")" ] ;

informative-pipeline-chain =
  postfix-expression,
  { ( ".filter" | ".map" | ".reduce" | ".forEach"
    | ".sortBy" | ".groupBy" | ".fold" ),
    "(", argument-list, ")" } ;

(* ===== *)
(* 12. Literals and lexical grammar *)
(* ===== *)

```

```

(*)
DECISION-LEX-001:
Source text is UTF-8. A U+FEFF byte-order mark is permitted only as the
first code point and is otherwise an error.

A FoLang identifier begins with an ASCII alphabetic character. Remaining
characters are ASCII alphabetic characters, decimal digits, or isolated
underscores. Consecutive underscores and a trailing underscore are lexical
errors. The spelling "_" is a dedicated contextual token used for discard,
wildcard, or filename-derived declaration names; it is never an identifier.

DECISION-BACKEND-001:
A resolved user-defined FoLang identifier is lowered to C++ by appending
the suffix "_fo". The no-consecutive-underscore and no-trailing-underscore
rules ensure this lowering never creates a C++-reserved double underscore.

DECISION-LEX-002:
Horizontal whitespace, line terminators, line comments, and non-nesting
block comments are discarded between tokens. A line terminator has no
statement-termination meaning.

DECISION-LEX-003:
Lexing uses maximal munch. For example, <..< is selected before <.. and <,
**= before ** and *, and => before => and =. A comment introducer is
recognized before the / operator.

DECISION-LIT-000:
Built-in literal spelling follows the configured C++ backend dialect. The
frontend stores the literal category and complete raw source lexeme. The C++
backend emits that raw lexeme unchanged. The productions below mirror the
current C++ built-in numeric, character, and string literal forms used by
this revision. Backend-conditionally-supported suffixes are accepted only
when the configured C++ compiler supports them.

This decision does not import the C++ nullptr literal or C++
user-defined-literal operator lookup.

DECISION-LIT-004:
FoLang user-defined literals use FoLang-defined syntax and semantics. The
frontend records their target FoLang type, original source spelling, and
structured literal components. The C++ backend may lower them to a native
C++ literal when one is equivalent, or to a constructor, aggregate
initializer, factory, generated helper, or ADT/variant construction.
*)

literal = builtin-literal
        | folang-user-defined-literal ;

builtin-literal = integer-literal
                | floating-literal
                | string-literal-sequence
                | character-literal
                | boolean-literal
                | none-literal ;

(* DECISION-LIT-001: C++-compatible built-in integer literal spelling. *)
integer-literal = ( binary-integer-literal
                  | octal-integer-literal
                  | decimal-integer-literal
                  | hexadecimal-integer-literal ),
                  [ integer-suffix ] ;

binary-integer-literal = ( "0b" | "0B" ), binary-digit-sequence ;

octal-integer-literal = "0", [ octal-digit-sequence ] ;

decimal-integer-literal = nonzero-digit,
                        { [ digit-separator ], decimal-digit } ;

hexadecimal-integer-literal = hexadecimal-prefix,

```



```

        hexadecimal-digit-sequence ;

hexadecimal-prefix = "0x" | "0X" ;

binary-digit-sequence = binary-digit,
                        { [ digit-separator ], binary-digit } ;

octal-digit-sequence = octal-digit,
                       { [ digit-separator ], octal-digit } ;

decimal-digit-sequence = decimal-digit,
                         { [ digit-separator ], decimal-digit } ;

hexadecimal-digit-sequence = hexadecimal-digit,
                             { [ digit-separator ], hexadecimal-digit } ;

digit-separator = "" ;

integer-suffix = unsigned-suffix,
                [ long-suffix | long-long-suffix | size-suffix ]
                | long-suffix, [ unsigned-suffix ]
                | long-long-suffix, [ unsigned-suffix ]
                | size-suffix, [ unsigned-suffix ] ;

unsigned-suffix = "u" | "U" ;

long-suffix = "l" | "L" ;

long-long-suffix = "ll" | "LL" ;

size-suffix = "z" | "Z" ;

(* DECISION-LIT-002: C++-compatible decimal and hexadecimal floating forms. *)
floating-literal = decimal-floating-literal
                  | hexadecimal-floating-literal ;

decimal-floating-literal = fractional-constant,
                           [ exponent-part ],
                           [ floating-point-suffix ]
                           | decimal-digit-sequence,
                           exponent-part,
                           [ floating-point-suffix ] ;

fractional-constant = [ decimal-digit-sequence ], ".",
                      decimal-digit-sequence
                      | decimal-digit-sequence, "." ;

hexadecimal-floating-literal = hexadecimal-prefix,
                              ( hexadecimal-fractional-constant
                                | hexadecimal-digit-sequence ),
                              binary-exponent-part,
                              [ floating-point-suffix ] ;

hexadecimal-fractional-constant = [ hexadecimal-digit-sequence ], ".",
                                   hexadecimal-digit-sequence
                                   | hexadecimal-digit-sequence, "." ;

exponent-part = ( "e" | "E" ), [ sign ], decimal-digit-sequence ;

binary-exponent-part = ( "p" | "P" ), [ sign ], decimal-digit-sequence ;

sign = "+" | "-" ;

floating-point-suffix = "f" | "F" | "l" | "L"
                      | "f16" | "F16"
                      | "f32" | "F32"
                      | "f64" | "F64"
                      | "f128" | "F128"
                      | "bf16" | "BF16" ;

```

```

(* DECISION-LIT-003: C++-compatible character and string literal spelling. *)
character-literal = [ encoding-prefix ], single-quote,
                    c-character-sequence, single-quote ;

c-character-sequence = c-character, { c-character } ;

c-character = basic-c-character
             | escape-sequence
             | universal-character-name ;

basic-c-character = ? any translation character except apostrophe, backslash,
                    carriage return, or line feed ? ;

string-literal-sequence = string-literal, { string-literal } ;

string-literal = plain-string-literal | raw-string-literal ;

plain-string-literal = [ encoding-prefix ], double-quote,
                      { s-character }, double-quote ;

s-character = basic-s-character
             | escape-sequence
             | universal-character-name ;

basic-s-character = ? any translation character except double quote,
                    backslash, carriage return, or line feed ? ;

raw-string-literal = ? one complete C++ raw string literal token, including an
                      optional encoding prefix; its opening and closing
                      delimiters are identical, contain at most 16 valid
                      d-characters, and enclose any permitted r-character
                      sequence ? ;

encoding-prefix = "u8" | "u" | "U" | "L" ;

escape-sequence = simple-escape-sequence
                 | numeric-escape-sequence
                 | conditional-escape-sequence ;

simple-escape-sequence = backslash,
                       ( single-quote | double-quote | "?" | backslash
                         | "a" | "b" | "f" | "n" | "r" | "t" | "v" ) ;

numeric-escape-sequence = octal-escape-sequence
                        | hexadecimal-escape-sequence ;

octal-escape-sequence = backslash, octal-digit,
                       [ octal-digit ], [ octal-digit ]
                       | backslash, "o", "{",
                       simple-octal-digit-sequence, "}" ;

hexadecimal-escape-sequence = backslash, "x",
                              simple-hexadecimal-digit-sequence
                              | backslash, "x", "{",
                              simple-hexadecimal-digit-sequence, "}" ;

conditional-escape-sequence = backslash,
                              ? one basic character that is not an octal digit,
                              simple escape character, or N, o, u, U, or x ? ;

universal-character-name = backslash, "u", hex-quad
                          | backslash, "U", hex-quad, hex-quad
                          | backslash, "u", "{",
                          simple-hexadecimal-digit-sequence, "}"
                          | named-universal-character ;

named-universal-character = backslash, "N", "{",
                           n-character-sequence, "}" ;

n-character-sequence = n-character, { n-character } ;

```

```

n-character = ? any translation character except right brace, carriage return,
               or line feed ? ;

hex-quad = hexadecimal-digit, hexadecimal-digit,
            hexadecimal-digit, hexadecimal-digit ;

simple-octal-digit-sequence = octal-digit, { octal-digit } ;

simple-hexadecimal-digit-sequence = hexadecimal-digit,
                                   { hexadecimal-digit } ;

double-quote = ? Unicode scalar value U+0022 ? ;

single-quote = ? Unicode scalar value U+0027 ? ;

backslash = ? Unicode scalar value U+005C ? ;

(*)
    DECISION-LIT-005:
    FoLang booleans are co.const.true and co.const.false, and the null literal
    is co.const.none. These spellings also match qualified-name; the literal
    reading is selected. A bare true, false, or True inside an annotation
    argument is an ordinary annotation-value name, not a literal.
*)
boolean-literal = "co.const.true" | "co.const.false" ;

none-literal = "co.const.none" ;

(*)
    DECISION-LIT-004:
    This production intentionally remains contextual until FoLang's complete
    user-defined-literal declaration and invocation syntax is finalized.

    A recognized form is resolved against an in-scope FoLang literal definition.
    It is never interpreted through C++ operator"" lookup. The backend may emit
    an equivalent native C++ literal or lower the structured value through a
    generated struct/class/ADT initializer, constructor, factory, or helper.
*)
folang-user-defined-literal =
    ? one FoLang-defined token-like or structured literal form recognized by
    an in-scope user-defined-literal declaration and resolved to a target
    FoLang type ? ;

(*)
    DECISION-LEX-001 as amended by DECISION-LEX-004:
    An identifier begins with an ASCII letter, or with a single underscore
    followed by at least one alphanumeric character. The grammar still prevents
    consecutive underscores and a trailing underscore, and a lone _ is never an
    identifier. Reserved-word rejection remains a token-class check applied
    after the character sequence is recognized, using hard-reserved-word.

    DECISION-BACKEND-002:
    When an identifier begins with an underscore, the C++ lowering prepends fo
    in addition to appending _fo, so _x becomes fo_x_fo and never collides with
    a C++ reserved identifier.
*)
identifier = identifier-head, { "_", identifier-segment } ;

identifier-head = ASCII-letter, { ASCII-alphanumeric }
                | "_", ASCII-alphanumeric, { ASCII-alphanumeric } ;

identifier-segment = ASCII-alphanumeric, { ASCII-alphanumeric } ;

ASCII-alphanumeric = ASCII-letter | decimal-digit ;

ASCII-letter = "A" | "B" | "C" | "D" | "E" | "F" | "G" | "H"
              | "I" | "J" | "K" | "L" | "M" | "N" | "O" | "P"
              | "Q" | "R" | "S" | "T" | "U" | "V" | "W" | "X"

```

```

    | "Y" | "Z"
    | "a" | "b" | "c" | "d" | "e" | "f" | "g" | "h"
    | "i" | "j" | "k" | "l" | "m" | "n" | "o" | "p"
    | "q" | "r" | "s" | "t" | "u" | "v" | "w" | "x"
    | "y" | "z" ;

binary-digit = "0" | "1" ;

octal-digit = "0" | "1" | "2" | "3" | "4" | "5" | "6" | "7" ;

hexadecimal-digit = decimal-digit
    | "a" | "b" | "c" | "d" | "e" | "f"
    | "A" | "B" | "C" | "D" | "E" | "F" ;

digit = decimal-digit ;

decimal-digit = "0" | nonzero-digit ;

nonzero-digit = "1" | "2" | "3" | "4" | "5" | "6" | "7" | "8" | "9" ;

(* DECISION-LIT-005 removed true and false from this set. *)
hard-reserved-word = "co" | "let" | "this" | "for" | "forall" | "fo" ;

contextual-keyword = "self" ;

(*
    DECISION-OP-005:
    These tokens appear in the reference operator table but have no assigned
    meaning. The lexer recognizes each as one token and the parser rejects it,
    so a user-defined operator cannot silently claim it before the language
    assigns it a meaning.
*)
reserved-operator = "::~=" | "->>" | "<->" | "`" | backslash ;

reserved-future-operator = ? one character from the reference reserved glyph
    set, comprising the Greek small letter lambda and
    the other mathematical and modifier glyphs listed
    as reserved for future use; reserved outside
    literals, comments, and declared operator
    symbols ? ;

(*
    Informative token-class summary. The parser consumes tokens; separators are
    discarded between them.
*)
token = identifier
    | keyword-token
    | literal
    | reserved-operator
    | reserved-future-operator ;

keyword-token = hard-reserved-word | contextual-keyword ;

token-separator = white-space ;

line-comment = "//", { ? any Unicode scalar value except CR or LF ? } ;

block-comment = "/*", { block-comment-character }, "*/" ;

block-comment-character = ? any Unicode scalar value that does not begin the
    two-character sequence */ ? ;

line-break = "\r\n" | "\n" | "\r" ;

horizontal-white-space = " " | "\t" | "\f" ;

white-space = horizontal-white-space | line-break | line-comment | block-comment ;

```

Appendix B - Grammar Decisions and Rationale

FoLang EBNF decision register — revision 4

This register lists explicit lexical, parsing, and C++-lowering decisions used by `folang-complete-v4.ebnf`. These are proposed normative rules for incorporation into `language-ref.md`, not hidden parser assumptions.

1. Statement termination

DECISION-SYN-001 — mandatory semicolons

- Every **simple statement** ends with `;`.
- Every standalone built-in directive such as `@co.ddap.import(...)`, `@co.ddap.alias(...)`, and `@co.ddap.use(...)` ends with `;`.
- Newlines are whitespace only. FoLang performs no automatic semicolon insertion.
- A block statement and a block-bodied declaration do not require another semicolon after their closing `}`.
- A complete condition, loop, or iterator chain used as an expression statement ends with `;`.

```
value := calculate();
value = other = source;

condition.do({
    run();
}).otherwise.do({
    recover();
});

Employee co.lang.struct = {
    id co.lang.int;
    name co.lang.string;
}
```

The struct declaration itself has no semicolon after `}`; its field declarations do.

DECISION-BLK-001 — block tail expressions

A final expression without `;` is a block's value-producing tail expression. It is not a statement.

```
classify(n) => {
    n + 1
}
```

Writing `n + 1;` makes it an expression statement and the block has no tail value.

2. Assignment and operators

DECISION-OP-001 — built-in precedence

Precedence	Operators/forms	Associativity
100	call <code>()</code> , index <code>[]</code> , member <code>.</code> , postfix <code>!</code> , <code>++</code> , <code>--</code>	left
90	<code>**</code>	right
80	prefix <code>+</code> , <code>-</code> , <code>!</code> , <code>~</code> , <code>@</code> , <code>#</code> , <code>^</code> , <code>++</code> , <code>--</code>	right
70	<code>*</code> , <code>/</code> , <code>%</code>	left
60	<code>+</code> , <code>-</code>	left
55	<code>..</code> , <code><..</code> , <code>..<</code> , <code><..<</code>	non-associative
50	<code><</code> , <code><=</code> , <code>></code> , <code>>=</code>	left
45	<code>==</code> , <code>!=</code>	left

40	<code>&</code>	left
38	<code>^</code>	left
36	<code> </code>	left
30	<code>&&</code>	left, short-circuit
20	<code> </code>	left, short-circuit
10	assignment operators	right

Precedence determines grouping. It does not replace FoLang’s normative left-to-right operand evaluation or target-first assignment evaluation.

Exponentiation binds more tightly than prefix operators, so `-2 ** 2` means `-(2 ** 2)`. Both prefix and postfix forms of `++` and `--` are accepted.

DECISION-OP-002 — right-associative assignment

```
a = b = c;
```

is parsed as:

```
a = (b = c);
```

An assignment expression yields the value assigned. Runtime assignment operators are:

```
= += -= *= /= %= **= &= ^= |=
```

DECISION-OP-004 — increment and decrement fixity

`++` and `--` are accepted in prefix and postfix positions. Prefix form mutates the target and yields the new value. Postfix form yields the previous value and then mutates the target. Their operand must be mutable and assignable.

DECISION-OP-003 — definition operators are statement-only

- `:=` declares and initializes an inferred variable and errors when the name already exists.
- `?=` declares and initializes when absent; otherwise it reassigns the existing binding.
- Neither operator is a general expression operator or may be chained.
- `::=` remains reserved until its semantics are defined.

DECISION-EXT-001 — custom operators

- Overloading a built-in symbol retains that symbol’s built-in fixity, precedence, and associativity.
- A new symbol requires explicit `fixity`, numeric `precedence`, `associativity`, and `arity` metadata.
- Higher numeric precedence binds more tightly.
- Associativity is `left`, `right`, or `none`.
- The compiler collects operator declarations before parsing operator expressions.
- Maximal munch selects the longest valid operator token.
- A custom operator may not redefine structural delimiters, comment delimiters, arrows, assignment/definition operators, or other reserved grammar tokens.

3. Lexical contract and C++ identifier lowering

DECISION-LEX-001 — source encoding and FoLang identifiers

- Source files are UTF-8.
- An optional U+FEFF BOM is allowed only at the beginning of the file.
- Invalid UTF-8 is a lexical error.
- A FoLang identifier begins with `A-Z` or `a-z`.
- Remaining characters may be `A-Z`, `a-z`, `0-9`, or `_`.
- Two consecutive underscores are prohibited.
- A trailing underscore is prohibited. This prevents appending `_fo` from creating a C++-reserved `__` sequence.

- A lone `_` is a contextual wildcard, discard, or filename-derived-name token and is not an identifier.
- Identifier comparison is byte-for-byte and case-sensitive; Unicode normalization is irrelevant because non-ASCII identifier characters are not accepted.
- FoLang reserved words are emitted as their reserved token kinds, not as identifiers.

Equivalent regular-language description:

```
[A-Za-z][A-Za-z0-9]*(?:_[A-Za-z0-9]+)*
```

Valid:

```
employee
Employee1
employee_name
x_1_y2
```

Invalid:

```
_employee      // leading underscore
employee__name // consecutive underscores
employee_      // trailing underscore
1employee      // leading digit
_              // dedicated contextual token
नाम            // non-ASCII identifier characters
```

DECISION-BACKEND-001 — `_fo` C++ suffix

Each resolved user-defined FoLang identifier is lowered to the C++ IR by appending `_fo`:

```
employee      -> employee_fo
class         -> class_fo
employee_name -> employee_name_fo
```

This makes C++ keywords harmless after lowering. Built-in FoLang names such as `co.*`, reserved words, externally linked names, and compiler-generated internal names follow their separately defined lowering or linkage rules.

The symbol table and AST retain the original FoLang spelling. The C++ name is a backend lowering result; source-level name resolution never uses the suffixed spelling.

DECISION-LEX-002 — whitespace and comments

- Spaces, tabs, form feeds, and line terminators separate tokens.
- `//` introduces a line comment.
- `/* ... */` introduces a non-nesting block comment.
- Documentation comments remain ordinary comments until a separate documentation model is defined.
- Newlines never terminate statements.

DECISION-LEX-003 — token selection

The lexer uses maximal munch. Reserved multi-character tokens are chosen before shorter prefixes.

```
<.< before <.. or <
**= before ** or *
=>> before => or =
..< before .. or .
```

Comment openers are recognized before `/` is treated as an arithmetic operator.

4. C++-compatible built-in literals

DECISION-LIT-000 — pass-through literal policy

FoLang accepts built-in literal spellings supported by the configured C++ backend dialect. For every literal token, the frontend stores:

literal category
complete raw source lexeme

The C++ backend emits the raw lexeme unchanged. Digit separators, prefixes, suffixes, escape spelling, encoding prefixes, and raw-string delimiters are not reconstructed.

The grammar mirrors the current C++ built-in numeric, character, and string literal forms used by this revision. A conditionally supported C++ suffix is accepted only when the configured backend compiler supports it.

This policy reuses C++ **lexical spelling and literal value interpretation**, but the resulting value participates in FoLang's type and object model. It does not automatically import C++ array, pointer, storage-duration, mutability, or identity semantics.

The following are not introduced by this decision:

- the C++ pointer literal `nullptr`;
- C++ user-defined-literal operator lookup such as `operator""_km`;
- arbitrary C++ tokens that are not FoLang literals.

FoLang-specific user-defined literals are governed separately by `DECISION-LIT-004`.

DECISION-LIT-001 — integer literals

Supported C++-compatible forms include:

```
0
42
1'000'000
0b1010'0110
0755
0xCAFE'BABE
42u
42L
42ULL
42uz
```

Rules:

- binary prefix: `0b` or `0B`;
- octal: leading `0`, not `0o`;
- hexadecimal prefix: `0x` or `0X`;
- digit separator: apostrophe `'`, only between digits;
- suffixes: `u/U`, `l/L`, `ll/LL`, `z/Z`, in C++-permitted combinations.

Underscore is not a numeric separator.

DECISION-LIT-002 — floating literals

Supported C++-compatible forms include:

```
1.0
1.
.5
1e10
1.602'176'565e-19
1.0f
1.0L
0xC.68p+2
0x1p-4
```

Rules:

- decimal floating literals use an optional decimal exponent `e/E`;
- hexadecimal floating literals require a binary exponent `p/P`;
- apostrophes may separate digits;
- `f/F` and `l/L` suffixes are supported;

- `f16/F16`, `f32/F32`, `f64/F64`, `f128/F128`, and `bf16/BF16` are accepted only when supported by the configured C++ backend.

The range expression `1..10` remains unambiguous under maximal munch: it is integer `1`, range operator `..`, integer `10`, rather than floating literal `1.` followed by `.10`.

DECISION-LIT-003 — character and string literals

FoLang accepts the configured C++ backend's built-in character and string literal spellings, including:

```
'a'
'\n'
u8'x'
u'Ω'
U'Ω'
L'x'

"text"
u8"text"
u"text"
U"text"
L"text"
R"(raw text)"
u8R"tag(raw UTF-8 text)tag"
```

Supported syntax includes:

- encoding prefixes `u8`, `u`, `U`, and `L`;
- C++ simple escapes;
- octal and hexadecimal numeric escapes;
- universal character names `\\uXXXX`, `\\UXXXXXXXX`, `\\u{...}`, and `\\N{...}`;
- C++ raw strings with matching delimiters of at most 16 valid delimiter characters;
- adjacent string-literal concatenation.

C++ multicharacter literals are conditionally supported only when accepted by the configured backend. Prefixed character literals must satisfy the configured backend's C++ restrictions.

DECISION-LIT-004 — FoLang user-defined literals

C++ user-defined-literal operator syntax and C++ `operator""` lookup are not part of FoLang's source semantics.

FoLang user-defined literals use FoLang-defined syntax and semantics. The frontend resolves each such literal to a target FoLang type and retains:

```
literal category
complete original source spelling
target FoLang type
structured literal components
```

The backend may lower a FoLang user-defined literal to any semantically equivalent C++ representation, including:

- a native C++ built-in literal when the FoLang value has a direct equivalent;
- aggregate initialization for a generated struct;
- constructor invocation for a generated class or scalar wrapper;
- a static factory invocation;
- a generated helper function;
- an ADT or variant-case constructor;
- a `constexpr` or `constexpr` construction helper.

For a struct-like FoLang literal, lowering may produce:

```
Point_fo{
    .x_fo = 10,
    .y_fo = 20
}
```

For a class whose invariants must be enforced, lowering may instead produce:

```
Point_fo::create_fo(10, 20)
```

The selected C++ representation is a backend implementation detail. It does not define FoLang source syntax, overload resolution, type checking, or literal semantics.

The EBNF currently represents this category with a contextual production until the complete FoLang declaration and invocation syntax for user-defined literals is finalized. A parser must not approximate this category as a C++ token suffix such as `12_km` unless FoLang independently defines that exact form.

5. Collection and metadata punctuation

DECISION-COL-001 — canonical separators

- Enum variants use commas, with an optional trailing comma.
- Array and tuple elements use commas.
- Map entries use `key: value` and commas.
- Object constructors use `field: value` and commas.
- Annotation maps use `key: value` and commas.
- Named annotation arguments continue to use `name=value`.
- `=` inside an object initializer or annotation map is rejected.

```
employee := Employee{  
    id: 1001,  
    name: "Rao",  
};
```

6. Ambiguity-resolution rules

1. `{ ... }` is parsed according to its syntactic position: declaration/block body, map literal, annotation map, or object-construction body.
2. `|x| => expression` starts a lambda only where a primary expression may begin; infix `|` is otherwise bitwise OR.
3. A qualified name followed by `{` is object construction only when the name resolves in type position.
4. Type argument/derivation syntax is parsed in type positions; call syntax is parsed in expression positions.
5. Range operators are non-associative. `a..b..c` is invalid unless parentheses explicitly create nested ranges.
6. Collection/map/object trailing commas are accepted; semicolons are not substitutes for commas.
7. Planned constructs remain feature-gated even when represented in the grammar.
8. C++ literal maximal munch is applied before FoLang operator parsing, so hexadecimal exponents, suffixes, and raw strings remain single literal tokens.
9. A C++ user-defined-literal spelling is not resolved through C++ `operator""` lookup. It is accepted only when the same source form is independently defined by FoLang's user-defined-literal mechanism.
10. FoLang user-defined literals are resolved by FoLang type and literal declarations before C++ lowering; the selected constructor, aggregate, factory, helper, or native-literal representation does not affect source parsing.

7. Existing contextual rules retained

The generated grammar does not weaken the reference's contextual rules:

- ordinary package files contain exactly one primary top-level declaration;
- entry files permit their restricted entry-local declarations and executable statements;
- library surfaces permit only boundary declarations and adapter functions;
- units contain functions and companion ownership is checked semantically;
- no physical local or nested named declarations are introduced;
- `forall(T).type` remains type-position syntax;
- capability, visibility, realm, capture, and definite-initialization rules remain semantic checks.

8. Reference-document edits implied by these decisions

Before declaring the grammar normative, update `language-ref.md` to:

1. add the ASCII identifier contract and `_fo` C++ lowering rule;
2. explicitly prohibit consecutive and trailing underscores;
3. add the C++-compatible built-in literal contract and raw-lexeme preservation rule;
4. replace `_` numeric separators with apostrophes and replace `0o` octal examples with leading-zero octal;
5. add C++ integer and floating suffixes, hexadecimal floating literals, encoding-prefixed literals, universal character names, and raw strings;
6. explicitly exclude C++ `operator""` lookup and `nullptr` from built-in literal pass-through;
7. add the FoLang-specific user-defined-literal model and document its declaration and invocation syntax when finalized;
8. state that user-defined literals may lower to native C++ literals, constructors, aggregate initialization, factories, generated helpers, or ADT/variant construction;
9. add the built-in precedence table;
10. state assignment's right associativity and value result;
11. state the mandatory-semicolon/no-ASI rule;
12. normalize examples by adding missing semicolons to simple statements and standalone directives;
13. replace enum semicolon separators with commas;
14. replace `=` with `:` inside object initializers and annotation maps;
15. document custom-operator collection and precedence integration.